

Cognitive representations of social relationships and their developmental origins

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Target Article

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Short Abstract

In this paper, I use findings from infant psychology to propose that humans have core knowledge of social relationships. I propose that these representations allow even infants to categorize relationships into a model (following relational models theory, i.e., communal sharing, authority ranking, or equality matching) and compute their strength (based on factors like expectation of future interaction, obligations, and commitment). Throughout the paper, I review evidence in support of this proposal and consider alternative explanations.

Long Abstract

In the human mind, what is a social relationship, and what are the developmental origins of this representation? I consider findings from infant psychology and propose that our representations of social relationships are intuitive theories built on core knowledge. I propose three central components of this intuitive theory. The purpose of the first component is to recognize whether a relationship exists, the purpose of the second is to characterize the relationship by categorizing it into a model and to compute its strength (i.e., intensity, pull, or thickness), and the purpose of the third is to understand how to change relationships through explicit or implicit communication. I propose that infants possess core knowledge on which this intuitive theory is built. This paper focuses on the second component and considers evidence that infants characterize relationships. Following Relational Models Theory (A. P. Fiske, 1992, 2004) I propose that from infancy humans recognize relationships that belong to three models: communal sharing (where people are 'one'), authority ranking (where people are ranked), and equality matching (where people are separate, but evenly balanced). I further propose that humans, and potentially infants, recognize a relationship's strength which can be thought of as a continuous representation of obligations (the extent to which certain actions are expected), and commitment (the likelihood that people will continue the relationship). These representations and the assumption that others share them allow us to form, maintain, and change social relationships throughout our lives by informing how we interpret and evaluate the actions of others and plan our own.

From the moment we are born and throughout our lives, we depend on social relationships for our happiness (Buettner et al., 2020; Demir, 2010), well-being (Cohen, 2004; Kawachi & Berkman, 2001; McCrory et al., 2022), and survival (Alvarez, 2000; Silk, 2007). For most people, life is made meaningful because of our social relationships. We rejoice when babies are born or when people get married. We go into mourning when our loved ones leave us by choice or by death. Every day, we act to maintain or change our existing relationships and create new ones. For example, depending on our culture, we may 'know' it is acceptable to eat off our spouse's plate but unacceptable to eat off our boss's plate. We may wait for the boss to start a meeting to maintain a deferential relationship or do a favor for a coworker to maintain a cooperative one. We might bring a rose to a friend to create an intimate relationship or advocate that our book club vote on the next book instead of having one person decide. We care for and love our children more intensely than other children. We go into deep mourning when some people die but not when others do. To make sense of these situations, we cannot simply think of others and ourselves as individuals who have traits (e.g., 'good', 'evil') or individuals that belong to social groups (e.g., 'Armenian', 'Red Sox fan'). We must also think about social relationships between individuals. How do people acquire this shared knowledge of relationships? In this paper I propose that knowledge of social relationships is based on core knowledge: early developing or innate knowledge (A. P. Fiske, 1991; L. A. Hirschfeld, 2001; Kaufmann & Clément, 2014; Tatone, 2017; Thomsen & Carey, 2013). The structure of this core knowledge shapes how adults think about social relationships. First, I will characterize adult knowledge of social relationships and then I will present evidence that this intuitive theory is shaped by knowledge that appears in infancy.

Core knowledge of social relationships: a proposal

When humans, including young children, reason in domains such as physics, navigation, and psychology, we use intuitive theories, which are interconnected systems of knowledge that allow us to make sense of varied and unfamiliar situations and plan appropriate actions (S. Carey, 1987; Gelman, 2005; Gerstenberg & Tenenbaum, 2017; Gopnik & Wellman, 1994). These intuitive theories are proposed to be built on very early developing or innate knowledge called ‘core knowledge’. Core knowledge serves as inductive biases: it supports infants’ ability to organize the vast amounts of information in their environment and direct infants’ attention to relevant information (S. Carey, 2009; Kemp & Tenenbaum, 2008; Spelke, 2022, p. 20; Spelke & Kinzler, 2007; Tenenbaum et al., 2011; Ullman & Tenenbaum, 2020; Xu, 2019). Core knowledge therefore shapes intuitive theories. What are intuitive theories of social relationships and what core knowledge shapes them?

First, consider the core components of intuitive physics and psychology. The core components across these domains are analogous. When we reason about physics, we reason about entities (things in the world), properties of the entities (e.g., heavy, smooth), and forces (how entities change properties); when we reason about psychology, we reason about agents (a specific type of entity), states (psychological or physical), and actions (how agents change states) (refer to Spelke, 2022; Ullman, 2015 page 200; and Liu, 2022; Liu et al., 2024). I propose that our intuitive theory of relationships has analogous core components: connections (who is socially connected?), characteristics (what type of connection is it and how strong is it?), and communication (the way that connections are established, maintained, and changed). That is, much the same way we reason about properties of entities and states of agents, we reason about the characteristics of relationships. Much the same way that we represent forces to be the vehicle by which entities change or maintain their properties (e.g., a force causes a change in the location of a ball) and we represent actions as the vehicle by which agents change or maintain states (e.g., a person moves their hand to reach for an object), we represent communication as the vehicle by which a social connection changes or maintains its characteristics (e.g., accepting a hug from someone to communicate social closeness). This communication can be explicit (e.g., saying, ‘I declare my allegiance and loyalty to you’) or implicit (e.g., expressing comfort with using the same spoon as a family member). Like the core components in physics and psychology, each of these components have foundations in infancy. In this paper, I focus on the second component: characterizing relationships. I propose that two basic computations determine the way humans represent characteristics of social relationships: categorizing the connection by model and computing the strength of the social connection.

While the core components of each domain are analogous, and the domains interact, reasoning about social relationships goes

beyond the domains of physics and psychology (for a discussion about the interdependence of intuitive psychology and physics, refer to Liu, 2024). Consider first that intuitive physics helps us understand the actions of people (or agents more generally). We can use our knowledge of agents as physical bodies to know it is unlikely that a person will walk through a wall. However, we need more than an understanding of the physical world to make sense of many actions of agents: when we see a person turn left to reach a door, we cannot only use intuitive physics to understand the action—we also need a representation of the person’s goal. Likewise, some behavior cannot be understood by only representing a single agent’s mental state. Take the notion from intuitive psychology, in which people expect agents to act efficiently toward achieving goals (S. Carey, 2009; Gergely & Csibra, 2003; Jara-Ettinger et al., 2016; Liu et al., 2017; Spelke, 2022). If we observe someone bowing and moving out of the way for someone else and then moving forward on their path, without an understanding of relationships, the action seems irrational. We could explain this action as evidence that the bower is prosocial, but it would not tell the whole story. If we know that relationships are at play, we understand that bowing communicates a certain role in a relationship (A. P. Fiske & Schubert, 2012; Kajanus et al., 2019; Thomas, Radkani, et al., 2022 refer to Box 2).

What do core representations of relationships allow us to do, and how are they distinct from representations of traits or groups?

Like representations of traits and social groups, representations of social relationships constrain and affect the way we explain and evaluate behavior. The difference between representations of traits, relationships, and groups is that they each lead to different generalizations. To illustrate, imagine you are on the subway, and you see someone return a dropped wallet. Without other social cues, you may attribute the behaviors to traits—for example, the wallet-dropper is forgetful, and the wallet-returner is kind. This could lead you to change predictions about the behaviors of the actors toward most people. For example, the wallet-returner may open a door for someone, and the wallet-dropper may be late for a meeting. Now imagine that the two people are wearing Red Sox hats, and the wallet-returner says, ‘Go Red Sox’, as they return the wallet. Here you may explain the interaction in terms of social groups, leading you to predict, for example, that the wallet-returner will extend prosocial behaviors to other people wearing Red Sox hats. The wallet-returner might even be less likely to help a person wearing a Yankees hat. In other words, the representation of groups would afford predictions concerning people who belong to the group of Red Sox Fans. Finally, imagine you see the same interaction, but first, the two people kiss. Here you would likely infer that the pair is a romantic couple. This could lead to predictions that are constrained to the couple. For example, the wallet-returner might hold open a door for their lover but may not be more likely than others to hold open a door for a stranger. You would be surprised to see them hug or kiss seemingly unrelated people on the subway.¹ One way of visualizing the distinction between these representations is to imagine a social network, where nodes are individuals, and connections between nodes are interpersonal relationships. Trait inferences change your expectations about how nodes with act toward most other nodes, group inferences update your expectations about different aggregates of nodes and how nodes from these aggregates interact, while relationship inferences change your expectations about the lines

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between nodes and how those two nodes will interact (refer to Box 3). Thus, infants' representations of social relationships would lead them to make different predictions than representations of traits and groups. Next, I will describe in detail how humans, including infants, may recognize relationships.

How do we recognize relationships?

So far, I have described the components of a relationship representation: connections, characteristics, and communication. Next, I proposed their function—they allow humans to make sense of interactions between individuals, updating predictions about whether and how people will interact. What follows is that relationship representations, require us to track individuals. The ability to track individuals has deep phylogenetic roots. For example, chimps recognize group mates even after decades of separation and they better remember those with whom they had closer relationships (Lewis et al., 2023). This ability has also been found in elephants, horses (Proops et al., 2009), sheep (Kendrick et al., 2001), and even wasps (Sheehan & Tibbetts, 2008). Human newborns recognize their mother's face (Field et al., 1984), scent (Vaglio, 2009), and voice (DeCasper & Fifer, 1980). Humans as young as four months (Kosakowski et al., 2022, 2024), have specialized face regions in the brain whose connectivity to other regions suggest that this specialization is driven by social value, perhaps associated with forming relationships with individuals (refer to Powell et al., 2018).

But recognizing relationships goes beyond tracking individuals. How do we know that individuals are connected? It may depend on recognizing cues that people are mutually aware.² Newborn infants recognize many cues of social engagement: they distinguish direct gaze from averted gaze (e.g., Farroni et al., 2002), prefer infant directed speech (e.g., Cooper & Aslin, 1990), are aware of contingency, (e.g., Bigelow, 1999; Bigelow et al., 2008; refer to Csibra, 2010) and Spelke, 2022 for an in depth review). As observers, infants as young as 12 months differentiate when others look at each other compared to when they don't, and expect people to look at one another when communicating (Beier & Spelke, 2012). Thus, infants may use cues of mutual awareness, such as mutual gaze, contingency, or joint attention, to infer whether a connection exists at all (refer to Wolf & Tomasello, 2023).

How do we characterize relationships?

Categorizing Relationships by Model

While inferring a connection would allow infants to predict whether people will interact, characterizing relationships would allow infants to predict *how* they will interact. How might infants do so? Alan Fiske's Relational Models Theory proposes that humans understand nearly all aspects of social life by categorizing relations using four basic models. Fiske calls them models because he proposes they coordinate nearly all aspects of social life including predicting, planning, and evaluating behavior. The first model is communal sharing, where people are seen as 'one'. People in communal sharing relations feel they overlap or share something. This can include shared land, bodily substances like blood, or experiences like suffering or a love of a cuisine. In authority ranking, people are ranked. People differentiate themselves, taking on different responsibilities, levels of prestige, and/or privilege. Examples of ways that people are ranked include age, gender, birthright, knowledge or skill. In equality-matching

relations, people are seen as distinct but equal. People are motivated to keep an even balance and are aware of imbalances. Examples include voting, returning favors, and start-lines in a race.³ (refer also to (Clark & Mills, 1979; Graeber, 2012). In support of this theory, Alan Fiske and colleagues have amassed ethnographic and behavioral evidence that suggests that universally, humans think about relations using these four basic models (A. P. Fiske, 1992; A. P. Fiske & Haslam, 2005; Graeber, 2012; Rai & Fiske, 2011). In part, because there is evidence that relational models are culturally widespread, Fiske proposes that infants have innate knowledge of these models (refer also to (Kaufmann & Clément, 2014; Tatone, 2017; Thomsen & Carey, 2013).

Computing a relationship's strength

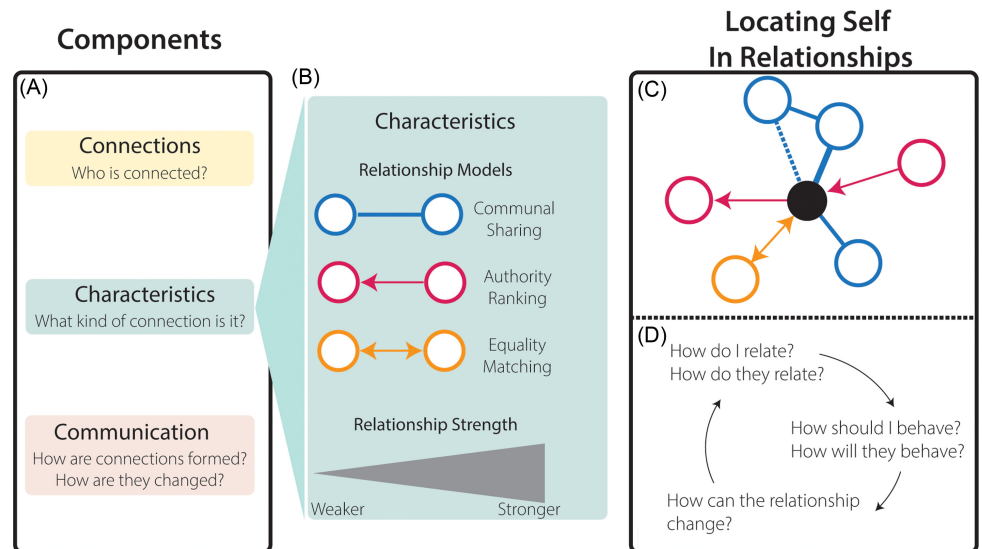
While models can tell us what to expect even in fleeting social interactions, computing a relationship's strength would tell an infant what to expect over many interactions, across time and contexts. This idea overlaps with proposals in many theories of social relationships, including proposals about pull (A. P. Fiske & Fiske, 2007), attachment (M. D. S. Ainsworth et al., 1974), interdependence (Rusbult & Van Lange, 2008), thickness (Margalit, 2017, p. 2), and strength (Granovetter, 1973). I propose that each of these proposed concepts relate to one underlying representation which I call "relationship strength". Relationship strength is a continuous computation: it informs the extent to which we are committed to someone (i.e., the strength of our intention to maintain the connection) and/or the extent we feel obligated toward someone (i.e., the extent to which certain actions are normatively expected). Relationship strength does not necessarily describe the 'strength' of the model (e.g., how highly ranked we see someone or how much overlap we have with someone, though this may be correlated). The computation occurs in relationships for each model. While people can be highly committed to many things, the extent to which they are committed to single individuals over time⁴ and across contexts⁵ determines the relationship's strength. For example, a highly devoted teacher may feel committed to teaching because she is generally motivated to impart knowledge on the next generation. However, if this commitment is not directed toward specific individuals the relationships would be relatively weak. By contrast, a teacher could have strong relationships with her students, if she is committed to individuals across contexts and intends to maintain relationships with them even after the course is over. Likewise, you might feel highly obligated to help victims of a hurricane, but if your obligation is not directed toward individuals, or if your obligation is not caused by a relationship, then it wouldn't be evidence of a strong relationship.

Relationships that fall under each of the three relational models can be relatively weak or strong. However, the likelihood of very strong or very weak relationships may vary depending on the governing relational model people use in a single relationship. Below are examples of relationships in each relational model that are relatively strong or weak. For example, you might feel much closer and more committed to your spouse than a teammate but use the logic of communal sharing in both relationships. The logic of communal sharing can even happen in fleeting relations, such as when people feel as one with strangers at a concert, sports, or religious event. Likewise, authority ranking relations can also be strong or weak: one mentee could be devoted to their mentor, while another may see their mentor as a temporary boss to whom they feel no allegiance and do not expect their relationship to continue

INTUITIVE THEORY OF RELATIONSHIPS

Figure 1. Intuitive Theory of Relationships

(A) The components of our intuitive theory of relationships are connections (which individuals are connected), characteristics (what is the relationship model, and what is the strength of the connection?), and communication (how are connections formed, maintained, or changed?). (B) People represent the characteristics of relationships along two dimensions: relational model and strength. (C) People (black dot) must locate themselves in relationships and learn about potential new social connections (dotted line). (D) This is a process of asking how people relate, what behaviors follow from different types of relationships, and what behaviors or actions might change these relationships or groups.



after the employment has ended. Authority ranking can also occur in fleeting interactions, such as when people feel they owe deference to a judge in court. Interestingly, Graeber suggests that equality matching relations encourage weak relationships since when 'debts' are fully paid, there is no necessity to interact in the future (Graeber, 2012). Thus, communal sharing relationships and authority ranking relationships may have a larger range of 'strength' as opposed to equality matching. Even so, equality matching relations can sometimes be strong and long-lasting, such as between neighbors who have long been doing favors for one another, or weaker, such as children taking turns at a playground. Each model may have different emotions tied with strength, which could serve as one cue to infer a strong relationship (e.g., attachment or 'kama muta' (A. P. Fiske et al., 2016) in communal sharing relationships; allegiance in authority ranking relationships; and trust in equality matching relationships).

One untested hypothesis about relationship strength is that the stronger a relationship (i.e. the stronger you feel attachment, allegiance and loyalty, the more you expect the relationship to continue, and the stronger you imagine the bond), the less replaceable the relationship and person. If this is true, then it might be one reason why we mourn more when stronger relationships end (Harris, 2018). Take, for example, one's relationship with their children. In almost all situations, a person's child is completely non-interchangeable. One would certainly notice if their spouse brought home the wrong child from school. The immense number of resources parents put into raising children is directed toward specific individuals. At least in most times and places, and possibly even for non-human primates (Goldsborough et al., 2020; Monsó & Osuna-Mascaró, 2021), the death of a child is often considered one of the worst things that can happen to someone, and parents do not feel that the death of their child can be solved by having a new child. Or consider that when people describe ending a relationship with one's child via other means, they often do so in terms of death (e.g., 'he is dead to me now'). This may be because death feels like the only way one can end a relationship with one's child. On the other extreme, you may not care much about which of your colleagues accompanies you to lunch or who helps you with an error in your code. If the colleague gets fired, and another

person takes that colleague's place then the person wouldn't think it's the *same* relationship, but it is conceivable that they would think the new relationship could replace the old one. The owner of a factory may see workers as individuals, but individuals who can easily be replaced by any worker who has the same skills. One can also think of friendships or acquaintance relationships that can easily be swapped or replaced⁶ (refer R. M. Carey & Markus, 2017 for evidence that the extent to which people think friendships can/should be replaced varies with social class in the United States).

Is this intuitive theory built on core knowledge of social relationships? If so, what would this core knowledge be?

I propose that our intuitive theory of social relationships is built on core knowledge. Elizabeth Spelke describes core knowledge as something between belief and perception; Susan Carey theorizes it to be iconic in form⁷ (S. Carey, 2009). Furthermore, core knowledge is innate, skeletal, largely automatic, and evolutionarily old. Core knowledge is abstract and cannot be computed from perceptual input alone. It persists throughout the lifespan and serves as the foundation of later developing representations, which are heavily influenced by cultural input, statistical learning, and explicit teaching.

Core knowledge of social relationships allows infants to track individuals and their social connections. Like other core knowledge, these representations act as inductive biases that enable learning. I further propose that infants recognize relationship models. These models are analogous to the ones found in Relational Models Theory: communal sharing, where individuals are like 'one', authority ranking, asymmetrical relationships where individuals are ranked, and, equality matching, reciprocal relationships where individuals are separate. They may be substantiated as edges of graphs, with nodes representing individuals and edges representing relationships that are undirected for communal sharing, directed for authority ranking, and bidirectional for equality matching (refer to Figure 1). These model representations could be innate. Another possibility is that they could be learned from the combination of more basic social relationship representations (i.e., connections between people: two

people are aware of one another) and abilities to recognize general relations (e.g., nominal, ordinal, interval, and ratio).

One way research could help us disambiguate these possibilities is to consider the literature on object recognition. When infants categorize objects they better individuate and remember them. A six-month-old infant is not surprised when a toy duck is replaced by a toy truck, but is surprised when a doll's face is replaced with a ball (Kibbe & Leslie, 2019). Finally, adults categorize objects at the same time they recognize them (Grill-Spector & Kanwisher, 2005). Thus, understanding how young infants distinguish relationships may help us understand whether they categorize them from an early infancy.

Whether the models are innate or early developing, infant representations of relationships are simpler than later-developing representations: they have the structure without the details. A key difference between infant and adult representations is likely that infants do not fully represent the minds of individuals within a relationship (Favre & Sornette, 2015). This is also supported by the fact that species with simpler representations of others' minds recognize relationships analogous to the three relational models (Haslam, 2013; Verbeek, 2006).

Using Authority Ranking as an example, young infants likely recognize asymmetrical relationships but may not understand the various ways rank is cued or construed. Relational Models Theory describes Authority Ranking relationships as those where one person is motivated to protect and guide, while the other is motivated to follow. These relationships are distinguished from dominance relationships, which occur when weaker individuals defer to avoid harm. By contrast, the analogous representation in infants is unlikely to be dominance or legitimate authority. As outlined below, infants do recognize relevant cues that occur cross-species, such as physical size that are associated with dominance, and do use these cues to predict behavior. However, it is unlikely they fully represent the motivations of the individuals until the second year of life. For example, toddlers distinguish between different types of asymmetrical relationships, which may include different motivations of high- and low-ranking individuals, a distinction that would require representing the minds of the individuals involved.

The skeletal nature of the infant representations allows learning of the vast cross-cultural variation in how relationships are established and marked. For example, gender, tattoos, hairstyles, clothing, and race have all been valid indicators of relative social rank in different cultures and times. Yet, they are not valid cues in all cultures, and different instances have meant different things. Or take an even more specific example. In some cultures, taking off a head covering in front of someone means that you have a family or intimate relationship with them. In other cultures, head coverings are only worn for fashion or utility and give little to no information about relationships. Infants in some cultures would need to come to know that head coverings can tell you something about social closeness, while those in other cultures would need to ignore it. Of course, this presents a learning challenge—if infants are born with abstract knowledge about relationships, they would need some way to connect perceptual input to this knowledge.

Some cues might be recognized with either very little learning or no learning (refer also to Fiske & Schubert, 2012; Thomsen & Carey, 2013). Cues that can be learned with very little or no input may be related to evolutionarily old mechanisms. Physical closeness, including touch and exchanging bodily fluids, correlates with communal sharing relationships and strength of relationships in many cultures and species, and it is also related to mothers

caring for their young. Physical size tends to highly correlate with formidability, which is what determines dominance rank in many species (Huntingford, 2013; Smith & Price, 1973). Infants may need very little or even no input to form expectations about the relation between differences in size and asymmetrical relationships (refer to Asymmetrical Relationships section below). Finally, reciprocal relationships are often, though not always, based on the exchange of goods; thus, equal resource distribution may be a salient cue of equality matching relationships. If there are innate inductive biases that relate to specific cues, then we would expect most humans to recognize them. Humans may take advantage of this to communicate about relationships (Astuti et al., 2004). Many cultures have ceremonies that feature saliva-sharing and physical contact, for example, dances where people hold hands or ceremonies where people drink from the same ceremonial cup. Likewise, many cultures make high-ranking individuals, whose bodies are not necessarily large, appear larger using thrones, headaddresses, crowns, monuments, etc. (A. P. Fiske & Haslam, 2005; Thomsen & Carey, 2013). On the other side of the spectrum, humans recognize and use many cues that are arbitrary and must be learned. Examples include some military rank symbols, specific labels (e.g., 'madame', 'sister'), and wedding rings.

Below, I review evidence that supports the hypothesis that humans possess innate or very early representations of social relationships. I organize the paper by relational model type and consider any evidence that infants infer the strength of connection within these types. While some of this proposal is supported by currently available evidence, other parts of the proposal have yet to be tested. Specifically, several studies support the proposal that infants represent relationships analogous to communal sharing and authority ranking relationships. In contrast, fewer studies support the idea that infants recognize equality matching relationships. Finally, existing studies only hint that infants represent the strength of relationships. This paper does not cover the vast amount of literature on children's and adults' ideas of social relationships, nor on how ideas about social relationships develop over the lifespan through cultural learning and first-person experience. Instead, it describes evidence that infants can recognize and categorize relationships. This knowledge lays the foundation for our adult intuitive theory of relationships.

Core Knowledge of Relationships: Evidence from Infants

Communal Sharing

In Relational Models Theory (A. P. Fiske, 2004), communal sharing relations occur when people understand themselves to be one and act in ways that reinforce this representation. In these relationships, people have expectations of care without expectations of direct reciprocation. They are mutual but not necessarily symmetrical. At least in stronger versions of these relationships, people understand their fate to be shared. People express communal sharing by accentuating similarities rather than differences. If infants have knowledge that allows them to categorize an interaction as communal sharing, they could quickly learn culturally specific cues such as whether it is normal to kiss on the mouth or spit on the forehead, when it is appropriate to hug someone, or which pieces of clothing tell you about social closeness. Below, I review evidence that suggests infants possess abstract representations of relationships that may support these more complex adult representations.

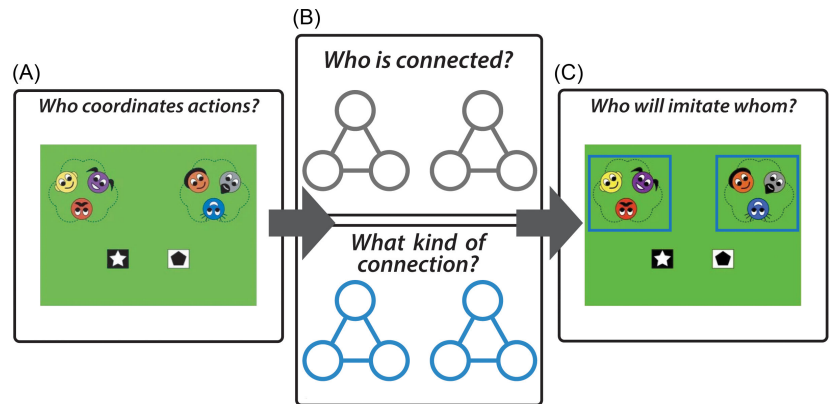


Figure 2. Stills of stimuli (green background) and diagram depicting the hypothesized underlying representation in Powell and Spelke, 2013. (A) What infants observed in the study. (B) hypothesized cognitive representation (C) What infants observed during the test trial and the prediction that they made. Diagrams were added by me.

Are non-human animals sensitive to communal sharing relationships?

This model has straightforward connections to affiliative relations and associated cues found in other species, including, kin, pair-bond relationships (Alcock, 2009) and ‘friendships’, which are characterized as long-lasting affiliative bonds with unrelated individuals who are not sexual mates (Cheney & Seyfarth, 2012; Proops et al., 2021). Primatologists have observed that close physical contact between individuals predicts long-term interaction. Examples include grooming, lip-smacking, or other group-specific behaviors, such as putting hands in one another’s mouths (Perry et al., 2003; Silk et al., 2013). Individuals in several species, such as baboons, vervet monkeys, and chimpanzees, also reason about these relationships as third-party observers. This suggests that they form abstract representations of social relationships (Cheney & Seyfarth, 2008, 2012; Seyfarth & Cheney, 2012). For example, when an individual expresses distress, they look toward that individual’s kin as though they expect that kin will be more likely than non-kin to respond (Cheney & Seyfarth, 1990; Seyfarth & Cheney, 2012). While the nature of these representations, as well as the process through which they are formed, is not yet known, the findings suggest that our ability to reason about these types of relationships could be supported by evolutionarily old mechanisms.

Are infants sensitive to cues of communal sharing?

Observing Social Interactions

During the first year of life, humans recognize cues of people acting as one, including when individuals imitate one another, act in rhythmic synchrony, respond to distress, eat together, or share bodily fluids. For example, in one set of studies, 8-month-old infants saw animations in which two triads of animated characters moved their bodies in rhythmic synchrony (Powell & Spelke, 2013) (refer to Figure 2). When observing a new scene, infants expected that the characters would imitate the actions of the members of their triad as opposed to the other triad. Importantly, infants did not match ‘same’ with ‘same’: infants expected imitation among triads when the characters had different physical characteristics (i.e., were different colors and had different styles of hair). Likewise, when the characters looked the same but failed to coordinate their actions, infants did not expect imitation. Infants could not have used trait inferences to make predictions. The characters do the same actions in the familiarization events, what differs are their social partners, thus infants likely see relationships. Why might these studies be evidence for representations of

communal sharing? In both the familiarization and the test events, the characters made themselves appear to be one unit. To adults, this coordinated action looks like a dance: each character does the same thing at the same time, moving in a circle. In the test events, infants expected that individuals would again make themselves alike by imitating each other’s actions. It is unlikely that the 8-month-old infants from the United States had much experience with people dancing, much less had the opportunity to compare subsequent behavior of individuals who dance together or not. Yet, infants recognized a cue that, in many cultures, identifies communal sharing relationships.

Expectations that arise from recognizing similarity may emerge even earlier. Four- and 5-month-old infants expect a character to approach another character it had previously imitated rather than one it had merely responded to (Powell & Spelke, 2016). However, these young infants did not expect imitation to predict mutual responses: they did distinguish scenes where a character approached someone who had previously imitated the character. Like young infants, twelve-month-old infants anticipated that a puppet who imitated an actress would respond to her distress. Unlike young infants, they also expected the targets of imitation to respond, suggesting expectations of mutual behaviors (Kudrnova et al., 2024; refer also to Pepe & Powell, 2023). Moreover, their expectations did not extend to individuals outside the initial interaction or to scenes where the actress laughed, suggesting they saw imitation as a cue of relationships. While there are several reasons why infants may have shown different patterns of results across the two studies, one explanation is that responses to distress are a better measure of relationship representations than approach behavior (for example, we approach objects we like but we can only comfort another person). It is also possible there is a developmental change such that younger infants see imitation as evidence that “A likes B” which would only affect predictions about A’s behavior toward B, while older infants see it as evidence that “A is connected to B,” which would also change predictions about B’s behavior.

The results reviewed thus far suggest that young infants use social cues in which individuals make themselves more like someone else to predict subsequent behavior. Converging evidence comes from another study, 9-month-old infants expected members of synchronized triads to help each other, rather than members of other triads, in a right-of-way conflict (Pun et al., 2021). Infants and toddlers also use other cues in which people make themselves more alike to predict future affiliative behavior. For example, toddlers (aged 2 1/2 years and 17 months) expect that individuals who refer to themselves by the same label will support

each other (Jin & Baillargeon, 2017; Ting et al., 2019; Ting & Baillargeon, 2021). Infants as young as 6 months expect people who announce the same preferences for food and express positive emotions when eating together to be friendly toward one another (Lieberman et al., 2016, 2021).

Interestingly these behaviors can be construed, at least by adults, as intentional—they are things people do to make themselves more like others. In the study with the dancing groups, intentional similarity was compared to happenstance similarity (i.e., individuals who happen to look alike). This suggests that infants use intentional actions to recognize communal sharing relationships. Infants also reach for puppets who express the same preference as the infant, but only when the puppet expressed their preference after the infant, not when the puppets expressed their preference first. This suggests that the communicative act of choosing the same mittens as the infant influenced the infant's evaluation (Mahajan & Wynn, 2012; Spelke, 2022).⁸ Likewise, 12-month-old infants expect that people who wear the same clothing will support one another (Ting et al., 2019). However, they no longer expect support when similar clothing has an instrumental purpose (Bian & Baillargeon, 2022). The distinction between intentional similarity and happenstance similarity is central to the way that many scholars in fields such as Anthropology and Sociology conceive of social relationships: relationships are created through intentional communication and actions (Ingold, 2018). While I do not propose that young infants need to represent the minds of the individuals to infer a communal sharing relationship, they may be more sensitive to actions that have cues of intentionality, perhaps because those cues have been more useful over evolutionary time.

In contrast to the proposal that intentional similarity matters, there is evidence that 9-month-old infants expect friendly interactions between people who speak the same language, which for people who speak one language could be construed as unintentional (Lieberman et al., 2017). Likewise, infants' first-person evaluations are affected by whether others share characteristics with their caregivers. For example, very young infants look more at faces who share the race of their caregivers (Bar-Haim et al., 2006; Quinn et al., 2008). Infants seem to prefer to interact with those who speak the same language as their caregivers (Kinzler, 2021). Interestingly, while infants look longer at same-race individuals, they do not prefer to take toys from them, suggesting that their visual preference might be based more on lower-level perceptual preferences than social preferences (Bar-Haim et al., 2006; Kinzler & Spelke, 2011; Quinn et al., 2008). In contrast to race, but agreeing with the results about language, 5-month-old infants preferentially look toward someone who had previously sung a familiar song (Soley & Sebastián-Gallés, 2015). (In this U.S. American sample, this was 'Twinkle Twinkle Little Star'; they used parental report to confirm the infants had heard the song before). Eleven-month-old infants also prefer people who sing a familiar song that had only recently become familiar because their parents are assigned to sing it to them, but not when their parents were assigned to play the song from a toy. This is true even after several weeks of not hearing their parents sing the song (Mehr & Spelke, 2018). Because language and music are learned over social interactions, it may be a good cue of a person's history of social interactions (Kinzler, 2021). In line with the other studies covered earlier in the section, this body of work could reflect representations of communal sharing relationships. That is, singing the same songs or speaking the same language could be seen as evidence that individuals have made themselves more alike in the past. They could also represent representations of groups (refer to Box 3)

Do infants think about relationship strength in communal sharing relationships?

So far, I have argued that infants recognize communal sharing relationships. A series of studies suggest that infants may also recognize the strength of communal sharing relationships. Both infants (8 to 10 months old) and toddlers (16 to 18 months old) use interactions that include saliva-sharing, as opposed to other cooperative interactions, to predict responses to distress (Thomas, Woo, et al., 2022). Across many cultures actions that require sharing bodily fluids through actions like kissing, drinking from the same cup, etc., are distinctively found in, and expected to be in, especially strong communal sharing relationships (e.g., Fiske, 2004; Hung et al., 2022; Miller et al., 1998; refer to supplementary materials of Thomas et al. 2022e). In this study, an actress and a puppet ate from the same orange slice. Then, the same puppet and a different woman passed a ball back and forth. After the puppet expressed distress, infants and toddlers looked first and longer at the food sharer, suggesting they anticipated that the orange-sharer would respond. Importantly, these expectations were not driven by inferring the traits of the characters: participants did not anticipate the orange sharer to respond to an unfamiliar individual's distress. Moreover, toddlers did not expect the orange-sharer to respond to requests for balls. This was true in a subsequent experiment that did not include food: infants saw a woman put her finger in her mouth, put her finger in a puppet's mouth, and then back in her mouth. The same woman was then shown touching her and another puppet's forehead. Here, infants and toddlers expected the puppet who had been the recipient of the 'saliva-sharing' action to respond to the woman's distress, suggesting that their inferences did not depend on sharing food and showing that they had expectations that responses would be mutual (refer to Figure 3).

Why might this be evidence that infants represent the strength of communal sharing relationships? As mentioned above, in many cultures, sharing bodily fluids is found in especially strong communal sharing relationships (refer to supplementary materials of Thomas et al., 2022), while the other actions may be found in many communal sharing relationships, both weak and strong. On the other hand, it is possible that the participants categorized saliva-sharing as communal sharing but categorized the other actions as evidence of some other kind of relationship. Future studies will be needed to disambiguate whether participants see the same model with ball-passing and saliva sharing, or different strengths. These findings leave open the possibility that infants represent relationship strength.

Toddlers make inferences about triadic closure when observing cues of communal sharing relationships (i.e., if A and B are family, and A and C are family, then B and C are family; Spokes & Spelke, 2017). When 15-month-olds saw the same large character respond to the distress of two small characters, they expected the small characters to be socially connected. They also expected two large characters who responded to the distress of the same small character to be socially connected. However, when the characters responded instead to laughter, toddlers no longer had these expectations. This suggests representations of strength: Responses to distress, unlike laughter, may be associated with stronger communal sharing relationships, and the logic of triadic closure may be especially true in networks of strong communal sharing relationships, such as in families. This idea is integral to Attachment Theory, where an infant's propensity to go to their attachment figures when distressed, but not in other situations, is a hallmark of a secure attachment (Bowlby & Ainsworth, 2013).

Figure 3. Stimuli (from experiment) and diagram of hypothesized underlying representation driving responses in Thomas et al. (2022e). (A) What infants observed in the study. (B) hypothesized cognitive representation (C) What infants observed during the test trial and the prediction that they made.

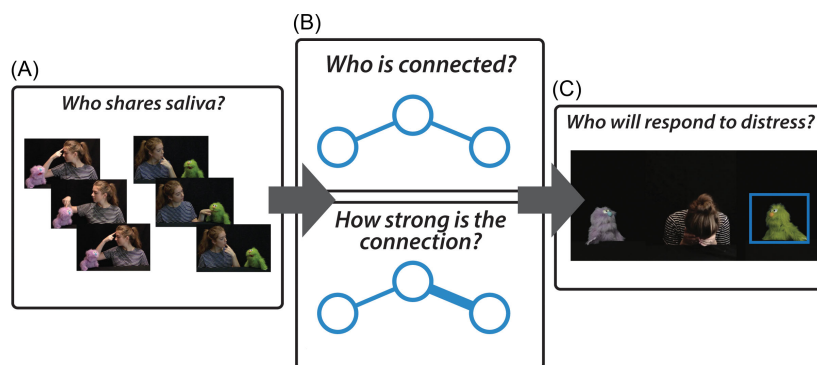
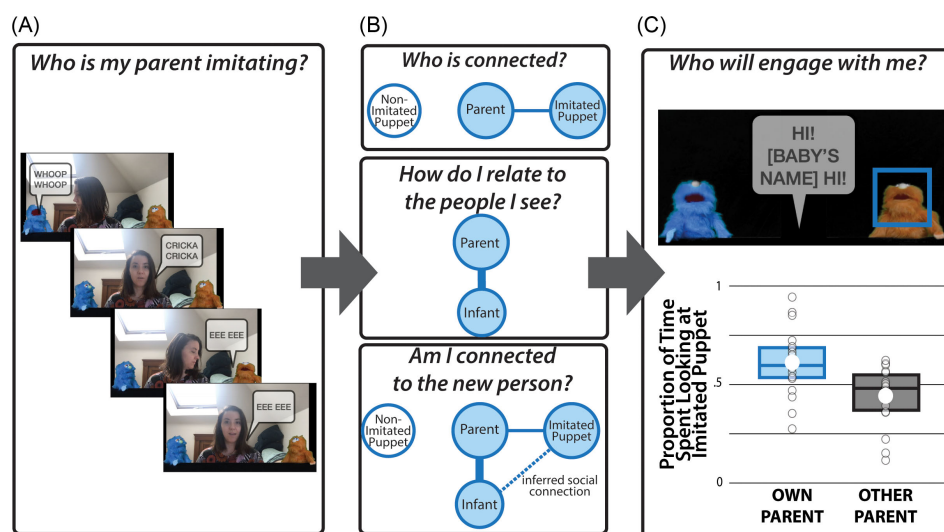


Figure 4. Stimuli from experiment and diagram of hypothesized underlying representations from Thomas, Saxe, Spelke, 2022. (A) What infants observed in the study. (B) Hypothesized cognitive representation (C; Top) What infants observed during the test trial and the prediction that they made. (C; Bottom) Data from one experiment in Thomas, Saxe, Spelke 2022, infants spent more time looking at the imitated puppet after they saw their parent's interactions (blue) but not after they saw another infant's parent's interactions (gray).



Do infants represent their own communal sharing relationships? In one study, infants used knowledge about their social relationships with their parents to infer whether they were socially connected to new individuals (Thomas, Saxe, et al., 2022) (refer to Figure 4). In this study, 12-month-old infants watched digitally edited videos of their parents interacting with two puppets. The parent showed affiliation toward one puppet by vocal imitation and had a friendly but non-imitative interaction with the other puppet by vocally responding. The question was whether the infant expected social engagement from the puppet their parent had imitated. During the test event, both puppets moved their mouths in synchrony while a friendly voice called the infant's name. Infants looked longer at the imitated puppet, suggesting they expected it to be the source of the voice. This inference was not due to attributing a friendly trait to the imitated puppet, as infants did not look more at the puppet when it appeared to have a friendly interaction with someone else off-screen. Crucially, infants' inferences were specific to interactions that involved their parents. When infants saw an unfamiliar adult (another infant's parent) interact with another set of puppets they no longer looked more at the imitated puppet. These results suggest that infants combine their knowledge of pre-existing relationships with observations of social interactions to discover new social connections. The study's details hint that infants may use cues of communal sharing relationships, such as imitation, to make these inferences. However, it remains an open question whether the strength of the infant-parent relationship was essential to their learning about new social connections.

Communal Sharing: Conclusions and Open Questions

In this section, I have argued that infants reason about observed interactions by using cues that occur in communal sharing relationships. They do this in situations they have likely not encountered before and about unfamiliar individuals, suggesting that they make predictions in these situations using abstract knowledge. The evidence here is suggestive that infants also represent the strength of communal sharing relationships. Many open questions remain.

First, many of these studies were originally interpreted as evidence that infants recognize groups. Why should we think about these studies as evidence of representations of relationships instead? In short, these studies do not provide any direct evidence that infants are generalizing beyond the interacting social partners (refer to Box 3 for a longer discussion). Second, in several of these experiments, infants may have used narrower representations of caregiver-child relationships, which are intimate and asymmetrical (refer to Thomas, Steele, Gopnik & Saxe, 2024), instead of a broader communal sharing representation. If caregiving relationships are the innate primitive it would agree with the argument in attachment theory that the quality of an infant's relationship with their caregivers acts as a model for all other relationships (M. D. Ainsworth, 1989). This is supported by findings that an infant's attachment to their parents affects their expectations about responses to distress (Biro et al., 2015; Johnson et al., 2010). Fiske argues that models can be used as a lexicon, so it is possible

that caregiving relationships are those that take on both communal sharing and authority ranking depending on the context. Third, what are the computations that infants use to infer the category and strength of communal sharing relationships? For example, do infants use naïve utility calculus, where one computes costs and rewards to infer the value someone places on things (Powell, 2021)? In short, I am skeptical that this is the only computation that infants use to represent relationships as the logic does not easily apply to all of these findings. For example, saliva-sharing has a less than straightforward 'cost' (refer to Box 2). Relatedly, how should we interpret the current argument, considering the claim that humans have an innate moral core? (refer to Box 1). Moreover, it is unclear whether infants think about obligation or commitment that goes beyond the time frame of the experiments, which may be necessary components of strong relationships. Far more work could be done to address these any many other questions.

Asymmetrical Relationships

In Relational Models Theory, asymmetrical relationships are authority ranking relationships, which occur in social hierarchies when people are ordered along a dimension (e.g., based on things such as physical size, age, skill, and military rank). In these relations, people have asymmetrical roles and often express the relation by accentuating asymmetries. For example, higher-ranking individuals are often made to appear larger (e.g., sitting on a throne, wearing a headdress, or sitting above others, such as a judge in a courtroom), brighter (e.g., wearing gold), or more important (e.g., being first or last in a procession, having a larger office).

Do non-human animals think about hierarchical relationships?

Many animals form dominance hierarchies, where rank is usually defined by priority access to resources, such as food, territory, and mates. Sometimes, dominance rank is established by one's ability to inflict harm, which correlates with factors such as body size or coalition size. Other times it is inherited or arises from group dynamics (refer to Tibbetts et al., 2022 review; refer to Mandalaywala, 2022 about whether prestige-based hierarchies are present in non-human primates). Many dominance hierarchies form because it is in the interest of weaker individuals to yield in conflicts that they would otherwise lose. Therefore, in dominance hierarchies, patterns of yielding in conflicts could be reliable cues about social rank (Smith & Price, 1973; van Vugt & Tybur, 2014).

Individuals in several species reason about relative rank when they observe patterns of yielding. Experiments show fish, birds, mice, and rats use transitive inference about dominance rank to make decisions about how to interact with new individuals (Grosenick et al., 2007; Paz-Y-Miño C et al., 2004). That is, after seeing that A is lower ranked than B, and B is lower ranked than C, the animals in the experiment are more likely to approach A than C. Field studies reveal that many other species, including hyenas, lions, dolphins, horses, and several species of primates, represent the dominance ranks of conspecifics in their group, for example, using dominance rank to predict support in conflicts (Seyfarth & Cheney, 2012). Chimpanzees also seem to connect spatial position with social rank (Dahl & Adachi, 2013). Finally, bonobos seem to prefer (they reach for) dominant individuals, even after observing interactions between simple characters (shapes with googly eyes) (Krupenye & Hare, 2018). While it is unclear how other species

come to recognize cues of dominance, this work leaves open the possibility that learning mechanisms that lead to representations of dominance relations are evolutionarily old and prevalent even in human infants.

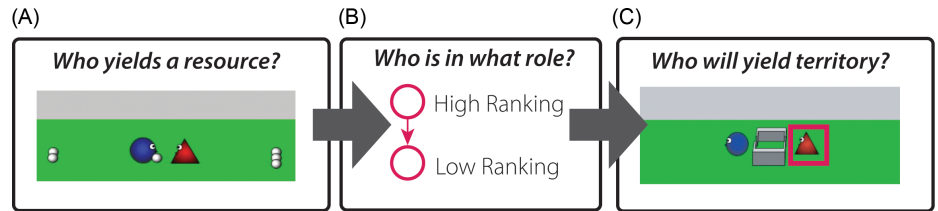
Contains Dot at the End Evidence that infants are sensitive to asymmetrical relationships.

Human infants do seem to represent dominance relations, using physical size and coalition size to predict who will yield in conflicts (Thomsen, 2019). In one study, 10-month-old infants from the United States expected smaller animated characters to yield to larger ones in a conflict. In this study, two characters (rectangles with eyes) had conflicting goals of crossing a platform in opposite directions, blocking one another's pathway. Then, one of the characters lowered its body (i.e., bowed) and moved aside, allowing the other to pass (Thomsen et al., 2011). Infants looked longer when the larger individual yielded to the smaller individual, as though it violated their expectations. In another study, Canadian 6-month-old infants were shown similar animations as described above, but the characters were the same size, here they expected a character with more allies to win (Pun et al., 2016). In another study, 12- to 16-month-old Japanese infants expected those who appeared lower in space to yield in conflicts over a resource (Meng et al., 2019). Why might these studies suggest that infants have abstract, and possibly, innate representations? Infants at these ages have likely seen conflicts. However, it is unlikely that infants at this age have observed a member of a dyad and a member of a triad in a right-of-way conflict and were able to compare the two outcomes. Together, these studies suggest that before the first year of life, infants compare the relative power of individuals (i.e., infants track who defers and who is larger, stronger, or has more allies). However, these studies do not provide evidence that infants understand these interactions in terms of relationships. Their predictions are based on the relative traits (larger, higher, or more-allied) of the individuals. In other words, there isn't evidence that infants use these interactions to update their belief about whether these individuals will interact again. Rather these studies provide evidence that infants have expectations about the outcome of an interaction given the individual traits.

Do infants use this information to inform whom they approach, as other species do? In one study 10- to 16-month-old infants were shown the right-of-way conflict described above, but this time the two characters were equal in size. Here, infants reached for the individual who yielded (Thomas & Sarnecka, 2019). Infants' preferences in these studies did not seem to be based on preferences for helpers: infants reached for the yielder even after a 'foot race' which differed only in that the characters moved in the same direction. While there are several interpretations of why infants reach for yielders, these results provide further evidence that they recognize these conflicts as social interactions potentially relevant to themselves.

Other studies provide some evidence that infants do represent asymmetrical relationships: they expect hierarchical relations to be stable across situations. In one study, 12- and 15-month-old infants in Hungary watched two animated characters compete for the same object, with one character yielding the object to the other. Later, during a conflict over territory between the same characters, 15-month-olds—and possibly 12-month-olds—expected the character who had previously given up the object to also give up the territory. Importantly, infants in this study did not expect past winners to prevail in a conflict with a new individual, suggesting

Figure 5. Stimuli used in Mascaro, 2012 and schematic (added here) of the proposed representations driving the inferences. (A) What infants observed in the study. (B) Hypothesized cognitive representation (C) What infants observed during the test trial and the prediction that they made.



that their expectations were not based on the traits of the individual but rather the relationships between the individuals (Bas & Sebastian-Galles, 2021; Mascaro & Csibra, 2012) (refer to Figure 5). At 17 months, toddlers expect rank to be related to resource distribution. When all else is equal, Italian and U.S. American infants expect equal distribution—looking longer when an individual gets more than another (Geraci & Surian, 2011; Schmidt & Sommerville, 2011; Sommerville et al., 2013). However, 17-month-old toddlers from the US seem to expect that those who prevail in conflicts will get more resources (Enright et al., 2017).

Infants also use transitive inference to infer asymmetrical relations. Since transitive inference is useful only if there are stable relations, this provides further evidence that infants represent individuals and their asymmetric connections. For example, in forceful conflicts over a toy, 10-month-old infants from the United States use transitive inference to predict who will defer (Gazes et al., 2015). When infants first saw a cow puppet forcibly take a toy from a bear puppet, then the bear forcibly take a toy from a pig, infants looked longer when the cow puppet yielded to the pig puppet, suggesting it violated their expectations (Mascaro & Csibra, 2014). Relatedly, in another study, French 14-month-olds expected dominance relations to generalize across allies, learning ‘pyramid’ structures of dominance relations but not ‘tree’ structures (refer to Mascaro et al., 2023 for a longer discussion about why pyramid structures occur more often than tree structures).

In summary, in the first year of life, infants use cues of formidability to predict how individuals will behave in conflicts. Around the end of the first year, they seem to make sense of social interactions by representing asymmetrical relationships: using prior behavior in social interactions to predict behavior between those individuals in new and different social interactions. However, none of these studies ask whether these interactions lead infants to predict whether people will interact with one another. For example, would they expect the individual who deferred over a resource to be more likely than someone who had not been in the interaction to help the high-ranking individual? Thus, it is not clear that they are representing relationships or relations between individuals.

Dominance or Authority Ranking?

So far, I have reviewed evidence that young humans understand cues of dominance found in other species and that at least around the age of one, infants may recognize asymmetrical relationships. However, especially in human relationships, rank is often based on more factors than the ability to inflict harm by being larger or by having more allies. In Relational Models Theory, authority-ranking relationships (A. P. Fiske, 1992; Henrich & Gil-White, 2001), higher-ranked individuals are expected to provide benefits, such as cultural knowledge, protection, and leadership. In these relationships, people are often made to appear larger by sitting on a throne or wearing a crown, which could be seen as ways to communicate formidability (Schubert et al., 2008). However, a person’s rank does not need to be the result of true physical

formidability. While the evidence so far suggests that infants recognize dominance relations, do they distinguish between dominance and other types of rank?

Some studies suggest that during the second year of life, humans begin to distinguish between legitimate authority and dominance. First, consider the finding described above where 10- to 16-month-old infants reach for a puppet who yields in a conflict. When toddlers aged 21 to 31 months (drawn from the same U.S. American population) were presented with the same puppet show, they reached for the winner (Thomas et al., 2018). This reaching did not depend on competence: they did not reach for a winner when it won a ‘foot race’ instead of a face-to-face conflict. While there are several explanations for this developmental change (e.g., refer to Thomas, 2018 for a discussion), participants may have understood the conflict differently. Infants see these conflicts in terms of dominance and toddlers see them in terms of authority or prestige. If so, the change in representation could be the result of a better ability to represent motivations. In support of this view, toddlers no longer reach for the winner when the winner uses coercive force to win.

Seventeen-month-old toddlers also expect leaders to provide benefits: they expect that a puppet whose directions were previously followed will be the one to intervene in conflicts and punish those who did anti-social actions (Stavans & Baillargeon, 2019). U.S. American and Italian toddlers form different expectations about future interactions, depending on the way that a high-ranking individual treats subordinates. Twenty-one-month-old toddlers saw one of two scenes. In the leader scene, a high-ranking character directed low-ranking characters, in the bully scene, the high-ranking character enforced its demands by hitting the subordinates on the head. When the leaders were absent, toddlers expected the subordinates to follow the requests of the leader to go to bed, but not the bully (Margoni et al., 2018). However, when the bully was present, toddlers expected the subordinates to obey. Interestingly, this study hints at inferences about the strength of an authority ranking relationship—the subordinates follow the leader’s requests in the leader’s absence, and thus seem to feel obligated or committed to the leader across time.

Asymmetrical Relationships: Conclusions

To sum up, these studies show that infants recognize many cues of asymmetrical relationships. One interesting question, like those raised in the previous sections, is whether toddlers form a general representation of asymmetrical relationships or a more specific representation of caregiving relationships. The 21-month-olds who reach for ‘winners’ or distinguish leaders and bullies, may have seen the high-ranking individuals as caregivers. In one study, the leader requested that the subordinates ‘go to bed’, something that the 21-month-old toddlers likely had experienced repeatedly from their caregivers.

Another interesting question is the extent to which dominance relations can be thought of as relationships at all. Certainly, in some

contexts, dominance need not be based on relationships (i.e., dominance can occur without any expectation of future interaction). For example, one person could defer to another person because they could inflict harm, without ever expecting to interact with them again. In the case of infants observing large and small characters in conflicts, their predictions can only be the result of comparing physical attributes. It is unclear whether observing the conflict and yielding leads them to form a representation of the individuals and their connection, which would lead to expectations of future interactions. Moreover, while dominance can be communicated by showing a 'commitment' to violence, for example, showing one's teeth or using threats, predicting behavior in these scenarios does not require a representation of commitment to the relationship itself. On the other hand, dominance can play a role in long-lasting relationships. However, an open question is whether people in the relationship think about their relationship in terms of dominance or are less likely to represent dominance in long-lasting or strong relationships. In many cases high-ranking individuals might feel the need to express their power but often argue that they are doing so to keep order or protect subordinates (refer to A. P. Fiske & Rai, 2014 for several examples). By contrast, while roles and power are asymmetrical in legitimate authority relationships, lower-ranked individuals may usually expect benefits in these relationships, such as knowledge, protection, care, etc.⁹ Thus, people may feel loyalty or allegiance more strongly when they see a high-ranking person as a legitimate authority.

It is worth pointing out that the distinction between legitimate authority, prestige, and dominance is not always cleanly disambiguated in relationships. Relationships are dominance relationships if the person yielding is acting to avoid harm and are legitimate if the person yielding sees the higher-ranking person as protective or just. Therefore, when humans evaluate asymmetrical relationships to be one or the other, we depend on inferences about the minds of those involved. This supports the idea that the early developing representation of asymmetry may be ambiguous regarding whether it is categorized as dominance or legitimate authority/prestige. Instead, the initial representation is more likely based on asymmetries that are easy to observe, such as physical size, patterns in yielding, etc., and subsequently may support learning of the different ways people can think about their asymmetric relationships. On the other hand, intentionality may matter to early representations of asymmetrical relationships. For example, if someone is physically larger but goes out of their way to make themselves seem smaller, who would infants or toddlers think is higher ranked? Moreover, the computation of strength and the understanding of loyalty or allegiance may only occur once children come to know that people can be ranked based on the benefits they provide. Future studies could investigate these questions as well as whether young humans recognize strength of asymmetrical relationships.

Equality Matching

In Relational Models Theory, an equality matching relationship is symmetrical and occurs when people keep track of what they give or receive and are motivated to maintain an even balance. Equality matching relationships can be out of balance, but people are aware of the imbalance and are motivated to return the relationship to balance. Examples of stronger equality matching relationships would be people in a long-term business relationship or neighbors who have been doing favors for one another for many years. Examples of weak equality matching relations include children taking turns at the playground and people waiting in a queue for coffee.

Evidence of Equality Matching in Non-Human Species

Some behaviors in non-human species indicate that there might be phylogenetic precursors to the logic underlying these relationships. Vampire bats share more of their blood meal with bats who have shared with them in the past (Denault & McFarlane, 1995). There is reciprocal grooming among bonded females of white-faced capuchins, chacma baboons, and bonnet macaques (Manson et al., 2004; Muroyama, 1991; Perry et al., 2003; Ventura et al., 2006). Experiments also show that captive brown capuchins may use the logic of reciprocity: previous sharing predicts whether food-taking is tolerated (de Waal, 2000). While it is unclear how these behaviors arise, it leaves open the possibility that this logic is evolutionarily old.

Contains Dot at the End Evidence that infants are sensitive to cues of equality matching.

Do infants represent equality matching relationships? I will first review studies about infants' expectations about resource distribution, though of course, each relational model can be applied to situations in which resources are shared or distributed. U.S. American and Italian infants expect that resources will be distributed equally when they are given no prior information about the relationship between the interactants. For example, at 19 months, 9 months, and 4 months, infants expect people to divide two resources equally rather than unequally between two animated puppets or characters (Buyukozur Dawkins et al., 2019; Sloane et al., 2012; Surian & Margoni, 2020; Ziv & Sommerville, 2017). Importantly, across these studies, these expectations depend on the recipients being agents as opposed to inanimate objects.

By 16 months, infants also seem to prefer those who distribute equally. They reach more often for a photo of a puppet who had distributed resources equally than for a photo of a distributor who had given all the resources to one of two characters (Geraci & Surian, 2011). Moreover, 16-month-olds expect a puppet who observes uneven versus even distributions to approach the even distributor. Likewise, 13- and 16-month-old infants choose to accept a toy offered by a person who allocated equally to two others as opposed to a person who allocated unequally (Lucca et al., 2018). By 25 months, toddlers are more likely to help a person who has distributed resources equally to two agents than a person who distributed resources unequally (Surian & Franchin, 2017). Each of these studies indicates an expectation or positive judgment of equal distribution. That is, given *no information about relationships*, infants seem to expect that two individuals will be given the same amount. But do these studies shed light on how representations of equality-matching relationships? While they do provide evidence that infants have early expectations about resource distribution, they do not tell us whether seeing equal resource distribution prompts infants to predict future interactions. Rather infants seem to infer traits of the distributor. (Mascaro et al., 2023) By contrast, an equality-matching relationships should be marked by even balance over time and across contexts between individuals.

Other studies suggest that infants have the cognitive tools to recognize these relationships. There are two studies in which toddlers use interactions in one situation to make predictions about interactions in another situation. In one study, 21-month-old toddlers were shown two characters who were asked to clean up toys (Sloane et al., 2012). When both characters helped, they expected equal distributions, but when one helped, they expected the more helpful character to get more. However, this study does

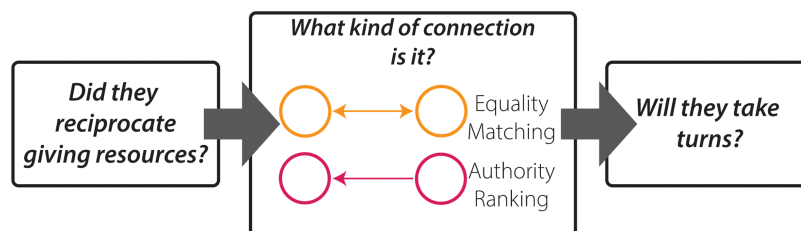


Figure 6. Logic of a possible study asking whether infants recognize equality matching relationships and use this representation to predict future interactions.

not directly test the relationship between the recipients of the resources, since the requester/distributor is a third party.

In another study with 17-month-olds, they saw that one character worked harder on a task, and then the characters distributed resources to themselves (Wang & Henderson, 2018). Here infants expected the one who worked harder to get more. Thus, toddlers tracked the extent of equality in one context (working) to make predictions in another context (resource distribution), suggesting that they may have inferred a relationship.

Denis Tatone and colleagues have more directly tested whether infants have the cognitive tools to represent equality matching relationships. They reasoned that infants should have different expectations about giving actions, where someone intentionally gives objects, as opposed to ‘tolerated’ taking, where someone still loses an object, but the intentions of the interactants are more ambiguous. In a series of studies, this was shown to be true. Here, Hungarian 12-month-old infants were shown interactions in which one character either gave an object to another character or took an object away from another character. The question was, would infants be more likely to encode the identities of the actor and the receiver when the actor gave compared to when the actor took? After watching giving interactions, infants seemed to expect that the ‘giver’ would continue to give to their social partner: they looked longer when the giver took something from, as opposed to when they gave something to, the original recipient (Tatone et al., 2015). Moreover, infants did not seem to encode the interactions by representing traits (e.g., ‘generous’) because they did not expect the giver to give to an individual who had not previously been in the interactions. The pattern of results was different after seeing that an actor took an object. Infants were first shown scenes where an actor took something from a character. Here, infants did not distinguish between scenes where the actor took again from the character and scenes where the actor gave to the character (Tatone et al., 2021). Together, these studies suggest that infants pay special attention to giving actions. They encode what object was given, suggesting that they can keep track of equality¹⁰. But they also track the identities of the interactants and the direction of these actions, suggesting they could encode the relationship. Since equality matching relationships require tracking equality, these are necessary components to recognizing these types of relationships. But do infants expect that the balances will be evened out?

In another set of studies, infants not only encoded these aspects of the interaction but also expected reciprocation of giving actions. For example, in one study, they saw one character (A) give an object to another (B). Then, they saw a third character (C) take an object from B. They then expected B to give to A, as opposed to give to C. They also expected B to give to A, compared to B taking from A (Tatone, 2017; Tatone & Csibra, 2020)¹¹. Further studies suggest that infants distinguish between characters who are given an item, as opposed to when those characters engage in ‘tolerated taking’ (i.e., take an item without protest from the other character). In both actions, a character ends up with one more resource, but infants

seem to see the interactions in different ways. When infants observe giving actions, they encode the direction of giving and receiving. However, when they observe tolerated taking, they encode neither. Tatone and Csibra argue that tolerated taking is more diagnostic of communal sharing relationships, in which are mutual, not symmetric and in which people do not track the direction or value of shared resources. Similarly, forceful taking should be more diagnostic of authority ranking, where one needs to track the direction but not necessarily the value of the object (since it will not be repaid). Equality matching is different—the interactants must keep track of the direction of giving and the object that is transferred, since part of the relationship is keeping an even balance (refer to Tatone, 2017; Tatone & Csibra 2020; and footnote for a longer discussion).

Taken together, these studies provide evidence that infants encode the necessary parts of social interactions that would enable them to represent equality matching relationships. They have at least rudimentary expectations of even balance. Importantly, these expectations are not the result of infants inferring traits of individuals, as they do not expect individuals to perform the same actions toward individuals not involved in the interaction. However, future studies could more directly test whether infants represent equality matching relationships, in which equality matching is maintained over time and across contexts. For example, one could ask whether infants expect even balance in one situation (e.g., giving resources) to generalize to even balance in another situation (e.g., taking turns). Moreover, that they expect that two individuals who maintain even balance will be more likely to interact with one another than those who had not previously interacted or interacted in a different way (refer to Figure 6).

Do infants compute strength for equality matching relations? So far, there is no evidence that this is the case. For equality matching relations, trust may be the important dimension: the larger the imbalance is allowed, the stronger an equality matching relation might be. For example, if I do many favors in a row for you and we are in an equality matching relationship, it could signal that I trust you will eventually ‘repay’ the debt. In many cultures, it is customary to give gifts whose value cannot be directly computed. This allows a sort of never-ending chain of gift-giving in which the debt can never be perfectly evened out (Carmichael & MacLeod, 1997; Graeber, 2012; Mauss, 1925; Fiske personal correspondence). Future studies could ask whether young humans make different inferences depending on the size of the inequality allowed in repeated interactions, which would imply that they are computing something about the size of the imbalance in an equality matching relation.

Conclusions

Humans depend on social relationships throughout our lives. It is in the context of social relationships that almost all learning, language, and cultural innovation occurs. Relationships are what people spend the most time thinking about and what people most

often point to when asked what makes their lives meaningful. Given the centrality of relationships to our survival and well-being, as well as the evidence that strength and models seem to matter in relationships cross-culturally and in other species, human infants may be prepared to reason about them, especially since unlike most other species, our survival from birth depends on a social network beyond our primary caregivers (Hrdy, 2009; Hrdy & Burkart, 2020). This early dependence on multiple relationships may have led to evolved capacities to recognize relationships, including the commitment of others, how others relate to one another and the infant themselves, and how new relationships form. The empirical work reviewed in this paper supports the claim that infants not only track relationships but also the characteristics of those relationships.

There are several caveats to consider. First, the study of infant representations of social relationships in cognitive development is relatively new. In other domains there are more comparative studies, brain imaging studies, and computational models that provide convergent evidence and better descriptions of their computational frameworks. Second, these results may be biased due to the culturally homogenous sample. Many questions remain about how early representations change because of culturally specific experiences.

A related question is, how infants learn about their place in their social networks. The question of how humans begin to understand themselves in relation to the world has been studied in the domain of navigation (refer to Spelke 2022 and Newcombe, 2019) suggesting that infants are able to understand themselves in relation to other things. I reviewed one study that touches on egocentric representations of relationships and there is a complementary study with 3 and 4 year olds which suggest they understand network connections – knowing for example that if a stranger knows the participant's favorite food, it is likely the stranger learned it from speaking to the participant's parent (Chuey et al., 2023). However, it is unclear the extent to which infants think about themselves in relationships and as part of their social world. Future work could investigate this.

Related to the question of learning is the extent to which people, including infants, think about relational models as either highly context-specific and variable within single relationships, or constrained and inflexible. Adults, for example, can see the same person as relating to them in different ways. For example, someone could be your boss at work and your teammate in a rec-basketball league. If relationships are highly context-specific from the start, then it would present an even more difficult learning problem. Fiske proposes that a single relationship can take on any of these models depending on the context. If so, how do people know which one to use and how would young infants learn cues? Being too flexible within single relationships would mean that the representations lose their predictive power which would add to the learning problem for infants and young children. One solution is that people may compute probabilities for each model (e.g., A relationship with a colleague could be a 3/10 on authority ranking, 9/10 on equality matching, and 1/10 on communal sharing). These probabilities could act as priors on acceptable or expected behaviors, used in combination with other contextual cues. This may also lead to learning new categories (which often have names, suggesting they are not core knowledge, which should be common to all Spelke, 2022). Take for example spousal relationships in the United States today. They often take on the form of strong communal sharing relationships. While divorce is possible, people explicitly commit to one another for life. Moreover, people do not

usually keep track of equality: just because one person gives the other a bouquet of roses does not necessarily mean that their partner “owes” them a box of chocolates. However, they can easily take on an equality matching model in some contexts: reminding your spouse that it is “their turn to do the dishes.” In diary studies, adults are more likely to mix up proper names of people with whom they have the same type of relationship (A. P. Fiske et al., 1991) and more so than people that they know who have the same characteristics such as gender, or race. This suggests that people represent governing relationship types for specific relationships. However, more work could be done on this front.

A related hypothesis is that people can view social interactions as occurring within frames. The linguist Ray Jackendoff, for example, describes bargaining as a competitive relationship (i.e., people are trying to get as much as they can from the interaction) that is framed in a cooperative one (i.e., acting as if they care about the other's well-being). It is possible that, for example, in some cases when a parent punishes their child, even though they take on the logic of an authority ranking relationship (in which the child should listen to the parent because of their rank), it is framed within a deeper understanding that the parent and child are in a communal sharing relationship (that the parent is ultimately demanding obedience because they feel that any harm that may come to the child would be akin to harm coming to the parent (Jackendoff, 2009).

Another hypothesis is that infants may be less flexible than adults about which model a relationship can take on. At first glance, this contrasts with the relationship that they likely have the *most* experience with parent-child relationships. In the current US, these relationships often take on both authority ranking and communal sharing models. However, it is unclear how infants and young children see parent-child relationships and their role in them. For example, U.S. American children only weakly expect preferential sharing between family members, as opposed to other prosocial relationships, and were just as likely to say that a child would preferentially share resources with their parent as they were to say that a parent would preferentially share a resource with a child (Thomas et al., 2023). It is an open question how infants see caregiving relationships (Thomas et al., 2025). Our lab has preliminary evidence that expectations about caregiving are weak in younger infants (C. Steele et al., 2024) but robust in toddlers around the age of 15 months. Fiske proposes these models are a lexicon, thus this developmental change may be due to constraints in combining them. The current state of the data leaves open several possibilities for how infants think about caregivers—both their own caregivers and a caregiver concept – and how this relates to relational models.

Likewise, rather than categorizing relationships into distinct models, we may understand relationships along two dimensions, which map onto relative rank and affiliation (e.g., (Leary, 2004; Wiggins, 1979), or warmth and competence (S. T. Fiske, 2018). Rank (or competence evaluations) would tell us whether we are higher, lower, or have equal status with someone, and affiliation (or warmth) would tell us whether we have the same goals.

Many open questions remain about how these representations change over development. Young humans learn an immense amount about how relationships are expressed in their cultures and countless studies show that older children have rich intuitive theories about relationships (Afshordi & Liberman, 2020; Astuti et al., 2004; Bedrov et al., 2021; Chernyak et al., 2019; Enright et al., 2020; Liberman & Shaw, 2017, 2018, 2019; Mandalaywala et al., 2020; Marshall et al., 2022; Olson & Spelke, 2008; Shaw & Olson,

2012; Spokes & Spelke, 2016; Thomas, Hernandez, et al., 2022; Woo et al., 2024). As discussed in the introduction, a key developmental change is likely the extent to which mental state reasoning is incorporated into our understanding of relationships. While adults likely use mental state reasoning in representing relationships, it may be relatively rare. We may be more likely to consider others' mental states when explicitly thinking about trying to change or maintain relationships (e.g., 'Did I offend my boss?'; 'How can I show I want to be colleagues instead of the boss?'). Likewise, the assumption that others share ideas about relationships is likely later developing given constraints in infants' ability to represent the minds of two people at once (Southgate, 2020). In general, it is an open question how much one needs naïve psychology to reason about relationships, especially since other species have apparent limitations in mental state reasoning but seem to represent relationships. Reasoning about intentions, beliefs, and desires may be crucial in making sense of social actions and relationships, but we may only invoke these computationally costly representations when needed (refer to Malik & Isik, 2023 for discussion and Box 2). A related question is how we use other domains to reason about relationships including biology (e.g., (C. M. Steele & Thomas 2024) and the physical world.

Even considering these caveats, the evidence I have presented points strongly toward the possibility that from birth, humans are prepared not only to reason about the physical world or the minds of individuals, but we are also prepared to reason about relationships. We do so in ways that suggest we categorize relationships into models and compute their strengths. Future work could uncover the exact nature of the computations underlying these representations and how they may change throughout infancy, childhood, and adolescence.

Box 1: Innate Moral Core or Innate Representations of Relationships?

How should we reconcile the claims I have made in this paper with the claim that infants have an 'innate moral core' (Woo et al., 2022)? This claim is supported by studies which have found that infants preferentially look toward and physically reach for helpers over hinderers, imitators over non-imitators, and those who distribute equally over those who do not (Hamlin, 2013; Woo et al., 2022). Arguments for an innate moral core implies that infants infer traits of the individuals (i.e., using the logic of, 'if they are nice to them, they will be nice to me'). But what traits do infants prefer? One way to interpret these findings is that infants reach for those who initiate or maintain prosocial relationships. Infants prefer those who help—which could be seen as one way to begin a communal sharing relation or an equality matching one; those who imitate, which could be interpreted similarly; those who distribute resources equally, which could be seen as establishing an equality matching relationship; and those who defer in a conflict, which could be seen as an individual who is establishing an asymmetrical relationship.

Consider that infants prefer imitators. As discussed previously, at least 12-month-old infants seem to see imitation as evidence of relationships. We had replicated the finding that 12-month-olds reach for imitators over non-imitators and unexpectedly found that infants reach for those who are *not* imitated over those who had been (Thomas et al., 2020). Given this asymmetry in preferences and the asymmetry in expectations of 4-month-olds (remember, 4-month-olds expect imitators to approach the target of their imitation but do not expect the target of imitation to

approach those who imitate them), we investigated whether older infants' expectations matched their reaching preferences.

First, we found that infants expected imitators to respond to their social partner's distress, as opposed to those who did not imitate. However, these expectations could be the result of either an inferred trait (the puppet who imitated is more responsive) or a relationship (the puppet imitated that individual and so will respond to that individual's distress). To disambiguate these possibilities, in the next experiment, rather than showing an imitating puppet and a non-imitating puppet, we showed infants a scene in which one puppet was imitated by the central actor and one was not. Being imitated does not give direct evidence of one's traits (one could be imitated without knowing it). To our surprise, infants also expected that the imitated puppet would respond to their partners' distress. In a third study, we found that infants did not have these expectations when a person uninvolved in the original interactions was in distress, suggesting that they did not infer traits. Finally, we found that infants do not expect imitative partners to respond to laughter, again suggesting they do not see imitation as a cue of who is more responsive, even toward their social partner. Taken together, these results present a puzzle: why do infants reach for imitators if they do not expect them to be more responsive to everyone? While infants may not expect the puppet to respond to all people's distress, they may still expect the puppet who imitates another individual to be more willing to create relationships. This would also explain the preference for non-imitated puppets, as the social initiation was not done by the imitated puppet and the non-imitated puppet is more socially available.

Admittedly, it is unclear why infants would prefer those who initiate relationships without inferring some sort of prosocial trait. These people might be seen as 'friendly', 'helpful', etc. If so, then it is unclear why infants seemingly generalize to themselves but not others. Another possibility is that infants generalize prosocial behavior toward themselves because of the way that experiments are set up. In many experiments, the people and puppets direct their attention toward the infant to keep their attention. Infants therefore may feel they are part of the social interaction. Their evaluations may, therefore, depend on these signs of awareness. There are two exceptions—one is in the hill paradigm in which the puppet's eyes are directed up the hill (Hamlin, 2014; Scarf et al., 2012). In this paradigm, however, the puppet may still appear to be looking at the infant, since both eyes are present (of note, reaching preferences of 6 month olds was not replicated in a multi-lab replication attempt; Lucca & Henderson, 2021). The other exception is in the right-of-way conflicts, in which infants reach for those who defer in a conflict. Here the puppets' faces appear to be looking away from the infant and toward its goal. While these two experiments raise questions about this proposal, it may be worth testing empirically. If so, some of the experiments reviewed in this paper only rule out broad generalizations. Of note, there are studies that suggest toddlers do generalize from a prosocial behavior in one context to a prosocial behavior in another context with new targets (Gill & Sommerville, 2023; Surian et al., 2018). For example, 14- to 27-month-old toddlers expect that a helper, but not a hinderer will distribute resources fairly to two new individuals. They also expect that someone who distributes things unfairly will hinder. Thus, it is possible that the ability to infer traits (which seems to be driven by tracking antisocial behavior) happens sometimes during toddlerhood or that different actions lead to different generalizations.

Box 2: Naïve Utility Calculus and Representing Relationships

Are relationships most basically understood in terms of shared or adopted utility (Powell, 2021; Tatone, 2017)? In a recent proposal, Powell argues that representations of affiliation compute whether people adopt the utility of others. This can happen when someone helps another person (i.e., takes on a cost so that another person can achieve their goal) or imitates another person (i.e., takes on a cost to achieve the same goal as someone else). One straightforward connection to the current proposal is that relationship strength is computed by adopted utility (Powell does not make this claim). While the connection between strength and utility is appealing, some examples call the connection into question. First, humans, including toddlers (Warneken & Tomasello, 2006; though see Barragan & Dweck, 2014) are happy to adopt the utility of strangers. These toddlers' expectations of future interactions is unknown, however adults help people whom we never expect to see again (e.g., helping a stranger pull their car out of a ditch). Young adults report taking on similar costs for strangers, acquaintances, and close relations (McGuire, 1994). Perhaps one way of reconciling this tension is that as adult observers, if we find out that a person adopts the utility of a stranger, then we shift our inference to be about traits instead of relationships (i.e., a parent who donates their kidney to their child is a good parent, but a person who donates their kidney to a stranger is a good person). The extent to which infants adjust their evaluations of prosocial behaviors based on different types of evidence could be tested.

We also fail to adopt the utility of even our closest relations. Take a parent who fails to adopt the utility of their toddler who wants another bowl of ice cream or a parent who fails to adopt the utility of an infant who reaches for a hot cup of coffee. An adult observer would assume that the parent's long-term goal is to increase the utility of the younger human. For example, the parent wants to help their child make healthy eating choices or avoid a burn. But do toddler or infants understand this? If not, it would lead to the uneasy hypothesis that infants and children do not see themselves in a strong relationship with their caregivers. Of course, infants and children are excellent statistical learners (Saffran, 2020), so it could be that unhelpful behavior from parents are overpowered by more helpful behavior. As a parent, I find this hypothesis to be less than straightforward, but it is an empirical question.¹² Interestingly, children do not seem to expect more altruism between family members than between friends. When asked whether someone will share with either a family member or friend, children do not expect that others will preferentially share with or help family members compared to other affiliates (Marshall et al., 2022; Spokes & Spelke, 2016; Thomas et al., 2023; Thomas, Woo, et al., 2022).

Perhaps a better candidate to infer relationship strength could be intimacy, such as physical or emotional intimacy (refer to Steele & Thomas; Thomas, Steele, Gopnik, Saxe for longer arguments). These cues may have become valid because of the costs and benefits associated with sharing bodily fluids and close physical contact. However, I think it is unlikely that infants recognize this. Disgust, especially social disgust, seems to develop relatively late (Rottman, 2014), suggesting young humans don't recognize overcoming 'disgust' as a cost. Infants differentiate actions that are difficult to distinguish based on utility alone (saliva-sharing vs. forehead touching, for example). Likewise, adults may see responses to distress costly because of emotional vulnerability, but it seems unlikely that infants see it this way. Ultimately these are open

questions. To sum up, it seems that adopted utility could be one way we compute strength, but perhaps not the only way and it is unclear whether utility computations occur in all representations of relationships.

Can adopted utility calculus be used in the computations that allow us to recognize models? One can describe the models in terms of adopted utility: a communal sharing relation is one where we adopt each other's utility, an asymmetrical relationship is one where utility is adopted for one person but not the other, and a reciprocal or equality matching relationship is one where we adopt each other's utility exactly (this is not Powell's proposal but refer to Tatone 2017).

However, some examples also call this description into question. Another proposal is that the communication of utility is more important. A distinctive feature of communal sharing relations is that individuals do not keep track of relative costs and rewards. People's skills, resources, and 'rewards' are pooled, and people don't communicate about how much people are contributing or how much people are taking. In fact, in some situations, when people keep careful track in what 'should be' a communal sharing relation, it can seem cold or even evidence that the relationship is not a communal sharing one. For example, in the novel *The Joy Luck Club*, a married couple tries to keep everything strictly equal, and the implication is that they are not intimate.

Second, in asymmetrical relationships, it is not whether one person adopts the other's utility that is the key distinction. At least with adults, helping itself can be interpreted as communicating both high and low status (Nadler & Chernyak-Hai, 2014). Subordinates often adopt the utility of leaders, but 'mansplaining' is critiqued because it implies that the man thinks they are higher status than the target of this explaining. Perhaps a better indicator is who decides whose utility is adopted. For example, take an academic mentor and mentee—both parties adopt the utility of one another in different circumstances. The mentee does not expect to get the largest office and so 'takes a cost' for their mentor. However, the mentor is expected to adopt the utility of the mentee in providing guidance. However, the mentor could probably give up their office and could decide guidance would do more harm than good with no penalty.

In equality matching relationships, a key marker is symmetry which requires that people keep track of whose utility is being adopted and communicate about imbalance. This is apparent when someone buys a coffee during one meeting and the other insists it is their turn to buy during their next meeting. There is an agreement, often explicit, that the utilities of both parties will be adopted at equal rates. Likewise in groups where voting occurs, the utility is matched in a way: each person has the same amount of influence. However, the utility can end up being asymmetrical in which some people's goals are adopted. Yet, this wouldn't necessarily change the relationship to be asymmetrical.

To sum up, it is possible, and perhaps likely, that the logic of naïve utility calculus plays a role in our representations of relationships in some contexts. However, it is less than clear that such computations could cover all relationship representations. An open question is what computational framework could.

Box 3. Groups and Relationships

In this paper, I have argued that there are related but distinct representations of traits, relationships, and groups, each leading to different generalizations. Representations of relationships are

constrained to the individuals involved, while representations of groups allow for generalizations beyond the individuals. Much of the work used to argue that infants represent communal sharing relationships was originally taken as evidence that they represent ingroups (Powell & Spelke, 2013; Spelke & Kinzler, 2007). This work was inspired by decades of research in social psychology showing that people and young children are prone to thinking about others in terms of ingroups and outgroups (refer to Dunham, 2018; S. T. Fiske, 2018; Shutts & Kalish, 2021; Tomasello, 2020). Indeed, the term 'folk sociology' was coined to argue that humans, including young children, are prone to thinking about aggregates of people (L. Hirschfeld, 2005).

However, in the work with infants that I review, there is little direct evidence that infants represent groups. Infants see two or three individuals in the same physical space doing something that makes them more alike. Then, infants' expectations about those same individuals are tested. There is no evidence that infants generalize beyond the individuals in the scenes. While these studies leave open the possibility that infants' expectations are based on representing groups, it is just as likely that they make predictions based on inferred relationships. Take, for example, (Powell & Spelke, 2013): in this study, three individuals coordinate their actions. We don't know whether infants see the three individuals as an aggregate or whether they see three relationships. Would infants expect affiliation when they get evidence that people speak the same language but not during their social interaction? Furthermore, would infants expect people who are part of an aggregate will speak the same language?

There are reasons to think that infants' ecological niche has led infants to represent relationships and social networks, as opposed to groups. Throughout our evolutionary history, care has mostly come from 'mothers and others' – that is, your primary caregivers and those with whom they have relationships (refer to Hrdy 2009). While it is true that these 'others' are also usually part of your mother's group, there are many people in the group who do not provide care. Therefore, it may be more important for infants to track individuals and their relationships. As discussed in the introduction, human infants, as well as other animals track individuals, a prerequisite in tracking relationships, but not necessarily a prerequisite in tracking group membership.

Thinking about larger coalitional groups may become important as children are exposed to more people in and outside of their social networks. There are several studies with older children in which they do seem to represent groups. They are sensitive to the 'minimal group effect' in which they generalize based on arbitrary cues (Baron & Banaji, 2006; Chalikh & Dunham, 2020; Dunham et al., 2011; Tajfel, 1982; Yang & Dunham, 2019). Importantly, these expectations and biases occur without direct evidence of interpersonal relationships (Chalikh & Rhodes, 2020; Yang & Dunham, 2019) suggesting that children represent groups as I have defined them. Representations of groups may arise from 'scaling up' from representations of relationships, but this process of scaling up would involve generalizations beyond the direct observation of behavior of or between individuals (refer to Figure 7). We may start by mapping out individuals and social networks, and over time, we notice or are told that certain shared characteristics help delineate these networks. This would be similar to the hierarchical learning theorized to be important to early learning in other domains (Ullman & Tenenbaum, 2020).

Exactly what leads people to infer group membership, and what kinds of predictions follow and when, is not a trivial question¹³ (Cikara, 2021; L. A. Hirschfeld, 2001; Pietraszewski, 2021). I add to

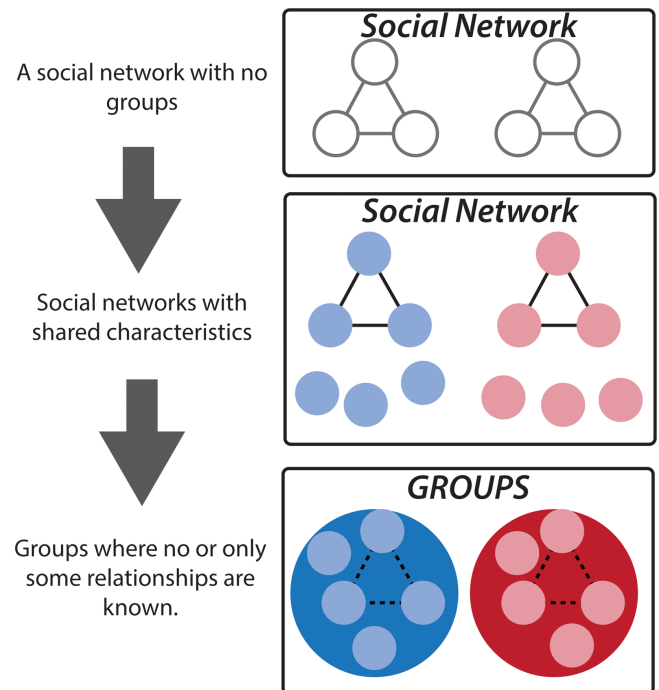


Figure 7. Here I illustrate what I mean when I propose that groups inferences may arise from making inferences about interpersonal relationships. One could map out relationships and networks, learn 'borders' of social networks, then learn (through explicit or implicit teaching) that characteristics delineate networks, including how people interact within and across networks (refer to Pietraszewski, 2021 and accompanying commentary as well as Cikara, 2021 for more detailed proposals about how that may happen.) More people could be added to the network until you are no longer representing nodes and lines but instead an aggregate with fuzzy borders.

the problem by proposing that people's representation of social groups includes more than group membership; they also represent the social structure of groups and the relative strength of those groups (Lickel et al., 2006). Children recognize group structure and use these structures to generalize about the behavior of group members without any direct evidence of their interpersonal relationships or the behavior of individuals (Thomas, Mitchell, et al., 2022). People may also think about relationships between groups, such as some groups being higher ranked than others (refer to work on social dominance theory and system justification theory (Jost et al., 2004; Pratto et al., 2006), as well as evidence that children connect identity and status e.g., Mandalaywala et al., 2018) or some groups being more likely to work together while others are more likely to be in conflict (refer work on coalitional psychology; Cikara 2021; Gershman & Cikara, 2020). People, including children, see aggregates as though they are entities with single intentionality (Bloom & Veres, 1999; Sheikh & Hirschfeld, 2019), suggesting that we may call on our intuitive theory of relationships when considering relations between aggregates.

Box 4: Navigation, Space, and Social Relationships

Many scholars have invoked the metaphor of navigation when discussing how we reason about social relationships (Banaji & Gelman, 2013). Considering recent research into the hippocampus (Montagrin et al., 2018; Tavares et al., 2015) as well as theories from linguistics (Jackendoff, 2009), this may be appropriate. Our reasoning about navigation allows us to represent where we are, where we can go and gives us an anchor for memories (Spelke,

2022). When I think about my hometown, I don't think of 'a small town in the central, south part of the west coast' I think about 'Ojai'. Likewise, when I reason about my daughter, I don't reason about a small child with blond hair, blue eyes, and an impish smile, I think 'Ursula' (Spelke, personal correspondence). There may be overlaps in the way we reason about nodes as locations in space and nodes as locations in social networks. First, in many languages, people use spatial terms to describe social relationships and social interactions. We can locate ourselves or others in a family tree or a social network. You can feel 'close' to someone or 'distant'. Someone can stand 'above you' or be on 'equal footing'. You can even be in the 'same circle' as someone. None of these descriptions need people to be in the same physical space to have meaning, yet they are easily understandable. In this paper, I have been vague about the formal computations that are involved in our representations of social relationships. However, graph theory may be one avenue of research that could be fruitful in describing our cognition of social relationships with formal computational models (Kemp & Tenenbaum, 2008).

Another way that our reasoning about navigation may connect to our reasoning about social relationships is the computational modeling work done about planning. A recent proposal connects intuitive psychology and spatial navigation (Ho et al., 2022). There may also be similarities in how we think about relationships. The proposal is that sometimes when we think about relationships, we think about them as objects in the world. Other times, when the status of our relationship is uncertain, or when we are trying to change a social relationship, we might go through the computationally costly process of simulating possible social actions, their consequences, and how this might affect the minds of our social partners and subsequently the relationship. For example, when someone knows that they are in a committed, communal sharing relationship it might mean that they go through actions the same way one does when they are in 'auto-pilot' mode in the car. You kiss your spouse goodbye without thinking twice about it. Errors such as calling your granddaughter the name of your daughter may be evidence of this kind of 'auto-pilot' mode where we represent the relationship with 'just enough' detail. On the other hand, when we think about changing or creating relationships, we might simulate many possible social actions or conversations that could take place. These ideas are tentative and speculative. More work could explore how the cognition involved in spatial navigation, planning our movements in space, and social 'navigation' may be connected.

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Notes

1 The question of attribution is a classic question in social psychology—*when* do people attribute a person's actions to traits compared to 'the situation'? (Kelley & Michela, 1980). One interesting question related to this classic work as well as the current proposal is: how does statistical learning over a lifetime, explicit teaching and other types of learning affect a person's tendencies to explain actions based on traits, relationships, or groups. While we know children use statistical reasoning to distinguish between context and trait causes about the cause of people's actions, more work could be done on this topic (Seiver et al., 2013).

2 I thank Herissa Lamothe for proposing this and thinking this through with me. She is currently working on computational models to test the hypothesis that mutual awareness is key to naive sociology.

3 Relational Models Theory proposes that market pricing and equality matching are distinct models, but it is unclear to me whether they need to be. In both types of relationships individuals are motivated to maintain balance, but in one people are required to think about ratios as opposed to 1:1 balance. For example, if you mow my lawn in the summer and I clear your snow in the winter, are we in an equality matching relation (we each do favors) or market pricing (we reason that mowing and clearing are relatively equal in by some fungible metric)? By the logic of relational models we are using market pricing, but the relationship itself seems like equality matching.

4 Imagine people in a summer fling saying, "I will love you very intensely for exactly 93.6 days" versus people avowing their love "forever". The later situation would likely be seen as indicative of a 'stronger' relationship.

5 In some sense this is embedded in the argument against nepotism – people assume that at work people will continue to care deeply and favor their family members.

6 Many songs and poems explore this topic. For example, in Prince's 'Nothing Compares to You' the singer tries to do things or interact with people that may replace 'you'. For example, 'I can put my arms around every girl I see, but they only remind me of you' suggests the singer has tried but failed to replace 'you' with others. Other parts of the song suggest that the singer tries to replace the relationship with other actions, such as, 'eating dinner at a fancy restaurant. The implication is that this bond was a strong one!

7 Iconic representations, unlike symbolic representations, bear some resemblance to the target of the representation, but are still more abstract than indices. For example, a footprint or a stop sign would be an indexical representation because it is physically connected to the target. Words are symbolic because they need not bear resemblance to the target. Diagrams or graphs are iconic, there is a direct translation between the meaning and the attributes of the sign. Note, Fiske proposes that different relationships are substantiated in the mind as different forms of representation. Specifically, following Peirce's semiotics, communal sharing is indexical, authority ranking is iconic, and market pricing is symbolic. He argues that Piaget's concrete operational stage is the best way to think about what constitutes equality matching in the mind (A. P. Fiske, 2004). I think it is more likely they all take the same form, but this is a very difficult thing to study.

8 An experiment could test this: ask whether infants still prefer puppets who make the same choice after them, when the puppet did not have perceptual access to the infants' choices.

9 If the hypothesis about stronger relationships being less replaceable is correct (i.e., that stronger relationships, regardless of the model, means that people are less interchangeable), then it should also apply here. The stronger a legitimate authority relationship is, the less interchangeable the high-ranking individual should be. In strong authority ranking relationships, people may mourn if a leader gets replaced by a new one, and feel that relationship will change if they do. By contrast, if an interaction or relation is based strictly on dominance, then the higher ranking person should be highly interchangeable with other individuals in as much as their level of formidability is the same.

10 It is an open question whether they can track 'value', though work on naïve utility calculus suggests they may be able to (Liu et al., 2017).

11 Note: Tatone & Csibra, 2020 argue that different types of exchange may cue different types of relationships. Specifically, tolerated taking should cue communal sharing relations, and the direction of resource transfer need not be represented; priority access and forceful taking should cue authority ranking relationships, and the direction of resource transfer needs to be represented; and

giving should cue equality matching, and the direction of resource transfer needs to be represented.

12 A potentially impossible one to answer since it depends on the reasoning of the babies. What counts as helpful? What counts as harmful? How do we know how infants/toddlers are interpreting the actions in their everyday environment?

13 Several recent papers have called for better characterizations and further specification of what we mean when we say, 'group'. These proposals, and others before (e.g., Hirschfeld 2005) point out that categorization based either on shared perceptual features or shared goals is unlikely to be the basis for inferring social groups. Take shared goals. We don't generally consider people waiting for a bus a social group, even though they all share a goal of getting on the bus. However, we might consider a 'food not bombs' chapter a social group, who share the goal of providing food for people and dismantling the US military. Or, take perceptual features: having freckles is a perceptually salient feature but is not (to the author's knowledge) used as the basis for defining meaningful social groups in any culture. In contrast, having darker skin is also a perceptually salient feature but has been used in certain times and places to define social groups.

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Open Peer Commentary

Relationships do not depend on prior knowledge, instead knowledge develops within relationships

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Abstract

Social relationships are crucial in human development, but we criticize Thomas's approach in terms of epistemology, methodology, genetics, neural development, and evolutionary theory. We critique the evidence Thomas cites to support claims of innate knowledge and instead propose that understanding relationships develops gradually through infants' interaction within developmental systems elicited by their embodiment, not from innate knowledge.

We agree with Ashley Thomas regarding the importance of social relationships in human development, but we disagree with how such knowledge should be conceptualized and explained. Despite her title referring to “developmental origins,” Thomas provides no real developmental account. She asserts that infants possess abstract representations of social relationships, built on core knowledge of relationships that is either innate or early developing, in the sense that it is already present before the infant engages with the domain of relationships and thus does not depend on experience.

Elsewhere, we and others have critiqued claims that knowledge is representational and instead argue that knowledge develops through action and interaction (e.g., Carpendale & Lewis, 2021).

A further problem is how such knowledge could be present before experience. There is a missing biological story here without references to indicate that this could be biologically plausible. Are we to assume that genes fill in the missing story in getting from the zygote to infants' innate knowledge of relationships? Although decades ago this claim was accepted, geneticists explain that genes do not determine outcomes and are part of a complex developmental system. Whether genes are expressed or not depends on many internal and external factors (e.g., Meaney, 2010; Moore & Lewkowicz, 2024). Furthermore, to get to the neural interconnectivity that is presumably required to support the claimed core knowledge of relationships requires experience to form and strengthen or prune away synapses (e.g., Stiles, 2017). Thomas also assumes a problematic conception of evolutionary theory that overlooks development (Moore & Lewkowicz, 2024; Witherington, Lickliter, & Moore, 2023). In addition to these problems, in this commentary, we focus on difficulties concerning the evidence Thomas draws on.

Thomas argues that infants possess abstract representations of social relationships (e.g., communal sharing, authority ranking, equality matching) and can also determine the strength of these relationships. However, the research cited is based on looking-time methodology, which can be over-interpreted and potentially misleading (Houston-Price & Nakai, 2004; Paulus, 2022). Looking-time studies often rely on the idea that if an infant looks longer at an event, this is due to a violation of expectations, presumably based on pre-existing social representations. For example, Powell and Spelke (2013, 2016) reported that infants looked longer when members of synchronized groups imitated out-group members. Thomas interprets this as evidence of an understanding of group-based affiliation. Looking-time paradigms, at best, provide correlational data and do not offer explanatory power (Aslin, 2007; Haith, 1998). Moreover, the interpretation of infant looking time can shift depending on the study. On the one hand, longer looking time is often taken as a sign of surprise at a violation of expected social behavior (e.g., Powell & Spelke, 2013). On the other hand, it can be interpreted as evidence of preference for an expected fair social outcome (e.g., Geraci & Surian, 2011). This inconsistency shows that looking time is susceptible to interpretive bias and lacks methodological clarity. In a meta-analysis of looking time studies, Hollich and colleagues (2000) concluded that infant looking behavior is influenced by multiple factors, such as perceptual salience and familiarity, yet these studies are used to claim infants' conceptual understanding despite minimal and ambiguous indicators.

A further difficulty with claims of innate knowledge based on young infants' looking time (e.g., Powell & Spelke, 2016) is that infants' cortical control of their movements develops slowly over their first year of life. Yet these actions are assumed to demonstrate knowledge that is continuous with that of adults, which is located in the cortex (Blumberg & Adolph, 2023).

If infants possess core knowledge of relationships, this should be evident in how they act. Yet studies based on behavioral evidence suggest a different developmental story. Play is an early form of self-initiated social activity, and its structure can reflect a child's capacity for relationship recognition. If infants understand communal or authority relationships early on, socially directed play should be evident early in life. But until about 12 months, play is predominantly solitary. Later, children begin playing side by side, but this parallel play is still not a shared or social activity. Around 24–30 months, children begin engaging in associative and cooperative play, with turn-taking, sharing roles, and explicitly

referencing one another's behavior (Howes & Matheson, 1992). Thus, real-world behaviors like play suggest that social relationships are not represented abstractly from birth, as core knowledge theory suggests, but gradually emerge through repeated interaction and shared activity.

Similarly, Thomas discusses infants' expectations of sharing behaviors (e.g., food sharing as evidence of communal relationships), yet when considering actual sharing, the developmental trajectory suggests that these behaviors emerge gradually in conjunction with growing socio-emotional understanding (Brownell, Svetlova, & Nichols, 2009), and sharing becomes relational as the child participates in social interactions and learns to recognize roles and intentions (Brownell, 2013; Carpendale et al., 2021). However, the implication of the author's core knowledge theory is that infants already possess models like "communal sharing" or "authority ranking," which attributes adult reasoning to infants who are still developing a sense of self and other. It also downplays the role of culture, where what counts as "sharing" or "deference" is shaped by social practices, not abstract categories (Gaskins & Miller, 2009). Behavioral studies of play and sharing, both grounded in action, reveal a developmental progression rooted in social interaction, cultural norms, and emergent understanding. The evidence supports a view of infants as active participants, not passive holders of pre-existing knowledge.

In contrast to Thomas's approach, we propose that infants are active participants, not passive holders of pre-existing relational templates. We argue that social understanding develops from participation in routines with others in the developmental systems elicited by infants' biological embodiment, and through these interactions, infants gradually develop an understanding of social relations, not the other way around (Carpendale & Lewis, 2021). That is, rather than Thomas's assumption that infants begin with cognition that makes relations possible, it is experience in relating to others in which communication and language emerge, which makes the development of cognition possible (Jopling, 1993).

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Early sociomoral reasoning encompasses more than interpersonal relationships

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Abstract

Thomas's account suggests that infants are limited to reasoning about interpersonal relationships. We argue that this account fails to do justice to infants' sociomoral reasoning.

We describe an alternative account that grants infants expectations of harm avoidance, ingroup support, authority, fairness, and reciprocity. We highlight differences between the two accounts and suggest research directions for moving beyond these differences.

Over the past decade, several accounts have been proposed for the core system that underlies infants' ability to reason about social interactions (Baillargeon et al., 2015; Spelke, 2022; Ting et al., 2020; Woo, Tan, & Hamlin, 2022). In Thomas's account, this system is limited to the representation of interpersonal relationships. It allows infants to detect when such relationships exist, to categorize them into types (communal, asymmetrical, and reciprocal), and to compute their strength (a strong relationship is one to which individuals remain committed over time and across contexts). Here, we suggest that this relationship-based account does not do justice to the available evidence on early sociomoral reasoning. We first introduce an alternative, principle-based account and then discuss some of its implications for Thomas's account.

According to the principle-based account (Ting et al., 2020), infants' social world is not a uniform landscape of interpersonal relationships but a varied landscape structured by several types of social distinctions, each laden with rich implications for how interactions may unfold. Do individuals belong to the same or to different moral circles? Do they belong to the same or to different social groups within their moral circle? Do they possess the same status within their group, or are some leaders and others followers? Importantly, at least five normative principles guide early sociomoral reasoning. They include *harm avoidance*, which regulates interactions within moral circles (Jin & Baillargeon, 2025; Ting & Baillargeon, 2021); *ingroup support*, which regulates interactions within social groups (Bian & Baillargeon, 2022; Jin & Baillargeon, 2017; Ting, He, & Baillargeon, 2019); *authority*, which regulates interactions between leaders and followers (Margoni, Baillargeon, & Surian, 2018; Margoni & Thomsen, 2024; Stavans & Baillargeon, 2019); *fairness*, which regulates social exchanges (Bian, Sloane, & Baillargeon, 2018; Buyukozur Dawkins, Sloane, & Baillargeon, 2019; Sloane, Baillargeon, & Premack, 2012); and *reciprocity*, which regulates partnerships or friendships (Jin et al., 2024). Together, these five principles and their attendant concepts form the "first draft" of human moral cognition (Graham et al., 2013). The first four principles are depersonalized and extend to all individuals who fall within the purview of their application: Individuals are expected to refrain from doing severe harm to any member of their moral circle, to help any ingroup member in need, to protect any of their followers from harm, and to distribute resources fairly between any recipients, whether or not they have formed interpersonal relationships with these other individuals. By contrast, the final principle, reciprocity, is personalized in that it regulates the development of interpersonal relationships between specific individuals.

The principle-based account overlaps with that of Thomas in that both encompass infants' expectations about reciprocal relationships. Beyond this similarity, however, differences abound, giving rise to several issues.

A first issue is that it is unclear how Thomas can account for the wide range of sociomoral expectations infants demonstrate. For example, she writes that while studies using resource-distribution tasks (Buyukozur Dawkins et al., 2019; Sloane et al., 2012) show that "infants seem to expect that two individuals will

be given the same amount, . . . they do not tell us whether seeing an equal resource distribution prompts infants to predict future interactions . . . By contrast, a [reciprocal] relationship should be marked by an even balance over time and across contexts" (p. 22). We agree that there is no evidence that infants in these resource-distribution studies are reasoning about interpersonal relationships and that reciprocal relationships should be balanced – but none of this helps explain why infants would expect equal distributions in the first place. In the principle-based account, such expectations are supplied by infants' sense of fairness, but there is little in Thomas's account to explain them.

A similar point arises with respect to leader–follower tasks. By the second year of life, infants expect a leader to intervene when a follower steals from another follower (Stavans & Baillargeon, 2019), and they expect followers to obey a leader's instruction even in the leader's absence (Margoni et al., 2018; Margoni & Thomsen, 2024). Thomas suggests that infants are reasoning about asymmetrical relationships between high-ranking and low-ranking individuals, but why would infants expect protection from high-ranking individuals and obedience from low-ranking individuals when they first see them interact? In the principle-based account, such expectations are supplied by infants' sense of authority.

A second issue is that in Thomas's account, infants are incapable of representing social structures such as groups; these representations develop later in childhood "as children are exposed to more people in and outside of their social networks" (p. 30). This claim is surprising given the extensive evidence that infants can represent a wide variety of social groups, including animal puppets that differ in kind (Bian et al., 2018; Rhodes et al., 2015; Ting & Baillargeon, 2021), geometric figures that differ in shape or color (Powell & Spelke, 2013; Pun, Birch, & Baron, 2021), and individuals who speak different languages or perform different conventional actions (Lieberman, Kinzler, & Woodward, 2017, 2018). Categorization of individuals into such groups, or even into minimal groups based on novel labels or outfits (Bian & Baillargeon, 2022; Jin & Baillargeon, 2017; Rhodes et al., 2015; Ting et al., 2019), elicits rich expectations of ingroup support.

Thomas argues (without marshalling evidence for her interpretation) that infants in these studies are representing communal relationships rather than groups; the latter characterization would fundamentally undermine her central claim that infants' core system is limited to representing interpersonal relationships. Although we reject this interpretation, we believe that more should be done to understand how infants identify and reason about interpersonal and group interactions. For example, would infants expect any member of a group to help an ingroup victim – but also expect a member who is a friend of the victim to do so more promptly? At the same time, would infants expect a leader in a group to ignore interpersonal ties when regulating access to a group resource among followers? Such investigations would help deepen our understanding of how infants represent, integrate, and rank-order the varied elements of their social world.

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Overcoming the shortcomings of the computer metaphor of the infant mind by embodying social relations

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Abstract

Thomas explicitly negates the Market Pricing relational model. However, her use of the computer metaphor of the mind implicitly renders Market Pricing as the universal framework through which all other relational models can be understood. Alternative metaphors could overcome this self-contradiction, allowing her to develop an embodiment approach to the developmental origins of the cognitive representations of social relations.

Fiske observed that “Market pricing is so pervasive in Western society and so important in the Western cultural conceptions of human nature and society that many theorists have postulated that all human social behavior is based on more or less rational calculations of cost-benefit ratios in self-interested exchange.” (Fiske, 1992, p. 706). It seems that Thomas’ article falls into this exact trap, with one difference: she replaces the market metaphor with a computer metaphor of the mind (Brette, 2019; Searle, 1984). In essence, following Fiske, we argue that her approach illustrates how *computer modelling* has become one of the mainstream Western cultural conceptions of human nature and society, leading to postulates that all human social behaviour is based on more or less *computational calculations*. Consistently, relationships are represented as the by-product of two “basic computations”: one that categorises the connection by model and one that computes the strength of the social connection. In other words, babies are portrayed as information-processing machines, equipped with the capacity to perform continuous computations of their relational environment. Indeed, Thomas’s approach to infants is predicated on the premise that infants possess the ability to organise vast amounts of information in their environment and to direct their

attention to that which is relevant. Since she sees this faculty as underpinned by “core knowledge,” something either innate or developed very early in life, she thus conceives of infants as natural-born computers.

Ironically, although Thomas rejects Market Pricing as a distinct relational model, using the computer metaphor leads her to place market logic at the heart of *all* relational models. A striking illustration of this is her “untested hypothesis” about the strength of relationships, which essentially states that the stronger the relationship, the less “replaceable” the relationship and the person concerned. This expresses the economic concept of “elasticity of substitution,” a ratio measuring how easy it is to substitute one input for another (Dixon, 2010). Unsurprisingly, this line of reasoning ultimately leads Thomas to illustrate her argument explicitly in market terms, namely the parental investment of resources put into raising children. This approach echoes the economic metaphor that children are “durable consumer goods” in which parents invest (Becker, 1976, p. 169), and is closely linked to the notion of “human capital” lifted straight from economics (Becker, 1993). Consistently, through the market metaphor that pervades some parts of Thomas’ argument, people are goods that can be “swapped” or “replaced,” whether it is a colleague for another colleague, a worker for the owner of a factory, or a friend.

We appreciate Thomas’ explicit acknowledgement that generalisations based on the economic notion of “utility” can be challenged. Nevertheless, although she may shy away from asserting market logic as underlying every aspect of social relations, it is noteworthy that the economic reasoning associated with Market Pricing can lie at the heart of every other relational model in her argument. Her proposed characterisation of the “computations” that infants supposedly employ to infer the category and strength of Communal Sharing relationships is one of “naïve utility calculus,” whereby costs and rewards are computed to infer the value an individual places on something. This corresponds to a standard utility-based model of economic reasoning and market behaviour (Becker, 1993). Likewise, Thomas’ approach to Equality Matching does not rule out infants using “naïve utility calculus” to “track ‘value,’” while Authority Ranking (or asymmetrical relationships) is understood through computations in terms of the “benefits” that people can provide. Crucially, such framing is antithetical to assertions of Fiske and colleagues that there is no common logic through which different relational models can be compared, meaning that conflicting priorities will result in confusion and negative affect (Fiske & Tetlock, 1997; Gallus et al., 2022). By contrast, Thomas’ approach implies that all apparent conflicts can ultimately be resolved through utility calculations.

We believe that Thomas’ market-based framing of all relational models has to do with her use of the “mind as computer” metaphor (Basso & Herrmann-Pillath, 2024a). This, however, could be circumvented by the use of other metaphors. Thomas’ argument could lend itself to an embodied approach to infant cognition, similar to her analogy of reasoning between the domains of physics and psychology. The embodied cognition literature, including Piaget’s (1952) description of how infants’ physical experience of the world provides the background for their psychological understanding of the world, obviates the assumption that infants are computers (Marshall, 2016) and promotes instead an embodied metaphor approach (Lakoff & Johnson, 1980, 1999). That way, Fiske’s (1992) relational models can be understood as the social expression of physical experience through kinaesthetic image schemas (Johnson, 1987, 2017; Lakoff, 1987). Of course, other schemas and additional

embodied mechanisms may be at play, but a close reading of Fiske’s relational models shows that Communal Sharing can be seen as emerging from the experience of the body as a container through consubstantial assimilation (IN/OUT); Authority Ranking from the body being in space, giving rise to the vertical metaphor (UP/DOWN); and Equality Matching as structured around the BALANCE metaphor, related to the experience of proprioception (Basso & Herrmann-Pillath, 2024b). Moreover, despite the assertion that Market Pricing is “not embodied” (Fiske, Thomsen, & Thein, 2009), we propose that it can indeed be viewed as derived from the embodied RESOURCE metaphor, in which everything, including people, is understood as a means to an end (Basso & Herrmann-Pillath, 2024b; Lakoff & Johnson, 1999).

By taking an embodiment approach, Thomas would not only move beyond the computer metaphor but also account for how different kinds of relationships are embedded within different types of social structures, promoting different metaphorical understandings of the relationship model. For example, an infant potentially comes to understand familial relations as being fundamentally different from friendships because the family is socially structured around containers (“in/out” family), whereas friendships are structured around balance (obligations must balance). This might resolve Thomas’ question as to why infants initially seem to confuse these two relationship types and yet later hold them to be quite different.

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Computational models of social cognition should incorporate social relationships as core primitives

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Abstract

Relational primitives can facilitate rapid inductive generalization beyond individual mental-state reasoning. Computational models of social cognition should incorporate social relationships as core primitives, enabling richer explanations for how minimal social observations lead to flexible, efficient, and culturally adaptive social predictions.

Thomas proposes that humans are equipped from infancy with an intuitive theory of social relationships, which supports recognizing distinct types of dyadic relationships and their strengths. In this commentary, we draw out the implications of this proposal for computational models of human social reasoning.

Traditional models of social cognition (or more specifically, Theory of Mind) focus on how people reason about individual agents' hidden mental states (such as their beliefs and desires), to explain and predict those agents' observable behaviors (e.g., Ho, Saxe, & Cushman, 2022). The fundamental unit of analysis here is the individual agent, whose actions are interpreted under the principle of rational action (Jara-Ettinger et al., 2016). Such models have successfully captured many aspects of human social reasoning, including goal inference, cooperative communication, and false belief (Baker, Saxe, & Tenenbaum, 2009; Goodman & Frank, 2016; Baker et al., 2017).

Such models have also been used to characterize social relationships in terms of one agent's preferences about another agent's preferences. In particular, cooperative and competitive interactions between agents have been modeled through the concept of *adopted utility*: the idea that individuals incorporate others' goals into their own utility functions (Powell, 2022). In a cooperative interaction, agents act to help each other reach their goals, and in a competitive interaction, agents act to prevent each other from reaching their goals (Ullman et al., 2009; Kleiman-Weiner et al., 2016). Such models have been tested primarily in grid-world paradigms, where agents navigate to reach goals in physical locations. Adopted utility models provide a good fit for observers' judgments of whether one agent is acting to promote or hinder the utilities of another agent (Ullman et al., 2009; Hamlin et al., 2013).

Thomas's proposal implies that computational models of human social understanding require a qualitatively different approach. Thomas highlights that social relationships go beyond cooperative and competitive interactions defined by adopted utilities. The distinctions between communal sharing, authority ranking, and equality matching relationships cannot be reduced to whether one agent adopts another's utility: in all of these relationship types, agents can act to promote each other's goals and preferences. Yet, as Thomas shows, the relationship types differ dramatically in the expectations they generate and behavioral predictions they support.

Even more dramatically, Thomas suggests that the basic representation that people use to understand behavior in relationships is not a representation of one agent with beliefs and desires about another agent. Instead, the key cognitive primitive that people reason over, when reasoning about social interactions, may represent the relationship itself. Humans may rely on a small set of structured priors, such as a set of cognitive primitives specifically designed to recognize relational structures, each corresponding to a certain set of internal dynamics governing intuitive theories of different kinds of social interactions (e.g., Pietraszewski, 2022). These priors can guide learning and inference in new social situations: people can use probabilistic inference to recover how likely each type of relationship is – as well as dimensions important to the specific kind of relationship, like its strength – given observed social interactions.

Incorporating relationship primitives in Bayesian models of probabilistic reasoning can contribute to testable theories regarding the development of relational social cognition. These models can not only capture how each social interaction can be described in terms of multiple kinds of relationships, but also how new social information can revise how people think about certain relationships in the first place. Hierarchical models and Bayesian nonparametrics (Kemp, Perfors, & Tenenbaum, 2007; Gershman & Blei, 2012) can capture how infants and children flexibly infer the number and nature of relationship categories (and sub-categories) from a combination of innate priors and social experience, as well as accommodate variability across different social and cultural contexts.

This relationship-focused framework explains how minimal social information can lead to rich generalizations and predictions. Knowing that a social interaction – such as sharing saliva – strongly fits a particular relationship type, in this case a communal sharing relationship, can activate expectations about subsequent interactions without necessarily invoking the computationally costly process of explicit reasoning about individual mental states (Thomas et al., 2022). Similarly, knowing whether a social relationship is equal or hierarchical leads to strong expectations

about whether generous acts will alternate or repeat (Chen & Saxe, 2023). In fact, Theory of Mind reasoning about individual agents' beliefs and desires may only become necessary when relational expectations are violated, or in special cases like agents with conflicting beliefs or desires about the same relationships. In relational contexts, Theory of Mind might be reserved for resolving ambiguities or social anomalies, rather than forming the core basis for routine social predictions (Bohl, 2015).

If relationships, and not just individual minds, are the primary units of social understanding, then there are many open questions. One question is what precisely constitutes these relational primitives in the first place. Are they simply about sets of actions that are consistent with different kinds of relationships, and if so, is learning about different kinds of relationships the same as a category learning mechanism? How is learning about relationships different from learning about other social structures that govern different kinds of social norms? How much of early social reasoning can be understood in terms of relational primitives, versus active mental-state reasoning?

Another open question is how we might operationalize the notion of relationships in computational models. What would it take to extend existing paradigms that focus on the individual agent to account for relationships? One approach, proposed by Jara-Ettinger & Dunham (2024), is to reify "roles" and their associated "norms" as additional entities in a causal Bayesian theory. Could this approach be applied to model infants' intuitions about dyadic relationships? And if so, how might we build such models concretely – how might we extend domain-specific programming languages like memo (Chandra et al., 2025) to capture the semantics of relationship reasoning?

In summary, we propose that computational models of social cognition should incorporate relational representations as primitives. Traditional inverse-planning models, which focus solely on individual utilities, overlook the critical relational structures that organize social behavior and can support fast and efficient social learning.

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Cognitive representations of social relationships: Innate structures or cultural shaping?

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Abstract

We concur with the author's hypothesis regarding innate knowledge of social relationships, but we suggest that (a) cultural learning and the social context are determining factors, as evidenced by cross-cultural variations in the perception of authority and reciprocity, and (b) infants' expectations may also reflect learning processes based on early statistical associations.

In the target article, Thomas proposes that human beings possess an innate relational knowledge from early infancy, which enables them to recognize, categorize, and evaluate the strength of social relationships based on shared cognitive models, such as those outlined by the Relational Models Theory (Fiske, 1992). This hypothesis is grounded in experimental evidence from developmental psychology, which suggests that even infants are capable of distinguishing between communal sharing, authority, and equality relations, displaying expectations consistent with these structures and differential reactions to interactions that either violate or confirm such patterns (Hamlin, Wynn, & Bloom, 2007; Spelke & Kinzler, 2007). The central thesis suggests that the ability to understand and evaluate social relationships emerges early and without the need for significant experiential input, relying instead on innate cognitive mechanisms that allow individuals to recognize the nature of interactions and predict their course.

However, although this perspective is supported by some empirical data, we argue that Thomas's innatist hypothesis overlooks the crucial role of cultural learning, sociocultural variability, and interindividual differences in the development of social understanding. A more nuanced account should consider the dynamic interaction between general cognitive predispositions and specific environmental influences, avoiding the interpretation of infants' early social sensitivity as unequivocal evidence of innate conceptual structures. Indeed, many of the infant responses observed – such as preference for synchronized agents, responsiveness to eye contact, or the distinction between cooperative and non-cooperative individuals – may reflect basic perceptual and attentional mechanisms, associated with biases toward coherence and regularity, rather than conceptual categorization of social relationships (Heyes, 2018; Tomasello, 2019). In support of a less essentialist approach, it is important to highlight the significant cultural variability in the conception and practice of social relationships. Cross-cultural research shows that the ways in which individuals experience and represent relationships of authority, reciprocity, or equality vary widely depending on the sociocultural context (Meristo & Zeidler, 2022; Gächter & Herrmann, 2009). For example, cultures that emphasize hierarchy and respect for authority foster relational expectations that differ from those in cultures that prioritize autonomy and equality. This evidence suggests that the categorization of social relationships cannot be regarded as a universal and innate process, but rather as the result of progressive cultural learning, mediated by mechanisms of socialization and natural pedagogy (Gergely & Csibra, 2020).

In this framework, statistical learning emerges as a plausible alternative explanation for early social sensitivity. Infants appear capable of detecting behavioral regularities and social contingencies through repeated observation of interactions between individuals, forming expectations based on the frequency and consistency of observed events. General cognitive mechanisms such as selective attention, sensitivity to temporal contingency, and associative memory may support the development of relational inferences without requiring the presence of innate conceptual

schemas (Csibra & Gergely, 2009). Furthermore, recent studies show that infants' social preferences can be rapidly shaped by experience: exposure to novel social contexts alters behavioral expectations, suggesting a crucial role for direct interaction and early learning in the processing of social relationships (Decety, Steinbeis, & Cowell, 2021; Thiele et al., 2021). Early interactions with caregivers and community members also play a decisive role in the development of social cognition. The concept of cultural scaffolding refers to the intentional actions of adults in facilitating the acquisition of social norms, roles, and categories by directing the infant's attention and structuring learning experiences that reinforce the understanding of relational dynamics (Williams, Mastergeorge, & Ontai, 2010; Kärtner, Schuhmacher, & Torrens, 2020). Overlooking these dimensions risks misattributing to innate competencies what is, in fact, the result of a shared construction of knowledge within social practices.

Considering these considerations, we propose an interactionist model of relational development, in which social cognition emerges from the interplay between general cognitive predispositions and culturally specific environmental input. Within this framework, representations of social relationships are formed gradually through the observation of others' interactions, participation in social practices, and exposure to behavioral regularities. This approach allows for the integration of the data reported by Thomas with a more nuanced and realistic understanding of the role of culture in the development of social competencies, avoiding reductionist explanations based solely on innatism.

Finally, to further clarify the relative contributions of biological predispositions and social learning, it may be useful to adopt computational models capable of simulating the acquisition of social knowledge based on empirical data. Longitudinal and comparative studies conducted with infants exposed to different cultural contexts could provide additional insights into the role of the environment in shaping social expectations. In conclusion, we argue that a more comprehensive account of the development of social cognition in early life must necessarily rely on an integrated perspective, one that combines cognitive predispositions with cultural learning in shaping the understanding of human relationships.

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Shift to a new paradigm: realizing the relational models

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Abstract

Children do not need socialization by adults, and in fully traditional societies, adults do not inculcate norms for children to internalize. From infancy, children actively seek to externalize the relational models they know. To function in the community in which they find themselves, children need to discover when and with whom to coordinate using which relational model in what ways.

Ashley Thomas's illuminating review demonstrates six revolutionary facts. First, preverbal infants *recognize* communal sharing, authority ranking, and equality matching relationships, readily distinguishing among them. Second, preverbal infants know how 'people' (humans, puppets, animated figures) *distinctively interact* in each of the three relationships. Third, infants have quasi-moral sentiments about what 'people' in each of the three relationships *should do*. Fourth, infants *recognize transgressions* of each relational model, *recognize punishment*, and 'approve' of punishing 'people' for not doing what they should do in the given kind of relationship (at least in equality matching). Fifth, infants *expect other 'people' to recognize* each of the three kinds of relationship and expect others to *know what people distinctively do* in each. Sixth, infants expect other 'people' to *recognize violations of each relational model, to recognize punishment, to approve of it, to and punish 'people' who fail to punish*

others for transgressions (at least transgressions of equality matching).

In my judgment, this body of research shows that humans are social beings from the start, evolutionarily prepared to use the relational models to coordinate with others – as I first theorized 34 years ago (Fiske, 1991, 1992, 2004; Fiske & Haslam, 2005). Human sociality does not depend on adults socializing children.

Furthermore, active socialization does not occur in 'traditional' communities without schools and without significant involvement in markets (Lancy, 2022, 2024; Paradise & Rogoff, 2009). In such communities, parents virtually never teach anything, explain anything, or inculcate the norms of social relations. Parents simply do not impart anything for children to 'internalize' (with the exception, in some communities, of occasional imperative utterances about manners). Indeed, in many traditional societies, adults do not even converse with children, only speaking to them to send them on errands.

So it is now time for a scientific revolution that takes an entirely new approach to explaining human social-relational competence and commitment. We need to adopt a new paradigm focusing on children's intrinsically motivated *externalization* of their innate social-relational capacities and quasi-moral sentiments. The new paradigm must also recognize human children's uniquely skilled, intrinsically motivated, unceasing observation, integrated with specialized imitation capacities that include role-playing based on cumulative episodic memories from previous days. Toddler play consists mostly of imitating adult social relations. Allied with this is children's intrinsic motivation to become proficient contributors to family well-being, which, in traditional societies, entails social coordination in subsistence activities.

The new paradigm for social development: realizing innately known relational models so they fit one's community

Infants recognize and understand at least three relational models, but an infant cannot know how people in their particular community will use them (Fiske, 2000). Infants absolutely need to discover when, how, and among whom their particular culture implements each relational model. Which relational model do people use to organize which activities, and who participates in what relationships? When do people in the infant's community use authority ranking to coordinate an activity, and when they do so, who has what rank, and how did they get that rank? When do people in the community use equality matching, and when they do, who has standing as an equal, and who is excluded? When people in the community use equality matching to allocate civic responsibilities, do they take turns in different offices, are they elected, or are they assigned by lottery, and in each case, how long does a person serve in a given office? Children need to learn that, and they can learn that without any instruction.

Participant observation in all sorts of traditional societies shows that toddlers and children observe intently, remember the episodes they observe, integrate them, and incessantly play-imitate the uses of relational models that they recall (Gaskins & Paradise, 2010). That is how they discover the local implementations of each relational model and practice doing what they have observed. Gradually, they begin to engage in more complex social relationships with family members and others. Like their mastery of their language, children are perfectly capable of developing their social capacities on their own, without any instruction or guidance from anyone. In traditional societies, they transition from passive

observation to peripheral participation in subsistence activities and, in small increments, gradually take on greater productive responsibilities as they improve their skills (Correa-Chávez, Mejía-Arauz & Rogoff, 2015). In most traditional societies, children receive no praise for any of this and are largely ignored until they begin to make real contributions – but still never praised or extrinsically rewarded.

Research in the new paradigm of Realizing the Relational Models should now explore children's inherent capabilities and intrinsic motivations to be social, their social relational emotions and moral sentiments, and their attentive self-motivated observation and imitative relationship play. It should explore infants' expectations, sentiments, and initiatives in their own relationships. Also, the new paradigm requires rethinking social systems, which evidently are built out of the four relational models.

Infant recognition, knowledge, and evaluation of at least three relational models, together with their expectation that others recognize, know, and evaluate the relational models, could only have evolved if, in fact, humans consistently used these relational models to coordinate most activities. Indeed, extensive ethnological investigation shows that the four relational models do organize social activities of all sorts in hundreds of disparate cultures and have done so for millennia (Fiske, 1991, 2025). We do not know how rapidly these relational model adaptations evolved, but the fact that they did evolve indicates that humans have been using the relational models for a long time and in most populations. That is an important insight into prehistoric social relations, about which little is otherwise known.

Coordination depends on all participants using the same relational model, which would be a huge, perhaps insurmountable hurdle if people did not recognize the distinctive modes of the constitutive representations of each relational model (Fiske, 2025). Without prior knowledge of the fact that people use just these four relational models to coordinate virtually everything, and without prior knowledge of their representation systems, it would be impossible for children to learn how to coordinate the myriad activities that humans coordinate.

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Infant representations of social relationships: the need to consider the social world

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Abstract

We present three challenges to Thomas' theory leveraging our expertise in social and cognitive development. First, Thomas' theory fails to capture defining relationship characteristics, necessitating consideration of relationship science. Second, we argue that declaring innate knowledge of relationships is premature. Third, we propose that in the core domain of social cognition, relationship and sociomoral representations exist simultaneously without need for replacement.

Thomas proposes a daring account of early relationships representations, which will lead to exciting future research in developmental science and beyond. However, Thomas leaves vague the definition of relationships. Indeed, Thomas includes discussions of transient interactions among (groups of) people that fall outside widely agreed upon defining features of relationships as *dyadic* interchanges that occur *over time* (Hinde, 1979), raising questions about the specificity of Thomas' theory to relationships. Moreover, Thomas draws exclusively on Fiske's (1992) Relational Models Theory. Although influential, this theoretical focus fails to incorporate rich theory and evidence from the field of relationship science (Berscheid, 1999; Finkel & Simpson, 2015; Fletcher et al., 2019; Simpson & Campbell, 2013), particularly as informed by social developmental research (Laursen & Bukowski, 1997; Sroufe, 2000; Underwood & Rosen, 2013). Here, we identify three critical issues with Thomas' account, highlighting the importance of considering infants' social world in formulating an intuitive theory of relationships.

First, Thomas argues that a core characteristic of relationships is strength. However, from a relationship science perspective,

relationships are defined in terms of their *quality*, with close relationships (e.g., parent-child attachments, friendships) distinguished from other types of relationships (e.g., peers, work relationships). One widely accepted definition of close relationships is that they are characterized by a high degree of interdependence that endures over an extended period (Kelley et al., 1983). Thus, close relationships are distinguished from other relationships by their nature or quality, which strength fails to capture.

Take, for example, attachment relationships. Attachments are typically the first relationships infants establish and thus, relevant to Thomas' proposal. Characterizing such relationships in terms of strength results in a fundamental misunderstanding of attachment (Sroufe & Waters, 1977). Attachments are defined as an enduring psychological bond between the infant and caregiver that serves protective functions and evolved to promote survival (Bowlby, 1969), highlighting defining characteristics that separate attachments from other relationships that cannot be reduced to strength. For example, having "strong" relationships with kin (e.g., parent, grandparent) does not indicate that such relationships are attachments. Indeed, an attachment relationship is evidenced by preferential direction of attachment signals toward a relationship partner in times of uncertainty (Bowlby, 1969).

Second, Thomas argues for a core knowledge system of relationships, analogous to core knowledge systems of objects and agents (Spelke, 2024). However, further information is needed to support claims of relationship representations as core knowledge that is "... innate, skeletal, largely automatic, and evolutionarily old" (p. 7). Although comparative studies relevant to specific relationship cues are reviewed, this falls short of providing an evolutionary account for the premise of core knowledge of relationships. Why would humans evolve to have innate knowledge of relationships *and* to organize it around the proposed core components? This is addressed in supporting knowledge in other core domains (Cosmides & Tooby, 1994). For example, innate knowledge of the physical and social worlds might have evolved to allow humans evolutionary advantages of locating objects, navigating, foraging, hunting, as well as facilitating living with others and in social groups (e.g., Hamlin, 2013; Völter et al., 2023; Wynn & Bloom, 2014). A core knowledge system of relationships might afford comparable adaptive benefits. However, situating the presence and organization of intuitive relationships knowledge in evolutionary theory is necessary, highlighting another way in which Thomas' theory would benefit from greater attention to social developmental and relationship science (e.g., Simpson & Belsky, 2008).

Moreover, the extant infant cognitive evidence cited by Thomas fails to support core knowledge of relationships. Instead, the results can be interpreted as infant sensitivity to relationship cues. For example, infants are shown one interaction between two individuals. The presence of certain cues (e.g., imitation, saliva-sharing) evidently leads to infants making predictions about the individuals' (inter)actions in a new context. Being able to interpret dyadic interactions based on certain cues, however, is not the same as having innate representations of relationships. Whereas it is possible that infants are born with a sensitivity to cues of relationships, it is premature to suggest that infants also have an innate concept of dyadic, interpersonal relationships guiding their reasoning about the social world. Rather, as infants are born into relationships with their parents, they may learn from early experiences (e.g., interpersonal synchrony and saliva-sharing

between parents and infants) the importance of certain cues and use them to make predictions about social interactions.

Third, Thomas implies that relationship representations can supersede or precede sociomoral knowledge (Baillargeon et al., 2015; Hamlin, 2013). As noted above, the extant evidence for infant relationship representations remains tenuous, while there has been a wealth of evidence for infant moral knowledge (Ting et al., 2020; Woo, Tan, & Hamlin, 2022). We therefore suggest that innate relationship representations, should they be present, complement moral understandings in the core social domain. For example, infant sensitivity to relationship cues might help them reason about dyadic interactions, as shown by Thomas and colleagues (2022a; 2022b). Group membership markers, even minimal cues (Jin & Baillargeon, 2017), can help infants reason about situations involving more than two individuals and evaluate interactions among them, based on morality (Ting et al., 2020). The interplay of group representations, relationship representations, and morality thus underpins social cognition.

In summary, we believe Thomas' theory will inspire more work that aims to examine multiple facets of infants' social cognition. However, just as we suggest that Thomas' theory would benefit from closer attention to relationship science, work on infants' social cognition would also benefit from closer attention to the social world (Tasimi, 2020). Socioemotional developmentalists have extensively studied how experiences and individual characteristics shape development (e.g., Borowski et al., 2021; Groh et al., 2014). These factors must be included when examining how infants construct representations of the social world, including their representations of relationships. As an example, we have started collaborative research on infant attachment representations, drawing on our shared expertise (Luo, vanMarle, & Groh, 2024). Collaborations such as this are necessary to chart the developmental course of infant representations as they pertain to the social world and critically evaluate theories regarding the role of experience and innate knowledge.

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The evolution and development of infant social relationship reasoning: a Tinbergenian analysis

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Abstract

Thomas (in press) proposes that infants possess core systems for reasoning about social relationships. Using Tinbergen's four questions – phylogeny, ontogeny, mechanism, and function – I critically evaluate this claim, focusing on infants' sensitivity to saliva-sharing cues. I argue that integrating evolutionary, developmental, neural, and adaptive perspectives, particularly those linked to cooperative breeding, provides a fuller account of relational reasoning's origins.

Thomas (in press) proposes that infants possess an intuitive core cognitive system for reasoning about social relationships. A Tinbergenian analysis (Bateson & Laland, 2013; Tinbergen, 2005) provides a valuable framework to deepen our understanding of this claim by examining how such relational reasoning evolved, develops, functions neurally, and serves adaptive purposes. Focusing specifically on Thomas et al. saliva-sharing experiments with infants, I evaluate the evidence and propose that integrating evolutionary, developmental, and mechanistic perspectives can strengthen and refine the proposed framework.

1. Phylogeny: how did it evolve?

Humans are unique among great apes in exhibiting cooperative breeding, wherein mothers and allomothers – grandmothers, siblings, and unrelated group members – actively contribute to infant care and feeding (Hrdy, 2009; Hrdy & Burkart, 2020). This cooperative system likely emerged around 1.8 million years ago with *Homo erectus*, profoundly reshaping human infant psychology, emotion, and cognition. According to Hrdy (2009), this socio-ecological shift selected for infants equipped with enhanced social cognitive skills, enabling them to infer others' intentions, preferences, and commitments – skills crucial for engaging multiple caregivers.

Thomas et al.'s finding that infants infer emotional closeness from food/saliva-sharing behaviors resonates with this evolutionary background. In ancestral environments, behaviors such as kiss-feeding – mouth-to-mouth transfer of food – served as salient cues of nurturance and caregiving commitment. Infants' sensitivity to food (and saliva) sharing thus plausibly reflects deep

phylogenetic adaptations shaped by cooperative breeding. The human-specific evolutionary context of communal caregiving and provisioning must be considered foundational when theorizing about infants' social cognitive abilities and their sensitivity to feeding-related relational cues.

2. Ontogeny: how does it develop?

Developmentally, it is important to consider the timeline of exposure to caregiving and feeding behaviors. Kiss-feeding practices, involving the transfer of food or honey-flavored saliva, have been observed among hunter-gatherers, beginning around 6 months, coinciding with infants' eruption of teeth (Hrdy, 2009; Hrdy & Burkart, 2020). Even in WEIRD societies, solid food introduction is encouraged around 6 months, often involving close physical feeding interactions (American Academy of Pediatrics, 2012).

Notably, the youngest infants tested in Thomas et al. were 8.5 months old – an age by which most would have already experienced feeding interactions. Thus, infants' expectations about relational closeness based on food/saliva-sharing cues may reflect accumulated social experience rather than exclusively core relational reasoning. To rigorously test Thomas's (in press) proposal that infants possess core relational reasoning abilities, studies should investigate younger infants, ideally before 6 months, prior to substantial feeding experience. Such research would clarify the extent to which sensitivity to food/saliva-sharing cues is experience-dependent versus reflecting early, possibly evolved, conceptual structures.

Moreover, Ciftci and colleagues (2025), while replicating Thomas et al.'s findings with infants, noted that the original study did not test whether infants' responses were driven by food sharing in general or saliva sharing specifically. They pointed out that alternative conditions – such as feeding without saliva sharing or saliva sharing without direct feeding – were not tested. Since feeding alone may signal a strong social relationship and is easier for infants to detect, it could independently explain infants' responses.

3. Brain mechanisms: how does it work?

Thomas's (in press) proposal pays limited attention to the underlying brain mechanisms supporting infants' relational reasoning. Recent neuroimaging work offers valuable insights. Farris et al. (2022) demonstrated that infants between 6 and 13 months engage the dorsomedial prefrontal cortex (dmPFC) when observing third-party social interactions characterized by physical contact and synchrony – cues relevant for communal sharing. These findings align with evidence from adult neuroscience, where the medial prefrontal cortex is implicated in mentalizing and social evaluation (Amodio & Frith, 2006).

The recruitment of dmPFC in infants suggests that brain specialization for processing social relational information emerges early. However, an important open question is whether the infant brain differentiates between relational categories beyond communal sharing, such as authority ranking and equality matching, as proposed by Fiske's (1992) Relational Models Theory. To date, no neuroimaging studies, even in adults, have systematically tested these relational distinctions within the same experimental paradigm. Future work should explore whether brain processes are differentially engaged by

distinct relational models and how the neural bases of social relationship reasoning develop during infancy.

4. Adaptive value: what is it for?

From an adaptive perspective, early relational reasoning likely provided significant survival benefits in human evolutionary contexts. Infants who could recognize supportive, nurturing alloparents – beyond the primary caregiver – would have been more likely to survive and thrive (Hrdy, 2009; Hrdy & Burkart, 2020). Sensitivity to behavioral cues such as food sharing, affectionate touch, and synchrony would allow infants to identify individuals willing to invest caregiving resources.

Relational reasoning thus likely evolved as an ontogenetic adaptation enabling infants to actively engage and recruit alloparents. Given the extraordinary costs of prolonged human infancy and childhood, the capacity to navigate complex social networks from an early age would have been crucial. Infants' ability to interpret food/saliva-sharing behaviors as indicators of emotional closeness fits within this broader adaptive framework.

More generally, early relational reasoning may form part of the broader suite of human-specific social cognitive adaptations supporting cooperation (Tomasello, 2020). Sensitivity to social relational information, including communal caregiving structures, may be an essential component of the evolutionary foundations of human sociability.

Conclusion

Applying Tinbergen's framework to Thomas's (in press) proposal enriches our understanding of infants' social relationship reasoning. Human infants' sensitivity to relational cues such as food sharing is likely shaped by both evolutionary pressures favoring cooperative caregiving and developmental experiences within socially rich environments. Brain specialization for processing relational information appears early but requires further empirical investigation to fully map onto relational models.

While Thomas makes a strong case for early-developing relational knowledge, future work should rigorously disentangle core cognitive architecture from experiential learning and probe the neural mechanisms supporting relational distinctions. A Tinbergenian approach reveals that infants' relational reasoning is likely not only an early-emerging skill but also an adaptive product of human cooperative social life.

Ultimately, understanding how infants reason about social relationships requires integrating evolutionary history, developmental processes, brain function, and adaptive significance. Doing so promises a more comprehensive account of the origins of the human social mind.

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Moral somatic marker: new directions for the mechanisms underlying morality in preverbal infants

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Abstract

Thomas theorizes that infants have cognitive representations of social relationships. However, this theory does not explain all the findings on infant morality. We highlight these discrepancies and propose that infants make moral decisions based on the perception of bodily responses (interoception), such as Damasio's somatic marker. This opens up new directions for research into infant morals.

Thomas theorizes that infants have cognitive representations that categorize social relationships into models from early development as a possible form of core knowledge. This theory explains and fits well with much of the existing evidence on infant social cognition. In contrast, there exist findings that cannot be explained successfully. Several studies have shown that infants exhibit early signs of morality. Many of these studies are grounded in social evaluation through preferences for agents with good characteristics (e.g., helping agents). Thomas interprets these with greater parsimony, for example, as preferences for imitators, without making assumptions about morality. However, Thomas' theory does not explain all the findings on morality in early development. Therefore, we explain the discrepancies and consider new directions for understanding the mechanisms underlying morality in early development.

First, as Hamlin (2024) notes, not all preferences for moral agents are necessarily explained by the preferences for imitative agents. Indeed, prosocial behaviors, such as helping behaviors, do not always stem from imitative behaviors (e.g., Kanakogi et al., 2017), and research has shown a preference for moral characters over imitators (Chae & Song, 2018). Second, infant studies on moral foundations, such as harm or care (Haidt, 2007), are also unexplained by relational models such as the authority ranking relationships (dominance) proposed by Thomas. For example, Kanakogi et al. (2013) observe that infants avoid the aggressor and approach the victim even when the aggressor and victim are compared to neutral agents. This tendency to approach the victim cannot be explained by the avoidance of the dominant agent, as claimed in an experiment with similar content (Thomas & Sarnecka, 2019), because there is no comparison with a neutral agent in the study. There is some concern, if not care, for the victim. In addition, Kanakogi et al. (2017) show that an agent who intervenes in a harmful interaction is preferred, unlike one who intervenes in neutral interactions; this implies a negative evaluation of harm, which represents harm aversion (although this finding may also be interpreted in terms of the authority ranking relationships [legitimate authority] of Thomas' theory). Third, and most importantly, a seminal study has demonstrated moral behaviors of infants attempting to punish agents who have harmed others: using a gaze contingency technique, Kanakogi et al. (2022) show that infants punish others who have attacked others selectively by their gaze. This behavior involves not only the categorization and evaluation of social relationships but also a decision-making process aimed at delivering appropriate punishment to the agent. Therefore, this behavior cannot be explained by Thomas' theory; we must assume some underlying motivation for driving the behavior.

How would motivation for the behavior be generated, and how would moral decision-making take place? Primitive moral emotions, such as harm aversion, may influence moral behaviors such as punishment. Emotional contagion for others' distress has long been reported in the developmental literature, with newborns crying when they hear others crying (Simner, 1971). It has also been reported that infants show similar expressions of distress when they see others in distress, suggesting that empathy for distress is present from the age of three months (Davidov et al., 2021). These findings indicate that from birth, infants are sensitive to others' state of harm. Through the experience of observing others' suffering, the feeling of harm aversion may be acquired by pairing negative affect (via emotional contagion) with actions that harm others (Blair, 2007). It is also possible that the principle of "not causing harm" exists as an innate core moral standard.

Regardless, it is likely that the primitive moral emotion of harm aversion is present from an early stage of development. Although it is still an open question as to how harm aversion leads to punitive behavior, harm aversion would be the most appropriate emotion for punitive motivation.

One promising mechanism for decision-making is Damasio's somatic marker. According to the somatic marker hypothesis, the feeling of bodily changes associated with emotions helps guide decision-making in situations of complexity and uncertainty (Damasio, 1994). The somatic marker hypothesis can be applied to moral judgment, during which strong emotional reactions arise when one considers sacrificing one person through one's own direct action to save other people; these emotions have been reported to significantly influence moral decision-making (Greene et al., 2001). Interoception (the perception of internal bodily signals) also plays a critical role in moral decision-making processes. For instance, Cui and Nakano (2024) showed that brain processing of heartbeats (heartbeat-evoked potentials: HEP) arising from the negative evaluations of harming others, regardless of the type of harm (direct or indirect), influences decisions not to harm others. From a developmental perspective, it has been shown that even three-month-old infants can discriminate between stimuli that move synchronously with their heartbeat and those that do not (Tünte et al., 2025). Furthermore, in five-month-old infants, cardiac discrimination ability is correlated with HEP amplitudes when viewing emotional facial expressions (Maister, Tang, & Tsakiris, 2017). These findings suggest that infants have sensitivity to their interoception. Given that HEP has been reported to influence moral decision-making in adults (Cui & Nakano, 2024), they also suggest that interoceptive sensitivity may be relevant to decision-making even during early development. Considering these findings, it is plausible that infants make decisions to punish those who harm others based on the perception of bodily responses arising from intuitive negative evaluations of harmful actions.

To date, no study on infant morality has examined the relationship between morality and bodily responses or interoception. However, if bodily responses and interoception are involved in morality in early infancy, it sheds new light on the underlying mechanisms, from the categorization and evaluation of social relationships to moral decision-making. This complements both Thomas' theory and Hamlin's moral core theory (Hamlin, 2013; Woo, Tan, & Hamlin, 2022). A more comprehensive understanding of morality during early development would open new research directions in infant morality.

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How to explain the strength of the bond? The normative missing link

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Abstract

This commentary highlights the importance of articulating Ashley Thomas's perspective on representations of social relationships with a reflection on the rules that constitute or sustain them. This approach may also help bridge the gap between developmental studies on *intuitive sociology* and both theoretical and empirical research in *scholarly sociology*.

Thomas's impressive article demonstrates that children identify social relationships and use them as a basis for inference and action. Showing that relationship-based reasoning emerges so early in life is groundbreaking. Until recently, the social domain was indeed viewed from a mentalistic perspective: it was thought to consist, ontologically, of mental states and, cognitively, of mindreading (for a critical discussion, see Kaufmann & Clément, 2014). Setting aside this mentalistic view, Thomas's research confirms that very young children can connect *perceptual cues*, such as large body size, to *abstract knowledge* about relational models – in this case, authority ranking.

Her studies on communal sharing provide strong evidence for this two-level cognitive processing. In Thomas et al., toddlers and infants identify concrete features of interactions (saliva sharing and food licking) to infer the nature of a social relationship (intimacy) and then use this inferred relationship to predict seemingly unrelated behaviors (helping someone in distress). Remarkably, children appear to set aside the individual traits of the characters involved (an actress's prosociality) and instead reason on the *relationship* itself. Furthermore, children interpret saliva sharing – but not ball playing or toy sharing – as a cue for intimacy. This suggests that they distinguish between strong, thick relationships, which involve bodily intimacy and emotional engagement, and weak ties, characterized by minimally engaging activities such as ball playing.

Thomas's fascinating line of research opens the door to a productive dialogue between developmental psychology and the social sciences, the latter of which regard relationships as their conceptual primitives. However, this dialogue seems to stall partway. One reason, somewhat paradoxically, is that developmental studies on *intuitive sociology* often fail to engage with a well-established insight from *scholarly sociology*: the intrinsic connection between social relationships and the norms and rules that, as I will argue, constitute the crucial *missing link* between the representation of social relationships and what Thomas refers to as “contextual cues.” To substantiate this claim, I will examine the rights and duties that social relationships entail, as well as the nature of the predictions that their cognitive representation allows.

Rights, duties, and social relationships

To Thomas, “the relationship's strength is the representation – and the mutual awareness – of *obligations* (the extent to which certain actions are expected), and *commitment* (the likelihood that people will continue the relationship)” (p.5). And yet, social relationships cannot be reduced to mere cognitive representations. They exist beyond a floating “in-between” of mutual awareness and possess an objective, impersonal, and public dimension that goes beyond the contextual cues individuals may be aware of. When a teacher meets a student at a university, their expectations are primarily shaped by their roles and the *rights and duties* defined by the institutional setting. The teacher is expected to transmit knowledge, and the student is expected to learn from the teacher. Even if the relationship becomes more informal, this shift adjusts rather

than replaces the underlying structure. What does this mean? It means that social relationships are anchored in the situations that “*afford*” them and in the normative rules that *reinforce* them. In this light, relational models may be more accurately conceived as *relational frames*, a term that captures both their cognitive dimension and their grounding in rule-governed situations.

The workings of predictions

As Thomas put it, “the function of relationship representation is to allow humans to make sense of interactions between individuals, updating predictions about whether and how people will interact” (p.4). However, there are two qualitatively different processes on which social predictions may be based. The first type of prediction is based on experience, statistical regularities, or commonly observed responses in specific social situations – that is, what usually happens. This is this *probabilistic* kind of prediction that Thomas seems to favor: relationship representation may be a computational process involving the assignment of *probabilities* to each relational model. For instance, a relationship with a colleague could be evaluated as “3/10 on authority ranking, 9/10 on equality matching, and 1/10 on communal sharing” (p. 25). The second type of prediction, based on *normative rules*, has a quality of obligatoriness or “strength” that probabilities and regularities lack (Clément et al., 2011; Rakoczy & Schmidt, 2012). While regularities pertain to factual observations and follow a logic of repetition and conformity, rules exhibit a higher level of constraint, sanction, and imposition. The distinction between these two social prediction mechanisms – probability vs normativity – has significant implications: it separates what *is* from what *ought* to be, what people normally do from what they should do according to social norms or moral ideals. As the sociologist Emile Durkheim ironically suggested long ago, this distinction is consequential: it allows one to distinguish the “average man” from the “ideal man.”

My point is that navigating the social world involves both descriptive and normative representations of social relationships. This is what my colleagues and I tested in an experiment we termed the *Robin Hood paradigm* (Charafeddine et al., 2016). When confronted with an asymmetrical relationship between dominant and subordinate individuals, children of different ages did not treat it in the same way. Children between three and five years of age favored the dominant individual, or the boss, who had already prevailed in previous games, thereby preserving the *status quo* – for example, by giving the dominant individual more goods. In contrast, children from five to eight years of age increasingly counterbalanced the relational asymmetry and favored the subordinate. Older children thus transitioned from normality to normativity, from a kind of social realism, based on observed *authority ranking*, to a normative stance, grounded in *equality matching*.

How to explain the “strength of the bond”?

According to anthropologist Marcel Mauss, it can be explained by “rights and duties” that constitute social relationships. Of course, this answer is only the beginning of the inquiry, not the end. The real challenge lies in articulating the impersonal, “objective” language of rights and duties with the subjective meaning the relationship holds for the “men, in their minds and in flesh and blood” who participate in it (Mauss, 2007: 89). Thomas's stimulating studies represent a significant step in this direction.

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What are the conceptual primitives of relationships? An agent-based modeling approach

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Abstract

The tripartite relationship model is unlikely to reflect how relationships are represented in the mind, given considerable difficulty in wresting clear conceptual distinctions from it. Instead, it offers a descriptive framework for organizing findings. Here, we present examples of conceptual difficulties of the model and elaborate on the functional importance of third-party inference based on our agent-based modeling work.

Social relationships shape how we behave toward others – whether to lead or follow, when to offer assistance, and from whom we can expect support and to what extent. Thomas does an excellent job tying together a wide range of empirical findings into a unified framework. And we agree with the central premise: the human mind, even from infancy, must contain relationship representations and procedures operating upon these representations in order to navigate the social world effectively. The strength of the model is that it is explicit, and like any good model, is bound to be

wrong in some respects; this is not a shortcoming, but entirely the point. Here, we take aim at one possible shortcoming, not to criticize, but to use the model as a scaffolding for pushing further inquiry, and preview some work we're currently doing in our lab that may speak to some larger issues.

Our constructive gripe runs as follows: Broadly, we wonder if the three categories of relationships presented are in fact distinct. In the process, we worry that fuzziness of the boundaries between categories may undermine some of the functional clarity that the model seeks to provide. Specifically, we worry that the lines denoting interactions between agents hide considerable ambiguity that warrants sussing out, to the point that at least one of the three categories may be an instance of a class of outcomes involving the other two.

In particular, equality matching seems conceptually different from the other two. Rather than describing a *type* of relationship, equality matching underpins *all* successful social exchanges. The ability to detect violations of fairness and punish free riders is a foundational adaptation that undergirds the evolution of “altruism” (direct and indirect reciprocity). In this sense, equality matching may not be a distinct relationship model but rather, may have the same status as what Thomas calls “strength”: adjectival modifications of the “types,” not the types themselves. That is, a hot pie and a cold ice cream describe only two foods, not four: hot and cold are adjectival modifiers, and so, we suggest, is equality matching. More literally, the human mind contains adaptations that are concerned with tracking the degree to which all relationships involve equality versus inequality, although not necessarily in kind. Equality matching is not a relationship type, but a particular value along this adjectival continuum.

Thomas' evidence supports this: in the absence of any relationship cues, infants default to expecting resources to be distributed equally. Their sensitivity to inequality (e.g., noticing when a character “steals” from another) reflects an expectation for fairness, not necessarily an assignment of a relationship “type” (see Tooby & Cosmides, 1996). Within non-kin relationships, roughly-speaking, humans' cognitive architecture should be sensitive to cues denoting some sort of reciprocation. Indeed, humans experience anger when they feel they are not being properly valued in a relationship and modulate the costs they incur for someone based on the costs the other incurs for them (Sell et al., 2017). Even within authority rank relationships, the expectation of reciprocity persists. Infants in Thomas' cited studies selectively deferred to “legitimate” leaders rather than bullies, suggesting that following a leader is contingent on some perception of shared or aligned interests. And Price & Van Vugt, (2015) notion of “service for prestige” is predicated on a reciprocal relationship between leaders and followers: meaning that a type authority ranking is caused by a kind of equality matching (just one not in kind).

Even leaving equality matching aside, the remaining categories are still not quite distinguishable. Communal sharing is defined by acting as one, having shared fate, and expecting care without direct reciprocity, but all of these features can also apply to hierarchical relationships that are cooperative rather than coercive. Leader-follower dynamics, when not based on dominance, still involve mutual obligation and shared goals. In coordinated group efforts, individuals take on asymmetric roles that collectively function as a single entity. From a macro-level view, they “act as one,” but the micro-level ties that support this unity are still built on reciprocal benefit – even if that reciprocity is uneven and role-dependent.

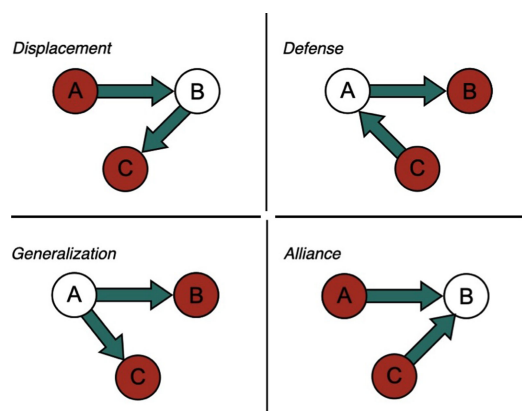


Figure 1 (Lavrador and Pietraszewski). Adapted from conflict interactions, above are the four ways in which groups might be transitively inferred through dyadic pursuits, with group-constitutive roles depicted in red. From left to right, A's pursuit of B is followed by: A also pursuing C; C pursuing B; B pursuing C; C pursuing A.

Soldiers follow a commander not just out of deference, but because the leader represents the group's interests, and the costs of obedience are incurred on behalf of a shared cause.

In sum, and with due deference to Fiske's model upon which Thomas builds, there is considerable trouble here, which suggests that while the model may hold adequate water for the descriptive purpose of organizing research findings (of which there is great value), the model does not quite decompose relationship types into their cognitive primitives; we should not confuse it for a computational theory of relationship types, but rather a framework to push on (as we are doing here) to get there.

On a final note, one key insight that we are now just beginning to appreciate in recent work in our lab is a point alluded to by Thomas – namely, that infants use third-party interactions to infer relationships. While this a well-worn observation outside of infancy, the function of using third-party relationships, we contend, has been underestimated and undervalued – even by ourselves.

While further refinement and empirical validation are ongoing, our results from a number of agent-based modeling simulations suggest that reasoning about third-party relationships may be *necessary* for the emergence of coordinated, stable social structures. In these simulations, agents form ties by scoring others based in part on how they relate in directed pursuits to other agents (four strategies based on the primitives introduced in Pietraszewski, (2022); see Figure 1). When agents use any of these triadic evaluation strategies (including any and all permutations of them), groups naturally emerge: densely connected cliques where every member is tied to every other – while direct reciprocity alone fails to produce group structures. It is no accident that young infants not only track their own relationship status with others but also track the relationship between others (third-parties), because without the ability to track others' relationships, our ability to pursue our own relationships falters, particularly if we want those relationships to be stable.

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Representations of equality or partiality? Unequal distributions as cues to social relationships

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Abstract

Thomas compellingly argues that infants are especially attuned to relationships. We agree but suggest her claims can reach further in her discussion of infants' reactions to distributional events. Thomas overlooks an interpretation of this literature in which *unequal* allocations might reflect relational inferences, challenging some of her claims regarding equality matching relationships and whether infants possess an "innate moral core."

Over the last decade, research suggests that infants have a rich understanding of others' social lives, potentially representing a heterogeneous set of core social domains. Thomas presents a bold thesis that could account for many of these findings from a unitary core capacity: Infants are predisposed to track relational cues and infer connections between others. We see this as a potentially transformative hypothesis with even broader implications than Thomas considers. Specifically, we suggest that findings interpreted as evidence that infants expect and prefer fair resource distributions might instead reflect a sensitivity to allocations as indicating a relationship status between the distributor and the recipients. On this view, *unequal* distributions might signal a relationship: unequal distributors favor one recipient over the other, suggesting closeness, whereas equal distributors treat both recipients impartially, which doesn't imply information about relationships. Research shows that unequal distributions lead children to think that the distributor is friends with the advantaged recipient (Liberman & Shaw, 2017). Although infants haven't been

tested on a similar paradigm, we argue that data on infants' responses to distributional outcomes may likewise reflect relationship inferences rather than moral evaluations. This possibility challenges prevailing interpretations and highlights the need for further research on infants' sensitivity to equality matching (EM) relationships.

Research on infants' reactions to distributional outcomes largely falls into two categories: infants look longer at unequal distribution outcomes – interpreted as a violation of expected equality (Meristo, Strid, & Surian, 2016; Schmidt & Sommerville, 2011; Sloane, Baillargeon, & Premack, 2012; Sommerville et al., 2013; Ziv & Sommerville, 2017) – and they show preferences for equal distributors over unequal distributors (Lucca, Pospisil, & Sommerville, 2018; Surian & Franchin, 2017) – choosing to engage with the former over the latter. This research has almost exclusively been interpreted as evidence for babies' sociomoral responses. However, longer looking times may reflect interest in relational cues, not violations of expected equality. And preferences for equal distributors may stem from seeing them as possessing an open alliance – open to forming *new* allegiances (including with the infant) – while unequal distributors may be inferred to already have preexisting ties reflected in their partiality.

Our proposal that infants are focused on relational information in resource distributions rather than violations of equality aligns with a body of research – much of which Thomas incisively summarizes – that infants are deeply attuned to social and affiliative cues like imitation, similarity, synchrony, and proximity (Kudrnova, Spelke, & Thomas, 2024; Liberman, Kinzler, & Woodward, 2014, 2018, 2021; Liberman, Woodward, & Kinzler, 2017; Powell & Spelke, 2013, 2018). It also challenges the idea that infants' responses to inequality indicate moral evaluations. In reconciling her relational account with claims about an “innate moral core,” Thomas speculates that instead of inferring moral traits, infants might favor agents who initiate or maintain prosocial relationships, for instance, reaching for “those who distribute resources equally, which could be seen as establishing an equality matching relationship” (pp. 26–27). However, she admits difficulty in explaining why without invoking trait inferences. Our relational interpretation sidesteps this issue: If infants interpret allocations relationally rather than morally, their responses might reflect reasoning about social connections rather than unfairness. Ziv, Whiteman, and Sommerville (2021) is, to our knowledge, the only study assessing whether infants directly reward and punish equal and unequal distributors, a hallmark of moral evaluation. While infants rewarded an equal distributor, they didn't punish unequal distributors, which aligns with our proposal that infants aren't viewing unequal distributors as immoral. And similarly, rewarding equal distributors doesn't necessarily indicate moral evaluations but possibly a desire to affiliate. Rewarding, reaching, and moving closer are all different ways in which infants may seek to affiliate with the equal distributor.

Thomas's argument that infants have some of the cognitive tools necessary to represent EM relationships rests largely on the work of Tatone and colleagues (Tatone & Csibra, 2020; Tatone, Geraci, & Csibra, 2015; Tatone, Hernik, & Csibra, 2021), which shows that infants especially encode features of giving actions relative to taking. However, tracking exchanges is not the same as representing equality. Infants may simply be attuned to the especially social nature of giving over taking. While Tatone's work emphasizes the relational importance of giving – supporting our

relational account – open questions remain about whether infants represent parity and reciprocal interactions, which are key to representations of EM relationships.

Ultimately, a theoretical account is only as good as the empirical evidence supporting it. Many tests could be conducted to differentiate between accounts of infants' social evaluations of distributional outcomes. For instance, a future study could test whether infants infer stronger relationships from unequal than equal distributions. Similarly, our account predicts that infants should expect unequal allocations after witnessing unequal affiliative information between a distributor and recipients, but not after witnessing equal or no affiliations. Additionally, research that probes whether infants generalize an agent's distributive tendencies to other moral or immoral behaviors or vice versa would help evaluate our proposal. As Thomas notes, some studies have been done suggesting that infants infer stable moral traits of individuals (Gill & Sommerville, 2023; Surian et al., 2018). But it's also possible that infants are reasoning about relationships in these studies. Future experiments could explore this possibility.

Our relational account also complements recent proposals of infant social affiliation such as adopted utility calculus (Powell, 2022) that similarly challenge trait-based views of infant socio-moral evaluations. Although a full comparison of these accounts is beyond the scope of this commentary, we note that both Thomas and Powell (2022) broadly accept the evidence that babies expect and prefer equal distributional outcomes. We reinterpret the evidence as reflecting infants' attention to relational cues embedded in distributions and their outcomes.

In summary, we agree with Thomas's broader claims and applaud her thought-provoking effort to reexamine disparate findings in infant social cognition under a relational umbrella. Building on her account, we argue that infants track resource distributions to infer affiliative bonds. This potentially explains why infants look longer at unequal distributional outcomes and why they show preferences for equal distributors without invoking moral considerations. In doing so, we aim to extend Thomas's broader thesis toward accounting for sociomoral evidence on relational rather than moral grounds.

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From pairs to patterns: combinatorial processes and relational reasoning in the processing of social relations

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Abstract

Ashley Thomas's account does not detail mechanisms for processing structural regularities, organizing sets of social relationships. I argue that combinatoriality and relational reasoning are essential for processing social structures. These abilities, foundational to processing social relationships, enable early, abstract, and flexible understanding of social hierarchies and networks, supporting adaptive social inference across species and development.

Ashley Thomas highlights identifying, categorizing, and communicating social relationships but does not fully address mechanisms for processing structural regularities – i.e., the systematic organization of social relationships among three or more individuals. These mechanisms are vital to a comprehensive account of core cognition of social relationships. I argue that processing such regularities requires (i) combinatorial mechanisms, which enable the integration of dyadic relationships into broader social networks, and (ii) relational reasoning that takes into account systematic relations between relationships. Both are essential to processing social relationships and their organization into larger structures.

Social structural regularities are widespread. Throughout the animal kingdom, dominance relationships tend to be transitive – if A dominates B and B dominates C, then A is likely to dominate C (Tibbetts et al., 2022), and to be organized in pyramids – one dominant over several subordinates is more common than the reverse (Mascaro et al., 2023). The prevalence of these regularities provides evolutionary leverage for the emergence of cognitive mechanisms designed to process them.

Mechanisms for processing social structures share key features of core cognition. They emerge early, persist throughout development, and are found across species. For instance, human adults, children, and infants draw inferences consistent with the

pyramidal shape of hierarchies (Hok et al., 2024; Mascaro et al., 2023). Similarly, transitive inferences over dominance relationships are found both in humans – from infancy (Gazes et al., 2017; Mascaro & Csibra, 2014), and across species including birds, fish, and insects (Grosenick et al., 2007; Paz-y-Miño et al., 2004; Tibbetts et al., 2019). Assumptions about social structures are abstract – they cannot be reduced to a purely sensory vocabulary. They also support inferences beyond the sensory input, about relationships between individuals who have never been observed interacting. Thus, core cognition might very well include mechanisms for representing social structures.

Assumptions about the shape of social structures have been observed in infants with months of social experience (e.g., Gazes et al., 2017; Mascaro & Csibra, 2014; Spokes & Spelke, 2017). Thus, it remains unclear whether these assumptions are innate or learned through experience. Indeed, computational models show how learners can navigate a hypothesis space of structural forms and choose the most appropriate one in a given domain, even with limited evidence (Kemp & Tenenbaum, 2008). However, even flexible models of structural learning require pre-existing cognitive abilities, such as the capacity to combine dyadic relationships into larger structures (combinatoriality) and to process structural shapes (relational reasoning). These foundational capacities appear to be prerequisites for learning about social structures, rather than products of learning, and are likely part of the start-up kit for processing social relationships.

First, processing social structures requires combinatorial processes to combine separate representations of relationships into larger networks. Evidence suggests they operate early. Infants effectively memorize hierarchical relationships presented sequentially (first $A > B$, then $B > C$, and last $C > D$), making it possible to build a unified representation of the hierarchy incrementally (first $A > B$, then $A > B > C$, and finally $A > B > C > D$). In contrast, infants struggle to memorize the same relationships presented in disjointed sequences (first $A > B$, then $C > D$, and finally $B > C$). This order requires holding separate relationships in mind before integrating three relationships at once into a unified structure (first $A > B$, then $A > B$ and $C > D$, and finally $A > B > C > D$). Thus, infants attempt to combine dyadic relationships into a unified representation of a social structure, rather than memorizing them in isolation (Mascaro & Csibra, 2014). Combinatoriality is key to processing social structures, which are rarely – if ever – directly observed. Instead, learners typically witness interactions between individuals, on the basis of which relationships might be inferred, and can, in turn, be combined in larger representations of social structures. Combinatorial processes allow learners to generate a wide range of social structure representations from a limited repertoire of social relationships – supporting flexible learning across diverse social environments.

The second key component for processing social structures is relational reasoning – the ability to process relations between relationships. This capacity is required to form assumptions about the way sets of relationships are organized, enabling learners to process patterns and regularities within social structures. For instance, assumptions about the transitivity or pyramidal organization of hierarchical relationships do not refer to relationships between pairs of specific individuals – they specify how hierarchical relationships in general are organized. Relational reasoning may be achieved through mechanisms that explicitly represent social structures (e.g., specifying structural forms and the

processes that generate them) or rely on more implicit mechanisms yielding computations consistent with structural regularities without directly representing them.

Mechanisms supporting relational reasoning are key to organizing and inferring social relationships not only within each relational domain but also across domains. They are required, for instance, to think about the systematic ways in which alliances and hierarchical relationships interact – a capacity evident from infancy (Pun et al., 2016, 2022). Moreover, they are likely to play a central role in helping learners infer their relationships to others, from observing how others relate to them and to each other (Thomas et al.).

Data on the processing of structural regularities offer key insights to test classic theoretical accounts. For instance, they reveal that infants' social inferences about dominance relationships do not align with predictions from the authority ranking model (Fiske, 1992). This model assumes a linear ordering in which everyone's rank can be compared with everyone else's, making transitive inferences obligatory. However, infants' inferences over dominance relationships are not obligatory (Mascaro & Csibra, 2012). Moreover, infants' assumptions about the pyramidal shape of hierarchies cannot be captured by a simple linear ordering (Mascaro et al., 2023). Infants' assumptions about the shape of social structures also illuminate why core cognition of social relationships is domain specific, complementing Ashley Thomas's argument. They suggest that the representation of social relationships cannot be reduced to either core agency cognition or, more broadly, to reasoning based on naïve psychology. The distinct patterns in social structures necessitate a specialized cognitive framework separate from other domains of core cognition.

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Intuitive sociology or intuitive social psychology?

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Abstract

Infants and young children appear to have core knowledge structures related to social dynamics. However, I think intuitions about group categories are too easily dismissed or ignored in the target article. More substantively, I argue for a theoretical distinction between intuitive sociology and intuitive social psychology.

Intuitive sociology or intuitive social psychology?

Fiske's Relational Models Theory is an excellent heuristic for understanding some of the computational dynamics likely underlying early social information processing.

I'll very briefly mention a few issues before diving into a deeper point.

First, there are likely other important forms of core knowledge that we should talk about. Boyer and Peterson (2018) and others, for example, might suggest that the intuitions underlying free-rider detection and exchange evaluations (e.g., fair/unfair, acceptable/unacceptable) are also present early in development. Additionally, in another work (McCaffree, 2022), I make the case that we should be seeking to measure intuitions about social norms given other theorists' emphasis on collective intentionality as a species-specific human trait (Tomasello, 2023). Intuitions about social norms might be evidenced by infants attributing agents' behavioral routines to the presence or absence of particular symbols (e.g., different shapes); that is, infants might be capable of understanding that others' behavior can change in accordance with symbolic cues. This basic capacity would then become

elaborated in early-middle childhood (e.g., Hardecker, Schmidt, & Tomasello, 2017).

With the space I have, however, I'd like to make a more fundamental point: I think what is being described in Thomas' target article is intuitive *social psychology*, not intuitive *sociology*, and I think this distinction is theoretically helpful for organizing research and generating hypotheses about the nature of social intuitions.

Let me be clear: I am *not* insisting on some silly disciplinary boundary between sociology and social psychology. I wish more sociologists today understood and contributed to the fascinating work coming out of developmental psychology and vice versa. My distinction, rather, has to do with *levels of analysis*.

What is sociology? Conventionally, sociology has been understood as the scientific study of "society." This formulation goes back to the institutionalization of the discipline in France in the late 19th century. This would suggest that intuitive sociology would relate to core knowledge about "society." Ok, so what is "society"?

In a recent forum article for *Behavioral and Brain Sciences*, wildlife biologist Mark Moffett (2024) summarized a wide swath of observational data related to animal social behavior. His conceptualization of society is thus inductive, drawn from how animals appear to be organized in nature. Of course, not all animals live in "societies." Orangutans, cheetahs, bison, and zebras are all social, but they do not appear to live in societies.

Moffett (2025) describes societies as "clearly membered, enduring, territory-holding groups," by which he means a group with a shared identity that endures over time and that exerts territorial control over a physical space. Wolves, lions, horses, hyenas, troop-living monkeys, chimpanzees, ants, and bees, for example, all live in societies. While social groups are common in nature, societies are not. Rubenstein (2018), for example, estimates that fewer than 10% of birds, 2% of insects, and 5% of mammals live in societies. Behavioral ecologist Bernard Crespi (2014) has referred to human beings, in particular, as "insectan apes," because of our history of living in both small-scale societies (characteristic of, for example, chimpanzees) as well as in large-scale societies (characteristic of, for example, some ant and bee species).

Crucially, there is evidence for core knowledge structures related to each of these aspects of "societal" group living: group identity (Chalik & Rhodes, 2018), territorialism (Nancekivell, Friedman, & Gelman, 2019), and intergenerationality/age-status intuitions (Moses et al., 2001; Harris et al., 2018). If we are talking about intuitive *sociology*, these are the core intuitions at play, while intuitions about specific relationship characteristics are an important component of intuitive *social psychology*. The former is about the core knowledge that is developing to enable the animal to be societal (sociology), while the latter is about how the animal interprets and navigates its immediate social context (social psychology).

Thomas (pg. 30), however, tells us, "There is little direct evidence that infants represent groups," and that "there are reasons to think that infants' ecological niche has led infants to represent relationships and social networks, as opposed to groups."

I doubt this. It is unlikely that infants represent relationships and social networks but not groups. From a computational and statistical learning perspective, perceiving agents as members of groups reduces the need to process the complexity associated with differences between agents and is thus more energetically efficient. By contrast, the constant tracking and memorization of specific relationships and networks is certainly something babies have

intuitions about, I'm sure, but where it is efficient to do so, we should expect infants to use simple in-group/out-group clustering heuristics (which will later be elaborated in various ways up to and including a "societal identity"). This is likely true for the same reason the brain collapses variation and complexity in favor of other efficiency-maximizing categories where it makes sense to do so: before/after, soft/loud, fast/slow, high/low, close/far, bright/dark, large/small, curved/straight, rough/smooth, inside/outside, same/different, etc.

Besides, the evidence for core knowledge related to group categorization is overwhelming. Thomas herself reviews much of the research supporting the existence of early intuitions about group categories and acknowledges that, "During the first year of life, humans recognize cues of people acting as one, including when individuals imitate one another, act in rhythmic synchrony, respond to distress, eat together, or share bodily fluids" (p. 10). Thomas just isn't correctly interpreting this work as evidence of intuitive cognitions about group categories. The capacity for group categorization is not the same as the capacity to track and recall specific relationship characteristics.

If sociology is the study of society, intuitive sociology is about the intuitions and core knowledge structures that infants and young children have related to society. Additionally, an emerging consensus of observations from ethologists and wildlife biologists suggests that animal societies are groups with a shared identity, which defend territory, and exist intergenerationally. The combined insight is that intuitive *sociology* is about core knowledge related to group identity, territory, and intergenerationality/age-status distinctions. Intuitive *social psychology* is, then, about core knowledge related to specific network contacts, their characteristics, and strength.

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Representations are constructed: where are the questions about the dynamics of the first interactions and the security of the attachment bond?

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Abstract

This commentary proposes a dynamic approach to the construction of relationship representations and mentalization. We argue that these processes result from early caregiver–infant interactions, particularly through mechanisms of synchrony, parental sensitivity, and attachment. This extends the author's model beyond cognitive and innate perspectives by emphasizing the role of early interactions in constructing representations that are embodied, affective, and dynamic.

Ashley Thomas' proposed model is an intuitive theory of relationships, which argues that our representations of social relationships, which enable us to predict how people will interact (what we define as mentalizing), are intuitive, abstract, and built on fundamental knowledge. Whether it is necessary to imagine an innate intuitive disposition – as suggested by Thomas – is an undeniably important question. However, the author takes a static approach to these representations throughout the article. In this commentary, we adopt a dynamic approach based on the construction of interpersonal bond and its intensity. We develop the idea that the nature and intensity of a relationship's representation and the resulting ability to mentalize are, in fact, constructed and depend on the quality of early interactions with caregivers, who are influenced by processes of synchrony and their secure or insecure responses to infants. We argue that Ashley Thomas' proposed model does not take into account attachment or mentalization construction processes. While the author presents a mainly cognitive and uniform approach, we argue that the construction of representations of social relationships is also fundamentally embodied, affective, and dynamic.

Approaches based on the development of attachment processes have already largely advocated the construction of relational models from the earliest interactions, or even from the gestation period (McNamara, Townsend, & Herbert, 2019). Early formulations of attachment theory assume the gradual construction of a relational

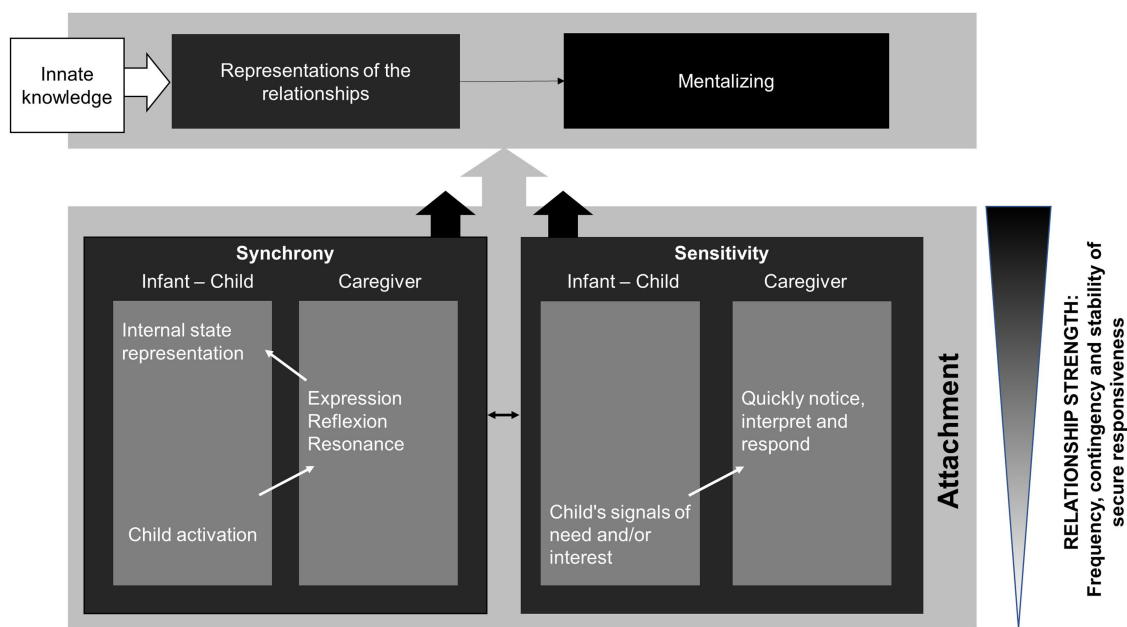


Figure 1 (Nandirino and Grynberg). Overview of the dynamic model based on synchrony and parental sensitivity proposed by Nandirino and Grynberg, which completes Ashley Thomas' model based on innate knowledge (white rectangle).

schema with a primary caregiver, allowing for the internalization of an internal model that will guide social or intimate interactions (Bowlby, 1969). From their earliest days, children have mechanisms at their disposal (e.g., joint attention and mutual gaze) with the function of reinforcing their caregiver's presence, specifically to maintain closeness and ensure security. It is through repeated interactions with an external co-regulator (caregiver) that a cognitive representation of the relationship is formed.

We argue that the construction of the relationship's representation is neither uniform nor static but intrinsically shaped by the nature of interactions and the skills of interaction partners at different stages of development. We use two core mechanisms to illustrate the dynamics of this construction. First, the *synchrony* between partners (caregivers and their babies) is a central precursor to the dynamic construction of the relationship, the relationship's representations, and the partner's goals (what we refer to as mentalization) (Davis, Bilms, & Suveg, 2017; Feldman, 2007). All maternal behaviours during the first hours after birth (such as looking at the infant's face, high-pitched vocalizations, positive expressions, and affectionate contact) suggest that humans are biologically prepared to engage in coordinated interactions. Similarly, the organization of the infant brain is already well prepared to function as an "intersubjectivity system" (Aitken & Trevarthen, 1997). These are precursors to *synchrony* (defined as the temporal concordance between simultaneous or sequential partner behaviours, such as eye gaze, facial expressions, and movement), which will gradually strengthen with the caregiver's responses, which are characterized by processes of contingency, congruence, and behaviour marking. Concurrently, as they gain experience, the parent and child adapt to each other's specific signals, and this bio-behavioural synchrony forms the basis of the parent-child bond and the establishment of *attachment* bonds (Feldman, 2015). In the same line, the theory of affect mirroring between child and caregiver states that the child needs the caregiver's emotional reflection to construct a representation of their internal state. This form of coregulation by the caregiver then

serves as the basis for the child's sensitivity to social cues and their capacity to mentalize the states of others (Fonagy, Gergely, & Target, 2007).

Another example supporting the dynamics of the construction of the representation of bonds concerns the role of certain parental abilities, such as *parental sensitivity* (defined as the ability to quickly notice, interpret, and respond to a child's signals of need and/or interest). Following Ainsworth's initial work, recent meta-analyses have confirmed that parental sensitivity is a key determinant in establishing a secure *attachment* between child and parent and developing the child's future socio-emotional capacities. This effect on the relationship varies according to distinct sensitive periods and the sensitivity of mothers and fathers (Madigan et al., 2024).

In addition to possible innate knowledge, the combination of synchrony, parental sensitivity, and their mutual interactions with attachment may contribute significantly to the representations of relationships in a dynamic, embodied, and affective manner (Figure 1). We also argue that these mechanisms are involved in the development of mentalization. Furthermore, and consistent with attachment theories, we suggest that the frequency of occurrence, contingency, and stability of secure responsiveness from the caregiver (a specific type of commitment) may be a more relevant way of characterizing the *strength of the bond*.

Our model offers a complementary perspective to the more innate and static intuitive psychology approach, according to which infants possess abstract representations of relationships and expect agents to primarily act efficiently to achieve their objectives. Although this expectation is undoubtedly central, it is crucial to emphasize how this mentalization competence is constructed, particularly within the context of proximal interactions (Fonagy et al., 2007). We, therefore, advocate the importance of integrating synchrony processes in interactions as a precursor to relationship building.

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Do infants recognize relationships indirectly?

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Abstract

The target article notes ways infants can use direct interactions between individuals to recognize relationships. We ask whether infants might also recognize relationships between individuals indirectly, by observing how they interact with other people. We suggest that indirect information might be used to distinguish between communal sharing and equality matching relationships and discuss how it might support inferences of relationship strength.

People’s interactions with one another are often informative about how they are related. We will draw a different conclusion about the relationship between two people if we see them passionately kiss, compared with if we see them dispassionately shake hands. In discussing how infants recognize relationships,

the target article focuses mostly on judgments that work this way – ones requiring that infants see agents interact with one another.

Sometimes, though, we may infer relationships between individuals indirectly, by observing their interactions and relations with other people. For instance, we may conclude two boys are brothers if they call the same woman “Mom” and this will work even if we never see the boys interact. Similarly, we might conclude two girls are likely to be friends if we know they have many friends in common, and again, this doesn’t require ever seeing the girls together. Recent experimental work shows that adults and preschool-aged children recognize relationships using this kind of indirect information (Sehl et al., 2025; also see Son et al., 2023). For instance, in one series of experiments, 5–7-year-olds and adults were informed about the friendships of two target individuals and then judged whether the targets were friends with one another (Sehl et al., 2023). Children and adults more often inferred friendship when a high rather than low proportion of the targets’ friendships were mutual.

The target article discusses two ways that infants might recognize *potential* future social connections (i.e., as opposed to recognizing existing relationships). First, if infants see their own parent imitate one individual rather than another, infants show signs of expecting the imitated individual to be more likely to engage with them (Thomas et al.). This suggests infants use the observed interactions of their parents as a cue to their own likely future relationships. Infants may also infer potential connections when making transitive inferences about dominance and subordination. If they see Alex dominate Bob, and Bob dominate Carl, they expect that Alex will dominate Carl (e.g., Gazes et al., 2015; Mascaro & Csibra, 2014). But, as the target article notes, infants might not recognize existing relationships between the individuals.

It is plausible, though, that infants might infer existing relationships between individuals by seeing how they interact with others. That is, infants might make inferences akin to recognizing that two boys are brothers from their separate interactions with their mother, and recognizing that two girls are likely to be friends, given their many mutual friendships. Moreover, the nature of the observed interactions could allow infants to distinguish different types of relationships. Our proposal here is speculative and might involve judgments well beyond the abilities of infants. If so, indirect inferences of relationships might develop later, perhaps in childhood.

Let’s first consider communal sharing relationships. The target article suggests infants recognize strong communal sharing relationships when they see interactions where individuals share saliva, for instance, by drinking from the same cup (Thomas, Woo, et al., 2022). Consider, though, what infants might conclude from seeing a woman and one boy drink from the same cup, and later seeing the same woman and another boy take bites from the same sandwich. It could be that infants would recognize only that the woman is in communal sharing relationships with each boy. Alternatively, infants might go further and conclude that the two boys are themselves in a communal sharing relationship, even though they did not share food and drink. Such indirect inferences might be especially likely for communal sharing, since this relationship type typically applies to bounded groups of people (Fiske, 1992). The conclusion that the two boys are related in this way might also be strengthened given additional evidence that the boys know one another – for instance, if the two episodes happened in the same place in close succession.

Now, let's consider equality matching relationships, which are marked by balanced reciprocal exchanges. In equality matching, if Alex gives resources to Bob, it's expected that Bob will later reciprocate and give resources to Alex. As reviewed in the target article, some evidence that infants recognize these relationships comes from findings that they expect agents to reciprocate in giving-and-taking. Here again, infants might infer relationships between individuals based on their interactions with others. Suppose Alex and Bob each separately reciprocate in giving-and-taking with Carl, Diana, and Earl, and these latter individuals also reciprocate with one another. Even without having observed Alex and Bob directly interact, infants might expect them to reciprocate. For instance, they might assume that all these individuals are part of an equality matching network.¹

These kinds of indirect inferences might also support conclusions about relationship strength. Numerical and statistical information could be relevant here in opposing ways. On one hand, a greater number of episodes observed could increase confidence both in the presence of relationships and in their strength. For instance, if two boys each frequently share food and drink with the same woman, this might suggest stronger communal sharing relationships with her, and possibly with each other. On the other hand, observing too many indirect episodes could push things the other way. If the boys frequently share food with each other but are not seeing or interacting with one another, this could suggest they are not strongly connected.

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Note

1 What about authority ranking relationships? As the target article discusses (and as noted above), transitive inferences about dominance could allow infants to recognize these relationships but might not. But indirect inferences might be more likely with evidence that the individuals are socially connected. If Alex dominates Bob, and Bob dominates Carl, infants might recognize an authority ranking relationship between Alex and Carl if given additional evidence that they are acquainted.

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Perceptual-cue-based mechanisms for recognizing social agents and their roles in social interactions

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Abstract

Thomas proposes the existence of core knowledge that supports an innate, or early developing, intuitive theory, where social relationships are understood in reference to three models: communal sharing, authority ranking, and equality matching. We consider possible precursor-representations that could support infants' discovery of relationships in the social world and how these representations relate to the basic models discussed by Thomas.

The first requirement for understanding relationships is to recognize that individuals are connected. We propose that this process relies on a set of perceptual-cue-based mechanisms underlying a distinctive representation of *social agents* (i.e., agents engaged in a social relationship) vs. agents in general and the assignment of social agents to roles in a relationship. Recent findings demonstrate that these mechanisms are already functional in the first months of life.

In one line of experiments (Goupil et al., 2022, 2024), we manipulated the interpersonal orientation of two individuals in a visual scene, so that it appeared that they could interact, either through *mutual accessibility* (face-to-face) or *transitive accessibility* (face-to-back: one faces another who faces away), or they could not interact (back-to-back: neither could access the other) (Fig. 1A). Taking differential looking times as an index of categorization of visual stimuli (Spriet et al., 2022), we found that 6-month-olds looked for a similar amount of time at face-to-face and face-to-back dyads and significantly longer at back-to-back dyads. These effects, ruled out for scenes involving inanimate objects, showed that, much earlier than previously reported (Beier & Spelke, 2012), infants are sensitive to visual cues (e.g., interpersonal orientation) that distinguish social vs. nonsocial agents or imply presence vs. absence of social relationship.

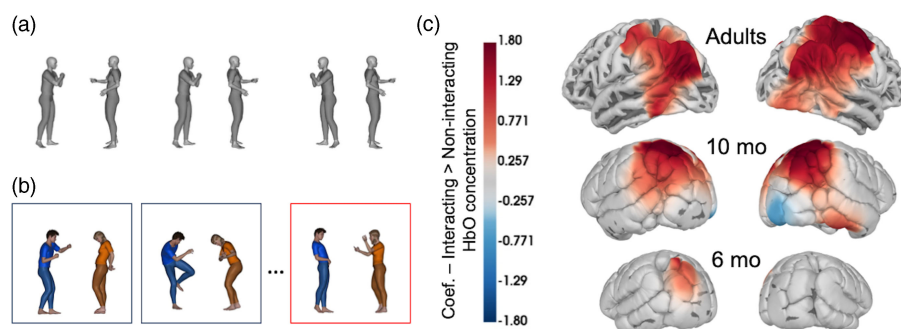


Figure 1 (Papeo et al.). A) Examples of face-to-face, face-to-back, and back-to-back dyads (stimuli adapted from Goupil et al., 2022). B) Examples of agent–patient dyads used in Papeo et al. (2024). Displayed is a sequence of scenes where the boy always looked like the agent and the girl always looked like the patient (gray squares), with oddball trials (red squares) in which the postures implied a role switch (the girl looked like the agent and the boy looked like the patient). C) Brain response to viewing third-party social interactions in adults ($n = 20$), 10-month-olds ($n = 21$), and 6-month-old infants ($n = 20$). Displayed are the group-averaged oxygenated hemoglobin (oxy-Hb) responses measured with event-related functional near-infrared spectroscopy during the presentation of animations of face-to-face (interacting) vs. back-to-back (non-interacting) body dyads. Red areas indicate significantly stronger response to face-to-face dyads; blue areas indicate significantly stronger response to back-to-back dyads. Face-to-face (vs. back-to-back) dyads recruited the bilateral middle occipital cortex, the middle/superior temporal cortex, and the temporoparietal junction, in both adults (16 out of 37 channels) and 10-month-old infants (8 out of 39 channels), and the middle occipital cortex and temporoparietal junction in 6-month-olds (1 out of 39 channels). Channels have been projected onto the cortical surface using MNI brain space for reference.

Further research showed that 7-month-olds use visual information not only to identify a social relationship but also to distinguish different roles in the relationship. In another study (Papeo et al., 2024), we manipulated body postures so that, in a sequence of different scenes featuring two people face-to-face, person A always *looked like the agent* (the one who acts) and person B always *looked like the patient* (the one to whom the action is directed). In a minority of “oddball” trials, roles, cued by the postures, were reversed: B appeared in agent-like postures, while A appeared in patient-like postures (Fig. 1B). Increased pupil dilation in oddball trials, measured by eye-tracking, suggested that infants were surprised by the role switch (see Papeo et al., 2024, for a replication with the habituation paradigm).

First, these results show that infants encoded the agent and the patient in each particular scene. In doing so, they assigned A and B to the *abstract* categories of *agent* and *patient*, which could generalize across the different situations depicted in different trials. Moreover, since the surprise response in role-switch trials disappeared when individuals were represented as unrelated (back-to-back), these results suggest that what infants were representing was not just each individual’s trait but one’s role *in relation* to another. In other words, although, based on postures, infants could represent an individual as agent-like or patient-like (or more or less dynamic), the effect specific to the relational (face-to-face) context indicates representation of the relationship rather than the individuals’ traits.

The results discussed so far are likely to characterize the mechanisms in the first of the three central components of Thomas’ intuitive theory of relationships, that is, the mechanisms to detect a social relationship (Thomas). However, infants’ representation of the agent–patient relationship also intrudes into the second component that specifies the relationship: communal sharing, authority ranking, or equality matching. *Where does the agent–patient relationship fit into the other three models proposed by Thomas?*

Roles such as agent and patient are regarded as universal components of core cognition (Carey, 2009; Rissman & Majid, 2019) and the agent–patient relationship, as a general structure to organize language (sentences) and thoughts (Strickland et al., 2012). Moreover, agent–patient exchanges (e.g., in physical or communicative interactions) can provide information about individuals’ dispositions and their social relationships (e.g., whether two are friends, rivals, etc.), which can allow predicting future interactions between the same individuals or with others (Stavans & Csibra, 2023). The agent–patient relationship is an asymmetric social relationship, similar to authority ranking. In effect, it cannot be excluded that in Papeo et al. (2024), infants’ representation of agent and patient overlapped with the representation of a dominant and a non-dominant individual. If so, this implies that certain postures (more dynamic and leaning forward) or certain orientations (those that allow perceptual access to another) can work as cues of formidability, analogous to physical size. Another possibility is that the agent–patient relationship provides a general framework, and the models of relationship discussed in Thomas fall out of this relational structure that emerges early in infancy. Hypothetically, different agent–patient dynamics might prompt the distinction between authority ranking on the one hand and communal sharing and equality matching on the other hand. In authority ranking, individuals tend to be assigned to fixed roles, either agent or patient. The other two models of relationships, where individuals regularly swap and alternate roles (e.g., in giving-taking exchanges), might be distinguished by the “level” of reciprocity (higher in equality matching) and additional intimacy cues (for communal sharing).

At the implementation level, in adults, the processing of visual events characterized by interpersonal cues (postures and orientation) has been associated with neural activity in visual areas along the lateral occipitotemporal cortex (LOT) to the posterior superior temporal sulcus (pSTS) (Abassi & Papeo, 2024; Gandolfo et al., 2024; McMahon et al., 2023; Walbrin & Koldewyn, 2019).

Recently, using functional near-infrared spectroscopy, we have observed a selective response to visual social interactions in LOTC/pSTS, in infants as young as 6 and 10 months (Mello et al., 2025; Fig. 1C). This is interesting as it suggests a model where social relationships – at least those that are visually available and display the canonical cues of social engagement – are initially extracted via bottom-up visual processing (see also Malik & Isik, 2023).

If the effects of postures and orientation are captured by LOTC/pSTS activity, and if this neural processing is available early in development, it is plausible that the literature reviewed here characterizes the initial components of Thomas' intuitive theory of relationships, where social relationships are detected – and initially understood in terms of agent–patient dynamics.

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Beyond the dyad: how do cognitive representations of social networks develop?

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Abstract

Thomas' paper provides a compelling account of the nature and development of mental representations of social relationships. Understandably, it focuses on dyadic and triadic relationships. We emphasize the need to elucidate the development of another key component of humans' understanding of social relationships: “stitching together” knowledge of social ties into representations of social networks that shape one's thoughts and actions.

Humans are keenly attuned to the structure of the social networks around them and frequently apply knowledge of such networks in day-to-day life. Adults use knowledge of the distance (i.e., “degrees of separation”) between people and various measures of centrality (i.e., individuals' importance) in social networks to predict how information will flow, then shape their behavior accordingly – e.g., when gossiping (Xia et al., 2025), assessing the veracity of gossip (Hess & Hagen, 2006), choosing “scapegoats” (Ellwardt et al., 2012), and deciding how to spread information (Banerjee et al., 2019). The selective sharing of information in social networks – and its strategic deployment to manipulate alliances – is perhaps further influenced by the relative strengths of the relationships around oneself (DeScioli & Kurzban, 2009; Krems et al., 2024). Monitoring the patterns of relationships that comprise social networks also allows people to recognize potential third-party threats to their relationships (Krems et al., 2021), strategically manipulate their social status (Faris, Felmlee, & McMillan, 2020; Krems et al., 2024), and decide whom to trust (Schwyck et al., 2024). People are so attuned to information about others' social network positions (e.g., “degrees of separation” from oneself, how well-connected they are) that they spontaneously retrieve such knowledge when encountering familiar others (Parkinson, Kleinbaum, & Wheatley, 2017; Zerubavel et al., 2015).

Despite a surge of evidence for and interest in the impact of social network knowledge on cognition and behavior, little is known about the development of this knowledge, or of humans'

ability to efficiently learn or strategically apply it. Of particular relevance to the target article, one significant open question pertains to which aspects of humans' understanding of social networks, if any, are rooted in "core knowledge" that emerges very early in development.

It is plausible that from early on in life, humans are prepared to perceive and reason about social networks. As Thomas points out, from birth, survival depends on such networks that extend well beyond one's primary caregivers (Hrdy, 2009); and as only some members of one's parents' social groups provide care, it would be beneficial to track specific relationships, not just group affiliations.

However, much remains to be discovered regarding how and when capacities to learn, represent, and use knowledge of social networks develop. For example, adults readily identify the people able to most effectively spread information in their networks due to the number and patterning of those people's ties (Banerjee et al., 2019). The measures of centrality that tend to predict ability to effectively spread information depend on not just how many friends or allies someone has, but also the people to whom that person is connected, and how well-connected those individuals are. How early on do the capacities to acquire, represent, and use such information emerge? Do inferences about such characteristics require granular social network knowledge, or could they be supported by simpler heuristics (e.g., tracking who one hears about most often or associated traits)? Related questions pertain to information-seeking – e.g., since central individuals are exposed to more people, they often know more about others in the network and may be more representative of its norms. When do we start to understand and make use of such phenomena?

Elucidating the developmental origins of the ability to track and infer others' centrality will also inform our understanding of how children reason about status and influence, as social network centrality facilitates influence and may be a particularly behaviorally relevant facet of status (Basyouni & Parkinson, 2022). Importantly, tracking and reasoning about measures of centrality may require different cognitive underpinnings than those that support analogous operations on other, more widely studied, facets of social status (e.g., reasoning about people's relative positions in linearly structured dominance hierarchies using transitive inference).

Research is also needed to elucidate the developmental trajectory of understanding and using information about distances in social networks. For example, when and how do our abilities to represent and apply knowledge of such distances develop, so that we can ascertain that people fewer degrees apart from one another in a network are more likely to know the same things, know things about one another, and be similar to one another?

Additionally, as Thomas notes, some results typically attributed to representations of social groups may instead reflect representations of social networks and their constituent relationships. Disentangling these possibilities will be informative in studies of both children and adults. This may be difficult, as some features of social networks and the schemas used to encode them can make representations of social networks and groups appear similar. For example, human social networks tend to contain many balanced triads and to have small-world structures (i.e., to consist of densely connected groups or "clusters" with few connections between them; Watts, 1999). Adults tend to exaggerate these properties in their mental representations (Brands, 2013; Kilduff et al., 2008). This tendency could arise from the need to compress representations of large, complex networks and/or the use of heuristics (Basyouni & Parkinson, 2022). It could also stem from a tendency

to represent coalitional groups rather than networks. Similarly, from infancy to adulthood, humans tend to infer affiliative ties between individuals who share certain properties (e.g., preferences, traits) even without ever witnessing them interact (Lieberman, Kinzler, & Woodward, 2021; Schwyck et al., 2024). In some contexts (e.g., paradigms where groups are defined by individuals sharing particular properties), this tendency could render it difficult to discern whether participants' behavior stems from representations of groups or from inferences regarding which individuals would be more likely to be connected by affiliative ties. Thus, more work is needed to elucidate interactions between mental representations of social networks and groups, and advancing understanding of social network knowledge and its development could reshape the way that certain existing findings are understood.

Understanding how and when humans begin to represent and reason about social networks will shed light on the development of skills that are central to everyday life. From deciding whom to trust to predicting how information will spread, network knowledge fundamentally shapes how people make sense of and act within their social worlds.

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Relational models are too elaborate for basic social relationship reasoning

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Abstract

Thomas proposes that infants use relational models to reason about interactions between relationship partners. However, many basic social interactions do not contain enough information to differentiate between models. We propose that both basic interaction reasoning and the formation of diverse, culturally relevant relationship categories are better accounted for by positing relationship features that can be flexibly combined.

We endorse Thomas's call to understand the development of systematic reasoning about relationships. We are not yet convinced, however, that relational models should be central to an account of infants' relationship reasoning. Thomas presents these categories as the basis for infants' ability to interpret and predict the social behaviors of interacting agents. However, many behaviors that are central to social relationships – e.g. giving, helping, cooperating, communicating, and imitating – occur regularly across many relationship types. This many-to-many mapping between behaviors and relationships suggests that immediately categorizing a relationship from observed behavior may be more difficult than Thomas proposes. This may be especially true for infants who lack substantial contextual and cultural knowledge, and for relationship categories defined by patterns of reciprocity that unfold over time.

Consider giving. Thomas suggests that infants use acts of giving to recognize equality matching relationships, thus explaining their expectation that one act of giving will be followed by another, reciprocal act (Tatone & Csibra, 2020). But giving is not exclusive to equality matching relationships. Close family and friends give one another gifts and resources; leaders and followers may give provisions or tributes, respectively. The fact that one act of giving ought to be insufficient evidence of an equality matching relationship casts doubt on whether infants' expectations of reciprocity stem from this specific relational model, rather than a

broadier expectation that social regard is often mutual. A broader reciprocal relationship representation would also help account for infants' expectation that the giver will keep giving to, rather than taking from, the recipient, even when exchange is unbalanced (Tatone, Geraci, & Csibra, 2015). Outside the lab, it would also be forgiving of fragmented observations and ignorance of the value of different resources (e.g., money).

Imitation is another behavior Thomas maps onto a specific relational model, proposing that it is an expression of “oneness” characteristic of communal sharing relationships. But imitation is ubiquitous in human social life: imitative alignment occurs in interactions between parents and infants, teachers and students, customers and salespeople, and research participants and confederates who have only just met (see Chartrand & Lakin, 2013, for review). Should observers conclude that all these relationships are to some degree communal sharing relationships? An alternative is that imitative alignment is seen as evidence of a valuing of others' goals or perspectives, which can be a feature of many relationship types (Powell & Winkielman, 2024). Of note, infants do not always expect group members to imitate one another (for evidence that strong imitative expectations depend upon the perception of behaviors as conventional or ritualistic, see Liberman et al., 2016; Hennessy, Spelke, & Powell, 2025).

A general alternative to Thomas's proposal is that people, including infants, can reason about the features and statistics of relationships without necessarily categorizing them first. This would be in line with other intuitive theories: observers can reason about the solidity of an object or goal of an agent without deciding what kind of object or agent they have encountered (Spelke, 2022). Similarly, people may infer features of a relationship – defined as the relation between any set of agents – and reason about how those features will impact the relevant parties' actions or emotions, without necessarily assigning the relationship to a particular relational model or other category.

As Thomas discusses in Box 2, Powell (2022) proposed that infants represent the relationship feature of care or “affiliation” as the weighted value that one individual places on another's rewards or welfare. This representation can be integrated with an intuitive psychology that explains agents' behavior and emotions in terms of what they believe and value (Jara-Ettinger et al., 2016; Doan, Ong, & Wu, 2024). This integration allows observers 1) to use others' behavior to infer strength of interpersonal care in the form of the magnitude of this weight, and 2) to use information about strength of care to predict interpersonal actions and emotions, in a graded fashion, within and across relationship categories (Jiang et al., 2025; Quillien, 2023; Smith-Flores & Powell, 2023; Tompkins et al., 2024). As Thomas notes, this need not entail relationship partners supporting all one another's goals; they may disagree on what is best for one individual's overall welfare, as in the case of a parent who hinders a child's dangerous or unhealthy goal. Children understand and approve of such protective hindering (Tompkins & Powell, 2025), and infants may in some cases as well. Finally, like communal sharing models, interpersonal care is inconsistent with exact exchange; this pattern suggests behavior motivated by expected reciprocal rewards, rather than valuing of the other's welfare.

While strength of care is an important feature for understanding many social relationships, we agree with Thomas that it does not define relationships alone. Another key feature that seems to be represented from infancy is relative difference in agentic capacity or power between relationship participants (Mascaro & Csibra, 2012). These and other features can be combined to flexibly form relationship categories, which are indeed a component of

mature social reasoning (Earp et al., 2021). Some categories could be innate, but many categories would be learned from the statistical co-occurrence of features, as well as events and contexts.

Relatedly, the role of experience with specific social interactions in early social reasoning is underexplored. For example, might infants' putative expectations about saliva sharing and comforting (Thomas et al.) be based on the association of multiple actions or events with the same relationship partner? Caregivers comfort and feed infants; they are also the most likely people to reach into infants' mouths, to feed them, clean them, or remove foreign objects. To fully consider associative explanations, studies should test for comforting expectations when an adult feeds or probes an infant or puppet without first putting the food or finger in their own mouth.

In sum, we propose that humans' intuitive theory of relationships is based on relationship features, supplemented by categories that combine these features, rather than core relational models. This multifaceted system can best explain how people flexibly reason about social information, spanning from isolated interaction observations to the histories of lifelong relationships.

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Early developmental origins of social knowledge

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Abstract

In addition to core knowledge, the prenatal environment, parental care, and early life experience play substantial roles in the establishment of social cognition. Given the evolutionary importance of early life care in humans and mammals in general, there are innate reward and stress based neuroendocrine mechanisms that are critical to maternal infant bonding as well as later social behavior.

Thomas begins with a clear truth that we depend on social relationships “from the moment we are born.” While we do not disagree with this stance, we levy two caveats. First, it is important to consider that social relationship formation is predicated upon influences that in fact precede birth, as the prenatal environment primes physiological mechanisms with direct implications for how we develop and react to social inputs. Second, while the theoretical framework explicated within the article is innovative, it is grounded in literature primarily conducted among children 6 months and older. Therefore, the developmental origins of social relationships, as outlined in Thomas, lack the critical nuances of the neonatal affective environment, including the development of the stress response, maternal mental health, and reciprocal parent–child affective processes.

In terms of prenatal exposures, maternal stress during gestation has been consistently linked with increased risk of self-regulatory and emotional deficits before the age of two (Korja, Nolvi, Grant, & McMahon, 2017), with particular implications for a child’s perceived strength of a given relationship, as outlined in Thomas’s framework. Much of this may be driven by neuroendocrine perturbations. The stress hormone cortisol is just one example, as a mother’s prenatal feelings of stress and anxiety specific to the pregnancy predict her infant’s cortisol reactivity throughout the first year of life (Tollenaar et al., 2011). Moreover, a mother’s own cortisol patterns have been similarly shown to prime stress reactivity in infancy (Irwin et al., 2021). These physiological patterns may also be borne out in infant behavior, as one study found that infants exposed to higher levels of maternal cortisol cried and fussed more and displayed more negative facial expressions during the first five months of life (de Weerth, van Hees, & Buitelaar, 2003). Alterations in hypothalamic-pituitary-adrenal axis responsivity and resultant cortisol profiles may have lifelong implications for social cognition and downstream relationship stability (Cheng & Kornienko, 2020; Ji, Flouri, & Papachristou, 2021). This of course would affect social relationship formation and learning.

Another factor related to prenatal exposures is postpartum depression, which we here use as an example of the importance of maternal care and related learning in cognitive representations of social relationships. Looking upstream, much prior work links cortisol profiles to maternal depressive states (Seth, Lewis, & Galbally, 2016). Similarly, our previous research has documented various prenatal stressors and other physiological factors – including DNA methylation of immunoregulatory genes, telomere length, and oxytocin levels in mothers – as related to postnatal mental health (Incollingo Rodriguez et al., 2022; Rodriguez et al., 2022; Sluiter et al., 2020). Moreover, oxytocin is not only implicated in buffering against the consequences of maternal psychosocial stress, but it also serves as a key modulator of early parent–infant contact and interaction and, in turn, attachment development (Scatliffe, Casavant, Vittner, & Cong, 2019). Not surprisingly, then, a strong body of evidence points to maternal depressive symptoms as a risk factor for impaired maternal–infant bonding quality (Tichelman et al., 2019). Because maternal postnatal depression can develop shortly after birth, it is critical to examine very early life pathways in bonding and relationship formation that may create enduring models for how humans go on to perceive the social world and develop relationships.

The target article’s Intuitive Theory of Relationships also overlooks **critical affective processes** that are foundational to understanding parent–child dynamics, particularly during infancy

and toddlerhood. While the theory incorporates behavioral and cognitive mechanisms, it neglects the roles of emotional reciprocity, maternal responsiveness, and dyadic regulation – factors that underpin socio-emotional development. This gap limits its explanatory power and practical utility in addressing early parent–child relational challenges.

The author can address this gap in various ways, but here are possible avenues to strengthen the theory with social and affective phenomena:

1. **Emotional Reciprocity and Validation:** Affective processes such as mutual emotional attunement, validation, and positive affect exchange are central to secure parent–child relationships. For instance, Gottman and colleagues’ well-known work on marital interactions highlights how de-escalation of negative affect and ratios of positive versus negative affect predict relationship stability. Similarly, parent–child interactions thrive when caregivers respond sensitively to infants’ cues (e.g., soothing distress, mirroring joy), fostering trust and self-regulation (Brown et al., 2025). The absence of these dynamics in the author’s model oversimplifies the emotional scaffolding required for healthy development.
2. **Dynamic Emotional Regulation:** The theory does not account for the bidirectional nature of emotion regulation. Infants and toddlers rely on caregivers to co-regulate their emotions through responsive behaviors (e.g., calming touch, vocal modulation). Maternal overcontrol or intrusiveness, conversely, disrupts this process, leading to poorer reactive control and effortful control outcomes (Brown et al., 2025). Affective processes like **empathy** and **affection** – linked to children’s later emotion regulation and social competence (Ornaghi, Conte, & Grazzani, 2020) – are critical yet missing from the author’s framework.
3. **Maternal Emotion Regulation Skills:** To enhance emotional reciprocity and validation and dynamic emotion regulation, interventions could be designed to strengthen parental (especially maternal) emotion regulation skills. Some research indicates the promise of this approach, especially those that address psychosocial stressors. For example, Ribaudo and colleagues developed a home visit program that reduced PTSD symptoms in mothers and improved co-regulation capacities (Ribaudo et al., 2022).

Affective processes are central to parent–child relationships. By foregrounding emotional reciprocity and dyadic regulation, Thomas’ framework could better predict and promote socio-emotional resilience from infancy onward.

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How to get from here to a relational social psychology

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Abstract

Integrating Relational Models Theory with Carey's core cognition theorizing is possible, promises to be productive, but requires clarifications. We sketch how such an integration can strengthen social psychological theorizing in three areas.

Thomas argues that knowledge of social relations should be considered *core knowledge* – innate or early-developing – following Carey (2009, 2011). This involves inferring enduring, agent-specific relations from behavior, separate from agent attributes. Carey's view that core knowledge persists into adulthood aligns with Fiske (1991) Relational Model Theory (cf. Thomsen et al., 2011).

Surprisingly, the foundational role of relational cognition is not mirrored in the research agenda of social psychology since the 1980s. Instead, social psychology, and especially social cognition, has favored a methodological individualism, extrapolating from conceptual models from cognitive psychology of object cognition (A. P. Fiske & Haslam, 1996).

We hope Thomas' review encourages social psychologists to treat the formation, representation, and navigation of relationships as central to social cognition. As Thomas argues, integrating Relational Models Theory with Carey's core knowledge framework is not only conceptually compatible but may also be essential for understanding how relational thinking develops and operates across the lifespan. Extensive evidence suggests that relational model core knowledge persists into adulthood, shaping social cognition, behavior, and emotion (A. P. Fiske, Schubert, & Seibt, 2025). This allows predictions for adult social cognition. Let us name two examples:

First, core knowledge structures can serve as foundations for further conceptual learning. For instance, the understanding of natural numbers emerges from the parallel individuation system via Quinian bootstrapping, aided by language (Carey, 2009). Complex social cognition may similarly arise through yet-undiscovered bootstrapping processes. Fiske's (2012) meta-relational models could be one such case, as could the application of relational models to intergroup relations. Second, adults may also rely on bootstrapped and potentially abstract social concepts that collapse under cognitive

load, revealing the enduring influence of core knowledge, much like how numerical reasoning reverts to the analog magnitude system, another core cognition module, under load.

For such inquiries to succeed, previous proposals for the involvement of core cognition in relational models cognition should be reviewed in light of the new findings. For instance, authority ranking has previously been argued to emerge from the analog magnitude system, because both exhibit Weber's law (Chiao, 2010; Schubert, 2020). The new proposal in the target article is that authority ranking is instantiated in its own system, not the analog magnitude system. As the number of identified core cognition faculties increases, it becomes more important to mutually differentiate them.

We now highlight three areas where Thomas' model can help strengthen social psychological theorizing.

Relative vs. absolute judgments

The Stereotype Content Model and similar models (Abele & Wojciszke, 2014; S. T. Fiske et al., 2002; Koch et al., 2016) exemplify an individualized approach, focusing on perceived goal compatibility. Even when applied to social interaction, these models don't capture relations in the sense of the target article. In contrast, Thomas rightly distinguishes between relational and character judgments: a good ally should be warm toward us, but cold toward our adversaries. An indiscriminately warm person may neither be a good ally nor be very competent.

Tracking states and transgressions

Social life is inherently relational. Most interactions occur within a shared history and defined roles that shape our expectations. Thomas emphasizes that even infants track relational histories; adults do too, across various relational models. In equality matching, we ask whose turn it is. In authority ranking, we scrutinize whether power is abused and whether respect is paid. In communal sharing, trust is key: has someone withheld help they could have given?

People have a long memory for transgressions. Resolving interpersonal or political conflicts requires understanding how each party frames offenses within a relational model, shaping views on revenge, repair, and justice (Kelman, 2008). Divergent interpretations of facts can be one source of disagreement. However, the bigger issues are often related to questions of responsibility, blame, entitlement, and duties, which are all inherently relational ideas.

Understanding the relational model

People often classify relationships by kinship or closeness (e.g., spouse, sibling), but Thomas' model highlights a different layer: the underlying relational model. This has practical implications for analyzing and intervening. For instance, while Western, Educated, Industrialized, Rich, and Democratic (WEIRD) societies idealize communal sharing in marriage, others may prioritize authority or exchange, reserving communal sharing for siblings or friends. Such mismatches can lead to cultural and interpersonal conflict.

Even within one culture, relational model mismatches are common and a source of conflicts. When expecting communal sharing, we can be outraged by someone keeping score or expecting

us to obey. While communal sharing may seem like the superior model for spouses, siblings, close friends, and others, the model does come with a cost: the lack of scorekeeping can lead to a strong sense of betrayal unless both partners are trustful and trustworthy. In moments of low trust, reverting to equality matching may help stabilize the relationship.

Thus, measuring relational model, type, and strength separately could clarify conflicts resulting from discrepancies between desired and perceived relational model and between own and partner relational model. If Thomas is right, relationships are mentally represented as entities in their own right. People often act for "the relationship's sake," e.g., repaying a favor not for the partner, but to maintain balance, even when neither party is concerned with immediate outcomes.

While exchange vs. communal relations (Clark & Reis, 1988), perceived partner responsiveness (Reis & Shaver, 1988), and other measures assess aspects of the distinctions Thomas discusses, her model can serve as a basis for a more comprehensive operationalization of relational models and relationship strength.

Conclusion

Thomas' review is a fruitful starting point for theorizing on the ontogenetic development of relational cognition and the interaction of relational knowledge with other core knowledge. It can also inspire inquiry into the relational nature of person perception, the relevance of relational history for current responses, and the consequences of mismatched relational models between partners.

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The evolution of core social knowledge

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Abstract

Core social knowledge is not limited to humans; it has evolved in other species. A comparative, evolutionary perspective prompts one to ask: what, exactly, has been favored by natural selection? A rigid, intuitive theory that matches each species' society, or a more general tendency to search for social patterns, leaving the details and theory to emerge during development?

The claim that humans possess a “core” system of social knowledge raises questions about its evolution. Either human core knowledge arose *de novo* – neither Thomas nor Fiske (1992) supports this view – or it should be found in other species and linked to fitness. Data from nonhuman primates support this view. The data are not essential to Thomas' argument, but they strengthen it in two respects: they suggest *why* knowledge of social relationships is “core,” and they raise questions about the evolution of social knowledge in different societies.

Nonhuman primates behave differently with different individuals, often forming close social bonds for months or even years (Seyfarth & Cheney, 2012). Primates not only have differentiated relationships, but also they recognize them in others. In species with linear dominance ranks, individuals recognize both their own rank and the ranks of others (Cheney, Seyfarth, & Silk, 1995a). Recognizing one's own rank could arise simply because individuals distinguish, from personal experience, “dominant to me” from “subordinate to me.” Recognizing other animals' ranks, however, is cognitively more interesting because it requires an individual to

organize her knowledge from the observation of interactions in which she is not involved.

Primates have expectations about how others will behave and, as with humans, these expectations depend on properties of relationships. When female vervet monkeys and baboons hear the distress cry of an immature, they look toward the immature's mother (Cheney & Seyfarth, 1980, 1999). When two capuchin monkeys exchange aggression and solicit support from an ally, they preferentially solicit allies who rank higher than their opponent and have a social relationship with the solicitor that is closer than their relationship with the opponent (Perry, Barrett, & Manson, 2004, 2008). After aggression, many primates “reconcile” with a victim by behaving in a friendly way toward her (Cords, 1992). Baboons reconcile by grunting (Silk, Cheney, & Seyfarth, 1996). Grunts increase the likelihood that the victim will interact in a friendly way toward her former aggressor (Cheney, Seyfarth, & Silk, 1995b; Cheney & Seyfarth, 1997). Moreover, grunts from a close relative of the aggressor can also function as “reconciliation-by-proxy.” victims treat the grunt as reconciliation if it comes from a close relative of the aggressor, but not if it comes from an individual belonging to a different matriline (Wittig et al., 2007a). The perception of other animals' alliances follows a similar pattern (Wittig et al., 2007b, 2014).

In baboon society, males leave the group where they were born, whereas females remain in this group throughout their lives, forming close bonds with their mothers and matrilineal sisters. Adult females can be ranked in a linear dominance hierarchy that determines priority of access to resources. From birth, daughters acquire ranks adjacent to those of their mothers. Rank relations among females are generally stable over time, with few reversals occurring either within or between families. Ranks among males change often.

Baboons' knowledge of other animals' relationships is embedded in their knowledge of society. In addition to recognizing other animals' ranks and kinship, individuals respond more strongly to evidence of a between-family rank reversal than to evidence of a within-family reversal. They act as if they classify females simultaneously according to both kinship and rank (Bergman et al., 2003; Schino, Tiddi, & Polizzi di Sorrentino, 2006). Similarly, although the relative ranks of males change often, individuals act as if they “update” the rank order when listening to competitive displays (Kitchen et al., 2005), recruiting allies (Silk, 1999), or monitoring sexual partners (Crockford et al., 2007).

These results are difficult to explain without assuming that baboons perceive others not just as a collection of individuals, or even a collection of dyads, but as an assembly of relationships organized according to an intuitive theory – an interconnected system of knowledge (Thomas, 2025). This knowledge is structured using implicit, abstract categories. The categories are implicit because they are not fully conscious, well-developed theories that the baboons can describe; instead, through processes we do not yet understand, baboons observe other animals and generate expectations about how each will behave in different contexts (Cheney & Seyfarth, 2007:118).

Social categories are abstract because they cannot be reduced to any one or even a few perceptual attributes. High-ranking baboon females are not older or larger than low-ranking females, nor do they live in larger kin groups. Family members do not always look alike, sound alike, or share distinctive physical features. Yet none of this variation affects category membership (Seyfarth & Cheney, 2015). Moreover, the classification of individuals into categories persists even when the individuals within them change. From the

moment an infant baboon is born, she is treated as a member of a high- or low-ranking matriline; from the moment a new male arrives in the group, he acquires a place in the male dominance hierarchy, with all the transitive implications that this implies. For baboons, social categories organize knowledge regardless of the individuals involved.

What can this zoological perspective contribute to Thomas' proposal?

First, it can tell us *why* knowledge of social relationships is “core:” because, for group-living animals, social relationships are a ubiquitous, inescapable part of life; because the members of a social group interact in predictable ways that allow an observer to deduce structured patterns; and because knowledge of these patterns affects fitness (Cheney & Seyfarth, 2007, Silk, 2007, Seyfarth & Cheney, 2012).

Second, it raises the broader question: how tight is the link between social knowledge and social structure? Baboons, chimpanzees, and gorillas have very different societies and would seem to require different intuitive theories. If we accept Thomas' view that core social knowledge is “innate” and “evolutionarily old,” how exactly has natural selection acted in different species? With a rigid, species-specific intuitive theory, such that a gorilla living among chimpanzees would struggle to predict their behavior? Or with a more general tendency to form *some kind of* social theory – like a universal grammar – with the details to be worked out during development? And which of these generalizations applies to humans in different cultures?

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Two modes of equality matching: positive and negative direct reciprocity

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Abstract

Equality matching describes a mode in which reciprocal relationships are maintained. While Thomas argues for core representations that can support equality matching, we argue that two forms of reciprocity must be considered and may require distinct representations: positive and negative direct reciprocity. We review both evolutionary and developmental evidence in support of this proposal.

Thomas argues that an intuitive theory of relationships is built on core knowledge that enables infants to categorize relationships into three types based on Fiske's models of social relations: communal sharing, authority ranking, and equality matching. We agree that infants are able to distinguish between relationships based on these models. However, we argue that equality matching should include both positive and negative forms of direct reciprocity, that is, responding in kind to actors who benefit and harm oneself. We review evolutionary and developmental evidence that suggests that positive and negative direct reciprocity may rely on distinct cognitive representations.

Evolutionary theorists have argued that both positive and negative reciprocity are important in managing social relationships, but that they serve different ultimate functions. Positive direct reciprocity supports mutually beneficial, long-term relationships with non-kin and may thus be a mechanism for the evolution of cooperation (Axelrod & Hamilton, 1981; Henrich & Muthukrishna, 2021; Nowak, 2006; Trivers, 1971). At a social level, positive reciprocity may be a human universal that helps human social groups to function (Fiske, 1992; Gouldner, 1960). Early theories of negative direct reciprocity proposed that it too could support the evolution or maintenance of cooperation (Boyd & Richerson, 1992; Clutton-Brock & Parker, 1995). However, more recent theories and empirical work suggest that it serves a more self-protective function that supports "revenge" (McCullough, Kurzban & Tabak, 2013) or competition to maintain or increase one's status (Raihani & Bshary, 2019). However, long bouts of retaliation are not beneficial to the individual or the group, and, therefore, humans and other species often opt to change partners after a tit-for-tat exchange or find a means to repair the relationship (McAuliffe, Marshall & McLaughlin, 2025). Although theoretical and empirical research in this area is ongoing, positive and negative direct reciprocity likely have different ultimate functions and possibly different phylogenetic histories. If this is true, then it is also likely that different representations subserve these two forms of reciprocity.

The developmental research reviewed by Thomas suggests that infants can track individuals and quantities, two important components of representing equality-matching relationships. Toddlers also distinguish between adults who intend to be helpers and those who do not intend to help, preferring to interact with the would-be helpers (Dunfield & Kuhlmeier, 2010). By the preschool years, children can engage more robustly in both positive and negative reciprocal interactions, and differences between the two forms of reciprocity emerge. Notably, when children's options are limited to either a choice between two partners or a response to a single partner, even 3-year-olds show evidence of both positive and negative reciprocity (House et al., 2013; Vaish, Carpenter, & Tomasello, 2010; Warneken & Tomasello, 2013; Wu & Gao, 2018). However, when children can respond in kind to multiple agents, a clear asymmetry appears: children between 4 and 7 years of age will selectively punish someone who took from them but will not give back to a giver (Chernyak et al., 2019). Instead, they give randomly to all the agents present, showing generalized positive reciprocity. Importantly, this is not due to selective memory – children recall perfectly well who took and who gave to them. Similar results have been found more recently but only among boys (Benozio, House, & Tomasello, 2024). In a follow-up experiment, children could quickly learn (or be reminded) to give back to the giver, but this suggests a stronger role for learning of positive than negative reciprocity (Chernyak et al., 2019). Combined, this evidence suggests that there are different developmental pathways for

positive and negative direct reciprocity. While basic components such as tracking individuals and benefits and rewards may be similar or the same, the cues that trigger a response, the emotions that motivate action, and the enculturation of contingent responses differ. More research needs to be done to determine the early ontogeny of positive and negative direct reciprocity, but we argue that distinct representations underlie these forms of reciprocity by the time children actively engage in ongoing relationships with peers.

In addition to our brief review of the evidence for distinct forms of equality matching, we suggest two empirical tests that can help to determine whether one or two representations are required. First, infant research can examine whether expectations for positive and negative reciprocal interactions are correlated, suggesting a single core representation of equality matching, or unrelated, suggesting separate core representations that underlie different forms of reciprocity. For example, studies could test whether infants who expect others to reciprocate help also expect them to retaliate against harm or whether these expectations emerge separately. Such findings could offer insights into whether infants' representations of equality matching are organized under a single representational system. Second, longitudinal research could map individual differences in infants' expectations of positive and negative direct reciprocity onto later first-person behaviors. Should infant expectations of positive direct reciprocity predict positive reciprocal behaviors at 4 years of age, but not negative reciprocal behaviors, or the opposite for negative direct reciprocity, this would signal distinct representations at work.

In summary, we argue that positive and negative direct reciprocity need to be considered as potentially distinct representations for equality matching. We are excited to engage in this discussion and by the empirical research this updated social relations model can inspire.

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Social groups include social relationships

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Abstract

In this commentary, we express enthusiasm for Thomas's article and research efforts, but question her proposed division between social groups and social relationships. Joining other social scientists, we view relationships as fundamental to – and sometimes synonymous with – social groups. Further, empirical studies focused on infants and young children highlight the tight link between social groups and relationships.

In her thought-provoking commentary, Thomas draws a line between social relationships and social groups. We appreciate Thomas's efforts to characterize how infants initially represent the social world, but we question whether separating relationships and groups is necessary or correct. Indeed, in looking broadly across literatures in the social sciences to define what a social group entails, sociologists and social psychologists often center – or at least reference – social relationships. For example, Forsyth (2019) defines a group as “two or more individuals who are connected to one another by and within social relationships.” Sociologists have also traditionally classified groups into types that vary according to features such as degree of intimacy and longevity (e.g., “primary groups” such as families vs. “secondary groups” such as work colleagues; Cooley, 1909). And, social scientists from many sub-disciplines recognize that groups often have hierarchical structures (e.g., Heck, Shutts, & Kinzler, 2022). Thus, it is not clear that there should, in principle, be a hard line between relationships and groups; a more parsimonious account may view social relationships as embedded within social groups.

More proximate to the psychology of infants and young children, empirical studies suggest that young children's developing representations of social groups include social relationships as a central component. Rhodes (2013) argues that categorizing humans into groups invites inferences related to 1) natural kinds and 2) social obligations. Another way to frame this distinction may be that a “social group” licenses two different downstream consequences: inferences about common attributes and inferences about social relationships (Lieberman, Kinzler, & Woodward, 2017a). Indeed, existing evidence with infants suggests that (at least rudimentary) aspects of both kinds of social-group inferences are observable. Infants' reactions to speakers of the same versus different languages provide a useful example. These studies suggest that infants both infer positive social relationships between same-language speakers (Lieberman et al., 2017b) and make inductive inferences about such individuals' common attributes (e.g., that two speakers of the same language will share food preferences; Lieberman et al., 2016). These two classes of inferences (i.e., “relationships” and “attributes”) may also inform one another: for instance, infants predict that people who share similar food preferences are also more likely to affiliate (Lieberman et al., 2014; Lieberman et al., 2021), and that people who affiliate are more likely to share food preferences (Lieberman et al., 2016).

Further, some research suggests that early inferences of shared relationships may be even more easily licensed by social group information than are those of common attributes. For example, in one study (Shutts, Roben, & Spelke, 2013), four-year-old children were presented triads of unfamiliar children. In each triad, two people matched in either gender or race, and one person differed in either gender or race (e.g., on one trial, children saw two Black girls and one Black boy). When asked to guess which two people were friends, children selected the people who matched in gender or race (e.g., the two Black girls; see also Roberts et al., 2017). When asked which two people shared the same attribute (e.g., liking to play a particular game), children were equally likely to select children from the same vs. different social groups. Thus, when reasoning about social groups, young children can privilege social relationships over shared properties.

The target article lays out a possibility that human social group notions are built from representations of social relationships per se. Alternatively, in studying infants' reasoning about the nature of

social connections, Thomas and her colleagues may be studying early social group cognition. In this case, Thomas's work could be seen as indicating that some of the social group phenomena revealed most often in studies of children and adults are also present in infancy – and that realization could guide further studies building on the creative methods Thomas has developed, but linking signature phenomena identified in children and adults.

Ultimately, of course, older children and adults come to know many more things about social groups than infants and young children do, including which properties (stereotypes) are associated with particular kinds of people (Sierksma, Brey, & Shutts, 2022) – and the nature and complexity of dyadic relationships within a group certainly grows. But the heart of social groups across development appears to include social relationships as a critical downstream consequence. Indeed, adults themselves have closer and more numerous social relationships with those who match them in gender, race, and other social group dimensions (McPherson, Smith-Lovin, & Cook, 2001).

Whether Thomas's work should be conceptualized as focusing on social relationships or social groups, there is one thing that is clear at present: we are likely to learn a lot about the origins of humans' extraordinary social capacities through the clever studies and methodology of Thomas and her colleagues in the coming years. We applaud these efforts and look forward to additional work testing ideas Thomas has advanced.

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Toward a unified theory of social relationships and social networks

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Abstract

Navigating the social world requires representing how people relate to one another and how people are connected in a network. Though both are forms of relational knowledge, it is difficult to unify them in a single theoretical framework. I hypothesize that the statistics of observation may be sufficient to learn both network structure and an intuitive theory of relationships.

As the poet John Donne noted, “No [human] is an island, entire of itself; every [human] is a piece of the continent, a part of the main.” The great drama of human social life is staged not only in interactions between dyads but in the larger context of our social networks. A piece of gossip shared with the wrong person can quickly spread to far-flung corners of the community; the shockwaves generated by a romantic couple's breakup might result in a cleaving of the network as people decide to take sides; an enterprising employee can try to climb the corporate ladder by strategically cultivating relationships with well-connected middle managers.

These examples illustrate that to adaptively navigate the social world, we must learn how people are connected to one another. Moreover, we must do so despite having imperfect memory and incomplete observations of others' social relationships. How can this be accomplished? It is tempting to assume that the cognitive mechanisms underlying social network representation are essentially the same mechanisms underlying the representation of social relationships. After all, social networks are, by definition, composed of social relationships.

In the target article, Thomas proposes that having an intuitive theory about social relationships can help children understand other agents' behaviors, much as intuitive theories can help us reason about the physics of an object or the psychology of an individual mind. I agree that the richness of an intuitive theory is useful for understanding how very young children and infants represent social relationships. However, it is unclear whether relational models (i.e., structured representations of different relationship “types”) provide the right cognitive foundation for representing complex social structures, like networks.

Due to limitations in both observation and memory of social relationships, learning a mental representation of a social network requires extrapolation from direct experience. Recent work has

shown that adults are able to use the statistical structure of observation to make principled inferences about the existence of unknown social relationships, including relationships they have never observed (Son, Bhandari, & FeldmanHall, 2021, 2023). The ability to make these inferences appears to enable adaptive social navigation (Son et al., 2024), including the ability to create a desirable social “niche” in real-world networks (Aslarus et al., 2025; Teoh et al., 2025). These capacities mirror at least two components of the intuitive theory outlined in Thomas’ proposal: inferring a relationship’s existence and changing relationships through one’s actions.

A simple learning mechanism, which does not necessitate any structured prior knowledge, is sufficient for explaining how adults jointly learn a probabilistic representation of both dyadic relationships and the larger-scale structure of a social network. Computationally, this mechanism works by chaining together direct observations of dyadic relationships. For example, imagine observing Abbie and Brooke eating lunch together. A week later, you observe Brooke and Charlotte going on a coffee run. By integrating over these observations, you can now infer a greater probability that Abbie and Charlotte will be observed together in the future, despite never having observed them together in the past. This co-occurrence probability acts as a useful signal for the existence of a relationship. If we think of these observations as “one-step” connections in a network graph, then the process of integrating over them produces a “multistep” representation in the mind. By integrating over many one-step observations, multistep inferences reveal the structure of an interconnected, relational system (Lynn & Bassett, 2020; Momennejad, 2020), including larger-scale social structures (e.g., communities, hubs, brokers, groups) reflecting the network’s topology (FeldmanHall, Son, & Bhandari, 2025).

If the statistics of observation are sufficient for learning about the structure of a social network, this suggests that core knowledge is not necessary for children to mentally represent social networks, particularly given infants’ aptitude for statistical learning. This is partly an empirical question to be tested, but it also raises an interesting theoretical quandary. Though mental representations of social relationships and social networks are both forms of relational knowledge, is it possible that they rely on different cognitive principles? Perhaps representing an intuitive theory of relationship categories is more akin to solving the problem of inferring mental states from behavior, whereas statistical learning about the existence and strength of a relationship is more akin to a structure-learning problem.

However, if we allow ourselves to be tempted by the possibility that the representation of social relationships and social networks is intertwined, then there are many interesting empirical questions and theoretical challenges for future work to address. In the spirit of “[intuitive] theories all the way down” (Gopnik & Wellman, 2012), I speculate that statistical learning could be used to discover distinct kinds of relationships: those that are mutually reciprocated, those that are sufficiently group-like that several individuals can be compressed into a single communal entity, and those that are asymmetric. Once a “proto-theory” is in place, this would provide the cognitive scaffolding needed to learn a richer intuitive theory specifying causal beliefs about (e.g.,) social roles, obligations, affordances, etc.

How could the statistics of observation be used to learn such a theory? A fully-fledged proposal has yet to be developed; here, I sketch

the outlines of what it might include. The cognitive foundation of equality-matching simply requires knowing that two people are related in some way, which is arguably present in the observation itself. If observations are sufficiently clustered in time and space, this could be used to discover group-like structures from dyadic, pairwise relations, serving as the basis for communal sharing. Directed sequences of observation could facilitate the discovery of a treelike structure, which could then be used to represent asymmetric relations and authority-ranking.

Of course, there are many other possibilities for developing unified theories of social relationships and social networks and in characterizing how statistical learning interacts with intuitive theories and/or core knowledge to produce rich relational knowledge. It is also possible that the representation of relationships and networks relies on such different cognitive principles that a unified theory is unworkable. But if there are common principles, discovering them will answer foundational questions in social cognition about how we humans navigate our complex and interconnected social world.

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Computing relational strength: an implausible component of early naïve sociology

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Abstract

Thomas proposes that two basic computations determine how infants represent social relationships: categorizing the connection by model and computing its strength. I argue that the concept of relational strength is ill-defined, its empirical evidence in early development unconvincing, and its putative operations better explained as categorical inferences about narrow relational schemas (e.g., caregiving).

At the core of Thomas' proposal is the claim that, beginning in infancy, humans are prepared to interpret the gamut of social relationships by appealing to three distinct models: communal sharing, authority ranking, and equality matching (as originally laid out by Fiske, 1992). Expanding upon previous attempts to frame early relational concepts within a Fiskeian framework (Thomsen & Carey, 2013; Tatone, 2017), Thomas expertly reviews a wealth of developmental evidence to argue that infants spontaneously form expectations consistent with the normative standards of these models (i.e., social equivalence, power asymmetry, and even balance). Ancillary to this proposal is the suggestion that, beyond categorizing a connection by model, infants may also compute its strength. Despite sharing the classificatory framework adopted by Thomas, I argue that the introduction of this second component into her account of naïve sociology is problematic for multiple reasons.

Thomas defines strength as a continuous representation of obligations (the extent to which certain actions are expected), and commitment (the likelihood that people will continue the relationship). Critically, strength does not define a continuum along which to organize relational models, but a parameter specifying the range of social expectations within a certain model. In other words, relational strength is computed only after interactions have been categorized under one of the three available models. Relationships falling under each model can consequently be weaker or stronger depending on their inferred strength. There are a few unstated implications of this proposal that are worth unpacking.

This account assumes that the observation of episodic interactions, like the ones that infants are presented with in the studies here reviewed, is sufficient not only to determine a model, but the extent of an individual's commitment to their social partner. This is an assumption that Thomas, albeit tacitly, is forced to make. If strength may be plausibly inferred via bookkeeping

asymmetries in social investment over repeated observations (e.g., by tracking the delay and magnitude of reciprocity between partners: Xue & Silk, 2012), neither the brief snapshots of social interactions that infants are provided within the course of an experimental session nor the limited opportunities for social monitoring in their everyday life appear suited for this kind of extended eavesdropping.

There must be, then, local proxies of relational strength that even isolated encounters may bear. The example that Thomas chooses to illustrate this point with is her saliva-sharing study (Thomas et al.): under her reading, infants expect comforting to occur between individuals shown to share saliva, rather than passing a ball, because, although both actions prime a communal frame, only the latter indexes a strong enough bond to license expectations of social support. As Thomas herself acknowledges, however, it is unclear why infants would categorize these two conspicuously different interactions under the same model. Considering that model identification is the first inferential goal that infants need to secure, and that the type of interaction observed is often the only stimulus feature differing between events in these experiments, it seems more likely that this information would be leveraged to classify interactions with respect to the relationships they token rather than to compute their strength. Corroborating this point, infants have been shown to categorize even superficially similar actions differing only in the direction of resource transfer (giving vs. taking) as instantiating qualitatively distinct exchange models (Tatone & Csibra, 2020). A second example that Thomas uses as evidence of strength computation in communal relation is Spokes and Spelke (2017), which showed that toddlers infer an unobserved bond between two agents if comforted by the same "adult", but not if comforted by different ones. It is unclear how this finding could support graded differences in strength, since the interaction patterns that infants compared were structurally different (one with a common partner, licensing transitive tie formation, and another without). It seems more plausible that infants interpreted these comforting actions as indicative of caregiving relationships, which, when combined in an open triadic arrangement, licensed inferences about siblinghood. Rather than exemplifying continuous strength computations, these findings instead suggest that, within the broader domain of communal relationships, infants may hold specific assumptions about kinship-informative acts and their combination (Lieberman, Tooby, & Cosmides, 2007).

Furthermore, the proposal that social interactions may bear two distinct classes of cues (i.e., model and strength identifiers), which infants are prepared to attend to, entails an implausibly large database of hierarchically organized cues, which sits at odds with the explanatory appeal of Relational Models Theory for cognitive development. Fiske's theory gained traction in the field due to providing a parsimonious solution to the "learnability problem": against the backdrop of infinite inductions about the meaning of interpersonal actions, a few relational primitives (and a corresponding sensitivity to their perceptual indicators) were proposed to suffice to constrain and bootstrap learning in the social domain (Fiske, Thomsen, & Thein, 2009). In this model, however, the in-built cognitive springboard is drastically expanded (to capture distinctions within each model based on connection strength), making its internal structure harder to justify on developmental grounds. This is not to say that cue integration does not occur in

early relational mapping, but it does so primarily to disambiguate models, not to compute the strength of their tokening interactions (cf. Steele et al., 2024).

Finally, in embedding the concept of strength within that of “interchangeability,” Thomas introduces yet another component in her model of naïve sociology which, while parcel of adult social reasoning, is unlikely to be an inferential target for infants. From the perspective of a naïve observer, it is unclear how observing an isolated interaction may engender considerations about the replaceability of the partners involved. For an observer to entertain said inferences, they would have to appreciate the trade-offs in bond formation that an agent faces so that their future social investments can be appropriately understood as redirected (i.e., A wants to be friends with B instead of C), rather than duplicated (i.e., A wants to be friends with B and C). This type of computation requires a network-level assessment of an agent’s standing, which is presumably beyond infants’ capabilities.

In lieu of graded computations of strength, I believe that narrow relational schemas (e.g., caregiving) may better account for the latitude of social expectations documented in the studies reviewed by Thomas without having to resort to implausible cue architectures.

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Relational core cognition is for understanding the meaning of social life, not just enduring interpersonal relationships

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Abstract

Understanding innate core cognition for social relations as a deep, generative social grammar that helps solve the learnability problem of discovering who relates to whom in what ways “amounts to asking how human social life is understandable, and hence possible” (Thomsen & Carey, 2013, p.17). Expanding the focus beyond enduring interpersonal relationships allows us to understand how social and societal meaning (e.g., in ideology) reflect fundamental forms of relational coordination ultimately grounded in cross-species affordances.

In their chapter *Core Cognition of Social Relations*, Thomsen and Carey (2013) proposed

that humans have evolved an innate, finite set of conceptual primitives that specify relational models for communal sharing, social hierarchy, and equality and are coupled with innate input analyzers that automatically identify some instances of each in the social world. Such a universal vocabulary of basic kinds of relational models would make it possible for young children to learn the relational makeup of their particular culture by specifying the kinds of social relationships they should be looking for and at least some initial ways of recognizing them (p.22).

The only empirical demonstrations from human infants at the time of writing were the discovery that preverbal infants represent social dominance, using the formidability cue of relative size (Thomsen, et al., 2011) and Over and colleagues’ (2009) demonstration that 18-month-olds more likely to help others if they have been primed with close, affiliative touch (i.e., communal sharing). This, then, is an exciting example of science moving on fast across many independent labs in a decade from the assumption that infants understand intentional agency (and whether agents help/hinder others in achieving their intentional goals), but ‘simply learn’ the other key aspects of the social world, to wide-spread hypotheses that some basic social relations manifest in the infant mind. Bringing a synthesis of much current evidence bearing on this general proposal before the scientific community for collective debate is an important contribution.

Thomas argues relational core cognition is for *concrete, continuing interpersonal relationships*. We argue instead that innate (shared) core cognition for abstract, skeletal relations coordinate social interaction at large and help solve the *learnability*

problem of discovering the structure of the social world. It's akin to a deep, *generative* grammar (Chomsky, 1967/2006) in which social meaning is created and experienced. This evolved relational 'grammar' specifies "rules of the games of social and societal life in intuitive, taken-for-granted ways" (Thomsen, 2020, p. 201), ultimately coordinating the distribution dilemmas of benefits and costs—resources, rights, obligations, and care—inherent in group life (Thomsen & Carey, 2013; see also Ho et al., 2025; Sheehy-Skeffington & Thomsen, 2020, 2023; Thomsen, 2020). Echoing at least the affordances of mammalian life, for instance, the impression of being one, categorical social body (*summary representation of unity*) is created by touch, spatial closeness, sharing food and bodily substances of breast milk, semen, saliva, and blood, moving in synchrony, and marking bodies to look alike (*cues*), in turn creating and prescribing the solidary sharing and caring (*coordinated normative interaction*) prototypically implemented among close kin, and the emotions that motivate it (Thomsen & Carey, 2013; drawing on Fiske, 2004).

Whether one thinks relational core cognition is for coordinating and understanding rule-based social life or for representing enduring, concrete interpersonal relationships have implications for the interpretation of the empirical record. For instance, Thomas suggests Ziv & Sommerville (2017) demonstrate expectations for even resource distribution, but not equality matching, because it is unclear if infants expect an enduring relationship. But these infants also gave away their own prized resource (toy), whereas infants expecting *inequality* refused to do so, reminiscent of the effects of social dominance orientation (SDO) on adult (in)egalitarian political ideology (Sheehy-Skeffington & Thomsen, 2020, 2023; Ho et al., 2025). This provides compelling evidence that core, abstract relational expectations motivate the navigation of resource dilemmas even in the infant mind.

Thomas also suggests there is no evidence that preverbal infants use relative size as cue for the dominance version of hierarchical relationships, because size is an individual trait that does not specify if people will keep on interacting. But size has no inherent meaning as an individual trait—like win-lose history and number of allies, it is a transitive formidability cue that only predicts the *mutual coordination* of yielding and prevailing in zero-sum conflict *relative* (i.e., 'relational') to others' formidability; repeated, coordinated interactions that over time will coalesce into an asymmetrical relationship of stable, transitive rank (which Gazes, Hampton, & Lourenco, 2017 confirmed that infants understand). Whether people *keep on* interacting is distinct from representing the adaptive affordances of this relational coordination, though its degree of asymmetry will reflect previous rank-based interactions. Following this, I doubt toddlers interpreted a relationship of *prestige* or a *caregiver*, rather than stable dominance rank, in our puppet version of the right-of-way dominance paradigm (Thomas et al.) because toddlers only chose individuals for whom others bow and move away in contexts of *zero-sum conflict*: Being bowed before without prevailing in conflict because the other yielded—a key indicator of stable dominance rank with critical adaptive benefits across species that stands in contrast to the definition of prestige as the ability to bestow benefits (Thomsen, 2020)—solicited no such preference. Conversely, even preverbal infants predict that others will join a more formidable, coordinated coalition (Fonn et al., 2025), and 14-18-month-olds prefer and help the highest resource-holding individuals (Eason et al., 2024).

This is not to say that toddlers don't also recognize authoritative, consensual instantiations of social hierarchy by dedicated cues. In fact, being bowed before—but *not* conferring

benefits, receiving tribute, imposing a cost, relative size, nor coordinated mutual movements—is a sufficient cue for Norwegian toddlers to predict that others will continue to follow an agent's directions in her absence (Margoni & Thomsen, 2024; modifying Margoni, Baillargeon, & Surian, 2018's paradigm which Thomas also suggests may show representations for caregivers). That was the case even though Norwegian toddlers have unlikely seen anyone bow, certainly not before their caregiver.

This points to the potency of evolved, core cues for predicting how relational coordination unfolds. Any particular social relationships are likely *structured* (Giddens, 1984) from a *learned* history of different relational interactions, represented as summary network structures reflecting cross-species affordances (e.g., close cohesion, vertical differentiation). Relational core cognition makes this *socializing learning* possible by specifying a coordinative social grammar grounded in critical, adaptive cross-species relationships (e.g., sharing bodily fluids/kin solidarity; relative formidability/rank), but unlikely depends on aspects that vary across ecologies, such as how long concrete interpersonal relationships endure.

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Attachment bonds: a foundation for understanding social relationships

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Abstract

Thomas proposes that infants classify social relationships using the categories of Relational Models Theory. Attachment theory and research describe an innate, early-emerging suite of psychological processes by which infants identify and bond selectively to caregivers. Classification of others' relationships might build upon these same mechanisms, with implications for future research on social cognition across development.

In the target article, Thomas proposes that infants' innate, primitive understanding (i.e., “core knowledge”) of the social world includes categorization of interpersonal relationships as communal sharing, authority ranking, or equality-matching, in accordance with A. P. Fiske's (1992) Relational Models Theory. At two points, Thomas briefly notes the possibility that identification of attachment relationships may be even more fundamental. Here, we argue that this is likely the case and develop that proposal further. Attachment theory describes a well-documented set of psychological processes potentially underlying infants' classification of relationships into these three models, with implications for future research on social-cognitive development.

Throughout the mammalian class, infants possess a suite of innate biobehavioral mechanisms for figuring out who is supposed

to care for them – their attachment figures – and optimizing the care received (Feldman, 2017). Attachment figures provide comfort, protection, and reassurance in times of uncertainty or distress (*safe haven*) and enable confident exploration of and engagement with the world (*secure base*; Bowlby, 1982). While it is not yet clear whether human infants define attachment figures categorically or along a continuum, they do so based on well-documented cues, and begin to show caregiver-specific proximity preference and separation distress at roughly six months of age – just as they begin to move around independently, and risk getting into trouble (Bowlby, Ainsworth, & Bretherton, 1992).

Fiske's relationship models map to features of attachment bonds in striking ways. First, the maternal-infant bond is arguably the primordial communal “we are one” relationship. Much of Thomas's evidence supporting infants' ability to identify communal sharing relationships focuses on synchrony – temporal concordance of biological and social processes between two or more individuals. Synchrony is the most widely studied predictor of attachment formation, and a foundation for healthy development (Feldman, 2003; 2007; 2012). As Thomas notes, infants are not only sensitive to cues of synchrony (Powell & Spelke, 2013), but also observing others' synchrony leads infants to expect that those individuals will be responsive to one another's distress (Kudrnova et al., 2024) and will help and support one another (e.g., Jin & Baillargeon, 2017; Pun, Birch, & Baron, 2021; Ting & Baillargeon, 2021). This suggests that infants may use synchrony as a cue to classify others' relationships in attachment-related terms, as well as their own.

Moreover, infants' attachment bonds are inherently authority-ranked. Attachment bonds are crucial for infant survival and development because the infant can defer to someone with more experience, resources, and physical capabilities – a useful “insurance policy” when exploring the environment. While we agree there is compelling evidence that infants can and do compare the relative rank (e.g., physical, material, and social) of those they observe interacting (e.g., Meng et al., 2019; Pun, Birch, & Baron, 2016), and expect those of higher rank to provide benefits to those of lower rank (Stavans & Baillargeon, 2019), this is consistent with our proposition that infants understand social relationships first in terms of attachment bonds. Beyond simply identifying others' relative ranks, infants expect these differentials to be stable over time (Bas & Sebastian Galles, 2021; Mascaro & Csibra, 2012). This expectation maps onto how infants track what they can expect from their attachment figures – the extent to which they can be consistently relied upon for resources and support – and calibrate their own behavior accordingly (Ainsworth, Bell, & Stayton, 1971; Shaver, Collins, & Clark, 2014).

Infant-caregiver attachment bonds are also inherently unequal, with clearly defined roles of caregiver and care-receiver. If infants do indeed categorize others' relationships in terms of attachment, equality-matching across interactions (or the lack thereof) may be one way to do so. With no prior information about a relationship, infants expect resources to be divided equally among interaction partners (e.g., Sloane, Baillargeon, & Premack, 2012; Ziv & Sommerville, 2017). However, after observing initial interactions in which someone gives and someone receives, infants expect the giver to keep giving in future interactions (Tatone, Geraci, & Csibra, 2015). Perhaps infants assume by default that others' relationships do *not* include attachment, and therefore should involve balanced give-and-take across interactions.

There is some evidence that infants generalize from their own attachment experiences to expectations for others' relationships. While viewing a video designed to evoke expectancies for caregiver behavior in attachment-relevant situations, securely attached infants expected the "caregiver" character to be responsive, whereas insecure-avoidant and insecure-ambivalent infants expected the "caregiver" to be unresponsive (Johnson et al., 2010). Our proposal raises a number of questions for future research. Is attachment-related core knowledge accessible in the first months of life, or only once the attachment system "turns on" and generates personal experience of attachment relationships? If the latter is true, we would expect greater homogeneity in infants' expectations about others' relationships prior to this point, and more diverse expectations and behaviors soon after infants are mobile.

Unfortunately, although mental models of attachment figures are central concepts in attachment theory, research explicitly examining infants' mental representations of attachment bonds is surprisingly sparse (Sherman, Rice, & Cassidy, 2015; Luo, vanMarle, & Groh, 2024). One study found that 4–12-month-old infants seem to expect adults to respond to infant crying – the most common cue of attachment distress – with care (Jin et al., 2018). As others have proposed (Luo et al., 2024), this suggests that mental representations of attachment bonds include behavioral scripts for and evaluations of attachment-relevant situations, which may be applied to understanding others' relationships as well as their own. Importantly, however, the emphasis on individual differences in the study of both infant and adult attachment has resulted in a dearth of research on how mental representations of attachment figures form. While these mental representations are thought to be malleable, it is unclear how stable they are. Is one schema-violating experience enough to alter mental representations, or do changes emerge only with new, but consistent experiences? Does this process change across development? If mental representations of attachment bonds are indeed a lens through which infants interpret their social world, empirical work documenting how they form and change is crucial to understanding socioemotional development.

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Social connection and mental state reasoning

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Abstract

Cognitive scientists have long investigated infants' abilities to reason about others' minds. Here, we identify a limitation to research on early mental state reasoning: that this research has mostly challenged infants to reason about the minds of strangers. In this commentary, we call on cognitive scientists to consider the role of social relationships when studying mental state reasoning and cognition more broadly.

Thomas reviews evidence that infants have core knowledge about social relationships: Infants reason about how people are connected, not only to each other but also to the infants themselves. At the same time that there is this growing body of research examining infants' capacities for reasoning about relationships, there is a large body of research examining infants' capacities for reasoning about others' minds (e.g., Scott & Baillargeon, 2017; Spelke, 2022; Woodward, 2009). The majority of research on infants' capacities for mental state reasoning has presented infants with individuals who are strangers (or near strangers) to the infants (e.g., Luo & Johnson, 2009; Onishi & Baillargeon, 2005; Southgate, Senju, & Csibra, 2007; Woo, Chisholm, & Spelke, 2024; Woodward, 1998). Yet, the people whose minds may matter most for infants' everyday lives may be the people in their social networks, especially their own caregivers.

Thomas reviews research that demonstrates that from birth, infants are attuned to their caregivers (e.g., DeCasper & Fifer, 1980; Field et al., 1984; Porter & Winberg, 1999), and that later in infancy, infants learn from their caregivers' social interactions about potential new social partners for themselves (Thomas, Saxe, & Spelke, 2022). It is possible that infants may be attuned not only to their caregivers' voice, smell, appearance, and social interactions but also to their caregivers' mental states. For infants, reasoning about their caregivers' goals is likely highly relevant for them to learn from and interact with their caregivers. Psychologists have long studied and debated the development of infants' abilities to

reason about different kinds of mental states (e.g., Baillargeon, Buttelmann, & Southgate, 2018; Phillips et al., 2021; Poulin-Dubois et al., 2018; Woo, Tan, & Hamlin, 2022): goals, knowledge states, and beliefs. It is possible that in presenting infants with strangers, we have underestimated their abilities to reason about certain kinds of mental states.

There is evidence that relationships matter for learning. For example, toddlers learn more effectively from their caregivers (van Rooijen, Bekkers, & Junge, 2019). Moreover, there is even evidence that toddlers learn more effectively from novel people who are socially engaging (i.e., making eye contact, smiling, and using infant-directed speech) (Breitfeld, Compton, & Saffran, 2024), and that toddlers more readily help people who have passed a ball to them (Cortes Barragan & Dweck, 2014). That is, infants do not only learn more effectively from and cooperate more effectively with people with whom they have long-established relationships (i.e., their caregivers), but also novel people who have sought to engage them. These findings may reflect an enhanced understanding of an individual's mental states. For children to learn, they need to understand the goal of the teacher (e.g., the goal of labeling an object) (Gweon, 2021). For children to help a person achieve a goal, they must understand the goal of the person (Hobbs & Spelke, 2015; see also, Woo & Spelke, 2023). How do children come to develop new relationships, identifying people whom they can learn from and cooperate with?

Thomas's research demonstrates that from their caregivers' social actions, infants learn about their caregivers' relationships, as well as how the infants themselves may be connected to other people (Thomas et al.): that someone could be part of the infants' own social networks. Such inferences could guide infants' learning. It is possible that infants may learn better from people within their social networks. When infants identify a person as being part of their social network, they may also be more motivated to learn from and cooperate with that person.

Educational researchers have examined the role that parents and guardians play in their children's learning. There is evidence that parental involvement in schools is positively associated with the development of skills relevant to literacy in preschool (Arnold et al., 2008), as well as with achievement in reading and mathematics in elementary school (Park & Holloway, 2017). These findings may reflect that, from increased parental involvement, children have inferred that their parents and teachers have a relationship: that the teacher is part of their social network. From increased parental involvement in schools, children may therefore learn more effectively from their teachers. Future research could test this possibility more directly and perhaps experimentally. We also note that education researchers, to our knowledge, appear not to have studied the role of parental involvement in learning in infancy. Given the findings that Thomas reviews, however, it appears that infants do track the relationships that their caregivers have with others. Thus, we call on psychologists and education researchers to investigate how parental involvement influences learning in infancy.

In sum, the evidence for core knowledge about social relationships should inspire researchers to consider how social connection shapes early mental state reasoning, learning, and cognition broadly. Children may reason about the minds of many people, but they may be especially likely to do so for people within their social networks. A comprehensive understanding of early mental state reasoning requires that we also understand children's relationships.

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Babies can be good at relationships and other things too: arguments for moral cognition in infancy

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Abstract

Thomas suggests that evidence suggestive of an innate moral core can instead be explained by relationship inferences and evaluations. We argue that the analogy between other demonstrations of prosociality (e.g., helping) and imitation is insufficient, and additionally highlight key findings that appear inconsistent with a relationship-centric view of infants' sociomoral capacities.

Thomas proposes that infants possess core capacities to identify and categorize relationships, and further suggests that such capacities can account for key findings in the study of early social cognition purportedly about other topics. Effectively, Thomas posits that infants are “pro-relationship” beings who (1) track and reason about the relationships between themselves and others as well as between third parties, and (2) prefer those who show evidence of initiating positive relationships with others. As part of this claim, Thomas argues that evidence suggestive of an innate moral core might be better explained by innate representations of social relationships: Rather than identifying who possesses positive *traits* such as helpfulness, infants

simply prefer those who “initiate or maintain prosocial relationships” (Box 1). Though we do not dispute Thomas’ claim that infants show impressive capacities for relationship inference, we argue that the body of evidence to date falls short of suggesting that infants privilege relationship inferences over moral/trait-based ones.

First, Thomas reviews evidence that infants expect imitators to later help their imitated targets, but not novel targets; that is, infants make relationship-level inferences for the further prosocial actions of imitators, but apparently do not make trait-level inferences (Kudrnova, Spelke, & Thomas, 2024; Pepe & Powell, 2025). They imply that these relationship-specific tendencies likely extend to other kinds of prosocial actors beyond imitators (e.g., helpers, fair resource distributors, etc.), and thus infants’ apparently moral inferences are actually about relationships. We find this suggestion puzzling. For one, although evidence does suggest that infants fail to infer broad prosocial traits from a single act of imitation even at 18 months, 14-month-olds do apparently infer broad antisocial traits from a single morally relevant act (Gill & Sommerville, 2023; Surian et al., 2018). We are also aware of no evidence that infants fail to make broader trait-level inferences about imitation (i.e., that an imitator of one target will not later imitate another); this evidence seems crucial to the claim that inferences about imitators are truly relationship-specific.

Second, Thomas’ central argument against an innate moral core requires that infants view imitation and helping as analogous (see Powell, 2022; Spelke, 2022). Yet, in studies including both imitation and helping, infants readily distinguish those who help from those who imitate, and prefer those who help (Hamlin, Wynn, & Bloom, 2007; Woo & Spelke, 2023); infants also prefer prosocial actors whose acts are not imitative (Geraci & Surian, 2011, 2023; Lucca, Pospisil, & Sommerville, 2018). Further, there is a well-documented “negativity bias” in early social cognition (see review in Vaish et al., 2008); this is presumably the result of limited cognitive systems choosing to privilege avoiding harmful stimuli. Indeed, negativity biases are present throughout the early “moral core” literature: 3-month-olds distinguish hinderers from neutral characters, but not helpers from neutral (Hamlin, Wynn, & Bloom, 2010), suggesting that negative social evaluation emerges first; relatedly, 6- and 10-month-olds expect a third party to avoid a hinderer but not to approach a helper (Chae & Song, 2018). Negativity biases are also observable neurologically: both 6- and 12-month-olds showed greater activity in the P400 ERP component (which selectively responds to social stimuli) after viewing hinderers versus helpers (Tan & Hamlin, 2022; see replication in Soleimani, 2023), suggestive of greater neural processing for hinderers. By contrast, we are aware of no empirical work within imitation demonstrating a negativity bias. Thus, although we agree that it would make sense to see imitators as better than neutral (they seem attentive to other agents’ actions and choose to reproduce them), it is unclear why humans’ cognitive system would view non-imitators – who in these studies merely produce a novel, neutral act that is irrelevant to infants themselves – as “bad” the same way they do hinderers. In sum, all of this work leads us to question whether the analogy between imitation and more morally relevant prosocial actions is appropriate.

Finally, Thomas posits that infants’ preference for prosocial over antisocial agents across diverse domains can be interpreted as being about relationships; that is, they desire to interact with those who initiate or maintain prosocial relationships. Although this pro-relationship view could account for infants’ basic preferences for prosocial agents, it is unclear how it explains findings from studies examining more complex questions. For instance, findings have shown that 4.5-, 8-, and 19-month-olds show context-dependent evaluations, preferring a character who hindered a previously

antisocial target over one who helped it (Hamlin, 2014; Hamlin et al., 2011). If infants simply prefer agents who initiate prosocial relationships, they should uniformly prefer helpers regardless of the recipient’s previous acts, because helpers always initiate prosocial relationships. The same explanation similarly fails to explain findings in which 8-month-old infants themselves punish third-party antisocial actors via selective attention (Kanakogi et al., 2022). If infants prioritize the establishment of prosocial relationships, they should avoid acting antisocially against third parties at all costs (as doing so is a sure way to establish an antisocial relationship). Further, a relationship-only framework struggles to explain why infants would selectively punish the aggressor instead of the victim, as punishing either agent could be interpreted as infants initiating a prosocial relationship with the other one. Instead, these findings seem better explained by more generic moral assessments: Infants view hinderers/aggressors as being “bad,” and thus it is seen as “good” to willingly engage in punishing them. This view is supported by evidence suggesting that by 13 months, infants associate antisocial actors with admonishment (DesChamps, Eason, & Sommerville, 2016).

In summary, though Thomas’ proposition of a purely relationship-based model of the social world is intriguing, we believe it currently falls short of fully explaining findings in the sociomoral domain. Indeed, it may be premature to extend findings from work with imitation to other prosocial interactions, and key findings in the broader “moral core” literature are difficult to explain via relationship inferences alone. Thus, although Thomas convincingly argues that infants can and do make relationship-specific inferences and evaluations, we look forward to future work that further explores the circumstances under which infants make relationship-specific inferences versus more general, moral trait-based ones.

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Author's Response

Cognitive primitives, learning, and the structure of early social relationship cognition

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Abstract

The broad range of commentaries addressed central debates in the origins, nature, and development of relationship knowledge. These debates mirror those that have played out in other domains: do infants possess innate, abstract primitives for representing relationships? Or is their understanding primarily the result of learning? Should we consider infants' behavior in experiments a reflection of abstract cognitive concepts or as reflexes or embodied knowledge? The commentaries also raise questions about other possible cognitive primitives as well as the evolutionary roots of such knowledge. What is the result of statistical learning, and what are inductive biases? Ultimately, these commentaries examine how innate representations of relationships may facilitate the resolution of other learning challenges, including the acquisition of cultural norms, institutional roles, group dynamics, and moral principles. Ultimately, I argue that advancing our understanding will require integrative approaches drawing on developmental, comparative, computational, and cultural research. This exchange clarifies many pressing empirical puzzles, setting the stage for a research program that I hope others across disciplines and theoretical standpoints will join in.

It was my honor and pleasure to engage with these thought-provoking commentaries. I am indebted to many of the authors whose previous work laid the foundation for this proposal and who inspired me to study this topic. Their diverse and thoughtful perspectives put forth many empirical questions, and I look forward to studies that will answer them. As I hope will be clear, there are several places where these commentaries either clarified or nudged my thinking.

R1. Preliminaries

I begin by clarifying some theoretical commitments. **Basso and Nolan** object to my claim that human minds (including infant minds) process information. However, decades of research strongly support the view that humans – including infants – and other animals recognize patterns, solve problems, and learn across diverse domains. This information processing helps us respond to diverse environments, innovate, and build cultural knowledge. My use of the terms “computation” and “information” is not meant to equate human beings with current technology. Instead, information can come from any process that is not random, and computation refers to the transformation of information. This does not mean that humans are maximally efficient, amoral, or self-interested, nor does it imply that people think about human relationships in the same way they think about bank accounts. I remain open to the possibility that people are intentional beings and that information can be processed not only within individual minds but also within bodies or between bodies, as when an orchestra collectively creates music (Vélez et al., 2023).

Following Susan Carey's definition, cognitive representations are states of our nervous system that process information and have content referring to concrete, abstract, or fictional entities, properties, or events (Carey, 2009). Importantly, the representations are not equivalent to the things to which they refer: relationships are not the same as cognitive representations of them. For example, just because actual relationships have fuzzy boundaries (as **Lavrador and Pietraszewski** correctly point out),

it does not mean that our understanding of them is not built from primitives that are categorical. Cognitive representations can, for example, change someone's behavior, allow them to make predictions, or help them imagine alternative possibilities. One type of learning occurs when cognitive representations change or become newly integrated with others, altering how we process information.

I am committed to the idea that looking-time measures are valid indicators of infant cognition. These measures yield reliable and replicable results across various study designs and stimuli, grounded in widely accepted theories of learning (Stahl & Kibbe, 2022). Available neural data suggest that infants as young as two months process information in ways similar to adults, in both cortical and subcortical regions (Kosakowski et al., 2024; Liu et al., 2023). This evidence supports the claim that looking-time measures can indicate higher-order cognition, challenging arguments that infants lack abstract representational capacity.

I am committed to the possibility that genes and natural selection could have endowed us with learning mechanisms, including representations, making such representations innate. This, of course, does not mean that all infants are identical at birth – a child may be born missing a limb – but that does not mean that humans are not genetically predisposed to have two arms and two legs. While **Aydin and Carpendale** raise important questions about the complex interplay of genes and experience, this complexity does not preclude the possibility of innate representations (refer also to Nettle & Scott-Phillips, 2023).

That said, my article did not deeply engage with learning. I covered a relatively large amount of developmental time (over 2 years), and many commentators were justified in questioning whether the evidence I provided supports the claims about innate representations. Any proposal regarding early representations of relationships must be placed in the context of the incredible amount of learning that occurs during infancy and childhood. Furthermore, any proposal about how infants come to know about social relationships should also consider how infants and children themselves shape the social interactions in which they participate (not just how they understand them). With these preliminaries out of the way, I will now address the specific issues raised.

R1. Is relationship understanding the result of embodiment?

In the target article, I claimed that infants have abstract cognitive representations. **Basso and Nolan**, along with **Aydin and Carpendale**, challenge this claim from an embodiment perspective, suggesting that we understand relationships from physical experiences. However, these embodied accounts face a problem: the concepts they invoke – such as “obligations,” “in/out,” and “balance” – are themselves abstract. If infants can represent obligations or containment relations, they already possess abstract representations. Moreover, how do infants generalize from their embodied experience to novel situations without an abstract representation? Consider my saliva-sharing study: when an infant observes a woman placing her finger in a puppet's mouth and then in her own, she must determine which features are relevant. An embodiment explanation may be that the infant understands the scene through her body's response and connections to past experiences. However, how does the infant know *which* experiences to connect it to in order to make predictions about what will happen next? Even if infants understand scenes through physical reactions, they must still organize these reactions; otherwise, we

would not observe the systematic behavior that we do. Abstract representations are particularly well-suited to organize diverse input (Goodman, Ullman, & Tenenbaum, 2011). These questions also apply to **Kanakogi's** argument that the foundation of moral cognition is somatic responses of infants (refer to the “Innate Moral Core?” section later in this document).

R1. Relationship representations and attachment

In the target article, I proposed that good candidates for primitives of relationship representations are Fiske's three models (communal sharing, asymmetrical relationships, and equality matching). Several commentaries show how this proposal contrasts with proposals about early relationship knowledge in attachment theory (**Nandrino and Gynberg**, **Groh and Luo**, **Rodriguez et al.**, **Vornlocher and Shiota**). The first way this proposal contrasts with attachment theory, on the surface, is the concept of an attachment figure, which is defined as individuals with whom infants form emotional bonds and from whom they seek proximity, comfort, safety, and protection in times of stress or threat. In the target article, I discussed the possibility that the representation, “care-giver” (similar to “attachment figure”), could be the result of an integration of representations of communal sharing relationships and asymmetrical ones. However, I also proposed that infants likely have representations that apply to a larger set of relationships than the specific definition of attachment figure in attachment theory. It is not that, as **Groh and Luo** suggested, that I misunderstood what attachment relationships are theorized to be. Instead, the target article was not clear enough about the idea that cognitive representations of relationships do not need to map onto “real relationships perfectly.” For example, while it may be true that attachment figures provide protection for infants, it does not necessarily follow that infants perceive these relationships in this way (albeit an interesting empirical question).

Second, according to attachment theory, infants do not have innate representations. Instead, the theory argues that human infants are born with predispositions for behavior (or “sensitivities”; refer to **Groh and Luo's** commentary) that give rise to emotional bonds with attachment figures. Specifically, Bowlby proposed that infants are *not* born with the internal working models, central to attachment theory, which are theorized to give rise to ideas about how relationships work. Instead, these working models are gradually built through experiences with their attachment figures (Bowlby, 1982)¹. The nature of these relationships and the corresponding internal working models is proposed to come from caregiver responsiveness and synchrony between caregivers and infants. This last point is supported by correlational and experimental evidence, which shows that caregiver responsiveness and synchrony predict behavior in the strange situation task, a method used to categorize the quality of infants' attachment to their caregivers (Juffer et al., 2017). Thus, while the commentaries that discuss attachment vary in the extent to which they agree that innate representations of relationships are possible, each of them argues that early experiences with caregivers significantly affect or give rise to representations of relationships².

Our disagreement centers not on whether early experiences affect representations – they clearly must – but when this happens in development, and on what level of abstraction. In other words, do infants' experiences with their caregivers create representations of relationships, or do such experiences update existing ones? And at which level of abstraction? Do they create the representation “relationship,” “communal sharing relationship,” “caregiving

relationship,” or “my relationship with this particular person”? I am inclined to think that infants learn a lot from interacting with and observing their primary caregivers, as well as the many other people they encounter. I am also inclined to think that, like other domains, this learning updates representations at different levels of abstraction (refer to Ullman & Tenenbaum, 2020, for a more detailed discussion on learning at hierarchical levels in development). However, the idea is that the starting state representations enable infants to organize the vast amounts of information they are bombarded with in everyday life, allowing them to learn an incredible amount over the first years of life.

Consider studies often cited as evidence of internal working models: securely attached infants, but not insecurely attached ones³, expect large characters to respond to small ones who display distress. Attachment theorists may take this finding as evidence that infants have learned through experience with their caregivers, the representation of “relationship,” and what it entails. However, it is not the only reasonable explanation of these findings. For example, infants may differ in terms of innate inclinations or temperaments that influence their behavior in the strange situation task, their synchrony with their parents, their parents’ behavior toward them, and their expectations in the experiment (Kagan, 2009; Kleppetto et al., 2024). Alternatively, my preferred explanation is that infants’ representations are updated, rather than created by experience with their caregivers. In other words, even if individual differences arise from experience with caregivers, it does not rule out the possibility of universal, innate, abstract representations. The proposal in the target article is that innate representations are likely to exist, not that they are static. Similarly, Rodriguez et al. raise questions about innate representations of relationships based on findings suggesting that prenatal experiences affect later relationships. Changes to emotional regulation, communication abilities, or physiology can certainly affect how humans interact and may create individual differences in relationships and people’s perceptions of them. However, again, none of this precludes innate cognitive representations that allow infants to recognize and categorize relationships. For example, in other domains, such as quantity approximation, there are known individual differences in accuracy, as well as extensive evidence that we are evolutionarily endowed with the ability to distinguish between quantities (Carey, 2009; Sarnecka, 2016; Spelke, 2022b).

The proposal that infants learn about relationships primarily through experience with their caregivers faces an additional challenge: infants demonstrate expectations about other kinds of relationships. For example, infants as young as 6 months expect that individuals with more allies will “win” a conflict. It is unclear how caregiver responsiveness, or synchrony with caregivers, would create such expectations. Moreover, in response to Groh and Luo’s proposal that infants’ behavior in the experiments is the result of sensitivities rather than cognitive representations, the question becomes: how do infants transfer experiences with caregivers to novel individuals and situations? What organizing principle(s) guide these sensitivities to the relevant social cues during their experiences with their caregivers? Finally, if the “attachment figure” is proposed as the representational framework guiding infants’ understanding of relationships, more empirical work is needed to show that infants distinguish this concept from broader categories of intimate, family-like, or communal sharing relationships.⁴

A third way the target article did not engage with attachment theory is the central claim regarding the evolutionary origins of attachment – namely, that attachment behaviors evolved as

adaptive mechanisms to ensure infant survival by promoting proximity to caregivers in the face of environmental threats. **Groh and Luo** ask, what evolutionary pressures would lead to innate abstract representations of relationships?

R1. What are the evolutionary origins of these representations?

In the target article, I only briefly covered the evidence that other species share our ability to represent relationships. However, I did not discuss the potential evolutionary origins of early cognitive representations of relationships. The commentaries by **Seyfarth** and **Grossman** consider these potential evolutionary origins. Seyfarth provides evidence that other species possess abstract representations of relationships. He also discusses how these representations improve fitness. For example, baboons have hierarchical (not in the social sense) representations of dominance in their social groups, where rank differences occur both across kin groups and within them. Importantly, these representations cannot have a 1:1 correspondence with perceptual features such as size, because rank is inherited. The ability to track specific individuals and their relationships increases fitness: recognizing social relationships informs animals’ decisions to interact with specific individuals. It has significant implications for survival and reproductive success. It is no surprise, then, that there are many pieces of evidence that several other species are endowed with abstract, implicit knowledge of social relations.

However, as **Seyfarth** points out, the question is how, and whether, ecological niches across species have led to different starting states. For example, would a gorilla, dropped into a chimpanzee group, come to understand the chimps’ social relationships? My prediction is that just as humans can watch chimpanzees and infer aspects of their social relations, so too could a gorilla observing a chimpanzee group (though perhaps not as enthusiastically as a bright-eyed graduate student). Ultimately, it is an intriguing empirical question. While putting a gorilla in a chimpanzee group would be unethical, one open and answerable question is the extent to which other animals perceive abstract animated shapes in a similar way to humans (Gergely & Csibra, 2003; Heider & Simmel, 1944). If they do, it would provide evidence that individuals across species could understand one another’s social interactions. This is not to say that there are no significant differences between what human infants and adults represent, and between those of other species and humans. The idea is that, like in other domains, human infants and other species share skeletal abstract representations of relationships that are starting states, from which learning is built.

Another question raised in these commentaries is how to think about human infants’ ecological niche. **Grossman** proposes a human and infant-specific evolutionary pressure: human infants are more likely to survive if they can find reliable caregivers in addition to their mothers, arising from many specifics of human reproduction and child rearing, including short interbirth intervals, early weaning, and hairless bodies that prevent human infants from clinging to their mothers (Hrdy & Burkart, 2022). This, he argues, may underlie infants’ ability to represent relationships and may shape which relationships they represent. Grossman uses kiss-feeding as a case study, which is prevalent across human cultures and is a valid cue of caregiving, because, as mentioned above, infants are weaned relatively early and depend on the provisioning of adults to survive. There are a few issues I see with this argument. First, non-cooperative breeders, such as

baboons discussed above, seem to have abstract relationship representations. Moreover, non-cooperative breeders engage in similar behaviors. While cooperative breeders, such as marmosets and wolves (Fentress & Ryon, 1982), do engage in kiss-feeding, non-cooperative breeders, including chimpanzees, orangutans, and bonobos, also exhibit mouth-to-mouth contact and lip-smacking as affiliative acts that help relieve social tension. Some monkey species put fingers in the mouths of conspecifics, possibly indicating social trust (one could lose a finger!; (Perry et al., 2003). In fact, Lameira and colleagues suggest that human kissing is “left over” from grooming practices found in other primates, which often end with a lip smack. When humans lost hair, we lost grooming, but we retained kissing as part of their bonding interactions (Lameira, 2024). Of course, the extent to which other species see these practices as cues of intimacy, as well as how much the concepts related to them overlap with those that may arise when infants understand kiss-feeding, are empirical questions, the answers to which could support or refute this proposal.

In summary, while I agree that cooperative breeding has shaped human inclinations, I am hesitant to conclude that it is the sole driver of our ability to represent relationships broadly. I think it is more likely that other proclivities of human infants were shaped by cooperative breeding. For example, infants’ ability to detect potentially human-specific cues of social attention (e.g., infant-directed speech, smiling, direct eye gaze) may be better candidates for human-specific mechanisms to ensure attention from mothers and others. Alternatively, human infants may be especially attuned to the social interactions of their caregivers in order to learn about the social relations of their caregivers, which may be a good cue about what other adults would be reliable caregivers.

R1. Underlying computations

I now turn my attention to the commentaries that disagreed with the specific underlying computations for relationship representations that I proposed in the target article. **Powell and Tompkins** agreed with the target article that infants may have core knowledge of social relationships. They disagreed with the target article that the primitives are likely to be Fiske’s relational models. Powell and Tompkins’ reasoning centers on my treatment of cues. In the original paper, I argued that infants use specific cues to recognize different relational models. For example, I proposed that imitation was a cue of communal sharing because it makes people unitary. Powell and Tompkins were correct to point out that, in everyday life, many social behaviors, including imitation, occur across the three models: Aside from communal sharing relationships, imitation can be indicative of asymmetrical relationships where low-ranking people often imitate high-ranking people, or equality matching relationships, where people may take turns doing the same thing, in essence imitating one another.⁵ While this may be true in actual relationships for adults or older children, young infants may understand imitation as especially indicative of communal sharing. This is an empirical question for future studies (But refer to Bas et al., 2022 for some evidence that 15-month-olds may perceive imitation as a cue of asymmetrical relationships).

That said, two commentaries, **Papeo, Mello, and Hochmann** and **Chen, Chandra, and Saxe** propose an alternative to the behavioral cues that I focused on in the target article: roles (refer also to Jara-Ettinger & Dunham, 2025; However, my treatment of them here is different than theirs, there is more below in the discussion of **Kaufman’s** commentary). Papeo et al. suggest that the primitives of relationships could include concepts of agent and

patient that loosely map onto linguistic concepts. Take imitation – infants may distinguish between interactions in which two individuals assume different roles (e.g., imitator and imitated) or the same roles (e.g., imitating one another). Both of these would be distinguished from non-interactions (e.g., people who happen to do the same thing without intending to interact, or perhaps for an instrumental purpose). The studies covered in their commentary, in which infants track the roles of individuals, support this argument. If so, infants should also be able to distinguish between social interactions that are asymmetrical and those that involve an agent and a patient, a question that is being actively tested in my laboratory. That being said, roles are unlikely to be the only way that infants make sense of social interactions. For example, infants distinguish across similarly structured interactions. Infants seem to see vocal imitation, as opposed to mere vocal reactions, as a cue indicating who will respond to distress. Likewise, infants seem to perceive saliva sharing, rather than ball passing, as an indicator of who will respond to distress (Thomas, Woo, et al., 2022). Across conditions within studies, the structure of the interactions between agents and patients remains the same; yet, infants distinguish between the two conditions. What allows infants to do so?

One possibility is statistical learning. For example, **Powell and Tompkins** argue that infants have learned a statistical association between physical intimacy and responses to distress. **Fabio et al.** propose that, while some innate predispositions may exist, statistical learning and cultural scaffolding account for much of infants’ social understanding through repeated observation of social regularities and contingencies. Son more radically argues that infants need not have any innate representations, but could, through statistical learning alone, acquire both relationship representations and knowledge of network structure. I agree that humans, including infants, are prodigious statistical learners. The question is not whether statistical learning informs relationship reasoning – it clearly must – but rather, what minimal starting state enables infants to organize the overwhelming complexity of social information they encounter. Ultimately, as **Chen et al.** argue, computational models that systematically vary initial constraints could help determine which combination of innate structure and statistical learning best explains the developmental trajectories observed (Griffiths & Chater, 2024; Lake et al., 2017; Ullman & Tenenbaum, 2020).

Another commentary critiques the target article’s proposal that Fiske’s models could be primitives from a different standpoint. In the target article, I proposed that equality matching is a distinct relationship representation. **Lavrador and Pietraszewski’s** commentary argued that equality matching is not a type of relation but rather underpins all successful social exchanges. The argument that all relationships involve reasoning about equality matching logic comes, at least in part, from the extensive evidence that, for example, humans are sensitive to free riders. However, this reasoning has most often been observed in studies involving strangers who are unlikely to meet again, as well as in games where the primary goal is to earn money. Thus, Lavrador and Pietraszewski’s idea may be truer for interactions outside of interpersonal relationships. Moreover, while pressures to ensure mutual benefits in relationships may make evolutionary sense, it need not follow that our proximate representations, especially the primitives, compute costs and benefits. Indeed, people expect reciprocal generosity in equal relationships, but not asymmetrical ones (Chen & Saxe, 2025). **Shore et al.** also engage with the equality matching section of the target article, proposing that intuitions about positive and negative reciprocity may follow distinct

developmental pathways due to the different evolutionary pressures that may have led to these behaviors. Their proposal implies that negative reciprocity intuitions may not be about relationships at all. Instead, negative reciprocity might operate more as a general response to harm or threat – potentially rooted in self-defense or status management rather than relational expectations per se. This may be true of *direct* negative reciprocity, but indirect reciprocity, including negative reciprocity, can certainly reveal something about relationships (refer to **Lavrador and Pietraszewski's** commentary and Pietraszewski, 2022). It would be interesting to expand on existing work on friendship (Afshordi & Liberman, 2021) to investigate whether young humans employ both direct and indirect reciprocity to infer relationships that they are not explicitly informed about.

In the target article, I proposed that infants may also recognize the “strength” of relationships, which I proposed was graded or continuous. **Tatone** argues that more specific relationship categories, such as “family” or “caregiver,” are more likely. He argues that adults compute strength by tracking repeated interactions over time, further suggesting that infants lack both the capacity and the opportunity to do so. First, I disagree that adults necessarily compute strength through direct evidence of repeated interactions. For example, as I argued in the paper, responses to distress may be especially indicative of strong relationships compared to responses to laughter or vocalizations. Second, infants can make graded distinctions (at least ordinally). There is empirical evidence that 10-month-olds track whether an agent incurs a high versus a low cost (Liu et al., 2017). There is also empirical evidence that 15-month-old infants use transitive inferences when reasoning about prosocial interactions (Pepe et al., 2025): when a protagonist helped A over B, and B over C, infants predicted the protagonist would help A over C, despite never having seen them interact directly. Thus, while I acknowledge that the evidence is limited, I disagree with the argument that infants cannot, in principle, track interactions over time or represent them in a graded fashion.

R2. Cultural norms and institutions

The target article focused on how core knowledge of social relationships underlie our ability to track and remember individuals and relationships in our social networks (i.e., “enduring relationships”). **Kaufman** and **Thomsen** show how these representations also likely underlie how humans come to understand cultural or institutional “rules of engagement.” Thomsen even argues that Fiske’s model is to cultural norms what universal grammar is to learning language. First, I agree that institutional and culturally specific norms affect how adults navigate and understand social interactions. Consider how a university student would think about two of his hierarchical relationships. The university student would expect his professor to teach and evaluate him but would be surprised if the professor asked the student to fetch her coffee. However, the same student working as an intern might reasonably expect to be asked to fetch coffee but would not expect academic instruction from his boss. Moreover, the student would negatively evaluate his professor if she demanded coffee, or his boss if she gave him an academic test. These differences are the result of cultural or institutional roles, not differences in the underlying model, both of which are asymmetrical (refer also to Chen & Saxe, 2025; Jara-Ettinger & Dunham, 2025). The adeptness with which adults navigate social interactions across different

contexts with varying rules of engagement presents a learning challenge for infants that may be comparable to learning a language. Thus, it seems reasonable that infants are endowed with representations that aid in this process.

These ideas fit in nicely with the proposal above, where the primitives answer questions about whether two people have the same role but are separate (equality matching), whether they have distinct roles (asymmetrical), and potentially whether they have the same role but act as one, or where the roles are fluid (communal sharing). These primitives could be flexibly built upon through statistical learning, where infants and children begin to recognize that the context affects how people in these roles behave.

One issue with the roles as primitives proposal is that empirical work suggests infants use other cues. Moreover, it is an empirical question when *infants* become sensitive to contexts that prescribe these rules of engagement. For example, it is hard to imagine an infant treating its parents differently because the encounter occurs at school instead of at home. What other patterns do children pick up? Does my two-year-old “know” that it is ok for her older sister to be naked in our house but not in “public”? How does she reconcile this with the changing room at the community pool? However, also, what tells her that we are in “public”? Would she be surprised if a guest were naked in our house? The broader questions about how infants and children learn about these rules of engagement are intriguing and pose important questions for future research.

That said, while I am happy to amend the target article to include how these representations help humans learn institutional norms, I remain convinced that infants also use them to represent specific relationships, which eventually allows them to understand their social networks. For example, something that is implied but not directly stated in **Thomsen's** critique (if taken to its extreme) is that humans and infants “forget” the characteristics of enduring relationships or are maximally flexible depending on the context. However, people do not seem to “forget” about enduring relationships, even when roles shift because of the context. For example, if you run into your professor at a dance club, you may feel awkward. The dance club would change the expected interactions, but memory of the relationship would remain. One empirical prediction that follows from this claim is that when a new context implies a *different* relational model, it causes a heavier cognitive load and also increased feelings of social awkwardness. For example, perhaps it would be easier to process if the employee joined an intramural basketball team where their boss was the captain. Moreover, as stated above, it is an open question when young humans first begin to grasp the implications of different contexts.

Kaufman also makes another important point about my omission of institutional norms. In the target article, I not only missed how institutions may affect how humans compute what other people *will* do, but even more importantly, what they *ought* to do. That is, as Durkheim pointed out over a hundred years ago, humans can imagine both an average person and an ideal one, and the target article failed to engage with this critical aspect of social cognition. Whether infants’ behavior in the studies covered by the target article reflects infants representing an “ideal” person is an intriguing possibility. In some ways, this would represent a radical departure from the way we have traditionally interpreted infants’ looking behavior in other domains. This interpretation would be akin to interpreting longer looking in violation of expectation studies about objects as a reflection of an esthetic evaluation of

unlikely events. However, in principle, infants looking across domains may reflect different things. Experiments about objects are typically not just improbable but impossible.

The possibility that infants discern what *ought* to happen relates to **Fiske's** commentary. In it, he argues that children are intrinsically motivated to engage in social relationships and do not need to be taught the value of social relationships, nor many of the prescriptive norms that surround them. This proposal contrasts with past models, where children gradually internalize prescriptive norms through explicit socialization. Fiske may underestimate the extent to which children receive input from others, since caregivers and peers can provide feedback in many implicit ways. However, the idea that from the start infants are motivated to learn how to “correctly” engage in social relationships instead of only passively understanding them seems right to me and is worthy of empirical investigation (refer to AP Fiske, 2010 for an example of clinical patients who can describe what is expected in social relationships but have severe social dysfunction, apparently because they lack intrinsic motivation to engage with others). Moreover, the idea that young children may use play to understand their culture's specific treatment of relational models is intriguing. An amendment is that children may even test out newly imagined cultural norms (Lew-Levy & Amir, 2024) or ways to navigate social conflicts.

R2. Moral core?

In the target article, I questioned whether we have evidence that infants make trait-based moral judgments. Several commentaries showed how my proposal could be expanded to include morality (**Baillargeon et al., Yuen et al, Kanakogi**). That said, defining “moral judgement” is inherently difficult (refer to Cushman, 2025, for review), and it is no surprise that each of these commentaries takes a slightly different angle on how infants may “have” morality. Baillargeon et al. argue for moral foundations (Graham et al., 2013) whereby infants have a “first” draft of moral principles. Yuen et al. argue for a trait-based morality, in which infants infer positive or negative traits from observing social interactions. Kanakogi suggests that infants somatically respond to moral violations, which lays the groundwork for the later development of cognitive moral judgments. Perhaps, one way to reconcile the three commentaries is to adapt Durkheim's distinction: do infants represent not just an average person but also an ideal one? This framing is helpful because it also allows us to integrate the current proposal; for example, we can ask whether infants represent both an average relationship of some sort and an ideal one.

As stated above, it is an intriguing possibility that infants' relationship cognition is prescriptive (as opposed to descriptive). However, in my view, we currently lack strong evidence that infants represent “ideals.” For example, **Baillargeon and colleagues** ask, if infants do not have prescriptive intuitions, “why would infants expect protection from high-ranking individuals and obedience from low-ranking individuals when they first see them interact?” The first answer is that infants do not have these expectations when they first see them interact. In these experiments, infants first see cues consistent with asymmetrical relationships, and then they look longer at some outcomes than at others. In other words, you have to *tell* infants about the relationship in these studies. The second answer is that their behavior in these experiments could be driven solely by expectations, without prescriptive reasoning. In other words, the

standard interpretation of the violation-of-expectation method is that the surprising events violate infants' predictions of what will happen, not what ought to happen. As mentioned above, an intriguing possibility is that these auxiliary hypotheses may be wrong in the social domain. This is a worthy empirical question. Third, in these studies, the participants are 17 or 21 months old. My own work suggests that infants under 16 months would interpret displays of asymmetrical interactions differently than these older participants, calling into question whether the behavior in these experiments should be considered a “first draft” (Thomas et al., 2018; Thomas & Sarnecka, 2019). Instead, infants may have innate representations of asymmetrical relationships that do not delineate dominance/prestige/authority, and do not include “ought.” This could enable infants to rapidly learn distinctions amongst asymmetrical relationships, as well as how people ought to act in legitimate authority relationships.

Putting aside asymmetrical relationships, **Baillargeon et al.** also discuss infants' longer looking at scenes in which someone gives more resources to one person compared to another. They argue that infants look longer toward unequal resource distributions because of prescriptive judgments. One issue with this comes from their own commentary, in which they argue that infants reason prescriptively about “moral circles,” that others ought to give preferential treatment to ingroup members. If infants indeed reason this way, why would infants think others ought to distribute equally, when they have no information about group membership? **Mackiel and Woodward** offer an alternative explanation to the longer looking toward unequal resource distribution. The longer looking toward unequal distributions may be because of an interest in understanding two different types of relationships: a positive one where someone is favored, and a negative/neutral one where someone is not.

When applying **Mackiel and Woodward's** argument to other social actions, we can see that they find support in findings from other studies that investigate helping and hindering. Repeatedly, researchers have found that infants *do not* look longer at hindering events compared to helping events. Based on Baillargeon's argument, we should expect that infants should look longer at hindering, which should violate the care principle – but infants do not. However, if Mackiel and Woodward are correct, then this looking pattern makes sense: both positive and negative actions update the infant's representation of the relationships between the protagonist and the helper or hinderer. By this logic, one might expect infants to look longer at a scene where someone hinders one person and helps another compared to a scene where someone either hinders or helps both people. While, in principle, it is possible that prescriptive reasoning does drive infants' looking, distinguishing between these two accounts is a question (albeit a difficult one!) for future research.

Kanakogi offers a different proposal regarding the developmental origins of moral judgment – specifically, somatic responses. This is a lower-level, non-cognitive explanation that would not require infants to have abstract representations of morality. Instead, infants would simply have negative physical responses when they see harm. Kanakogi focuses on his fascinating paper about third-party punishment. His explanation is that when infants see someone harming someone else, they experience a somatic response that leads them to punish the harmer in later trials. I have two questions about this paper. First, how should we think about the training trials in which infants learn to harm characters? If infants experience negative somatic responses to harming, why would they be willing to “learn” how to harm

innocent others? Do they stop harming as soon as they figure out that looking at an innocent character results in them being smooched? It might be worth examining those trials to see how long infants were willing to punish innocent others. My second question asks whether these studies distinguish between prescriptive and descriptive. While this study, along with others, suggests that young humans may recognize harm and even feel negatively about it, not all harm is considered immoral. If a tree branch falls onto someone's head, causing a concussion, we recognize harm has occurred. We might feel bad for the person or feel afraid of the tree. However, even if we update a descriptive belief about how likely branches are to fall from trees, that change is not necessarily accompanied by a prescriptive or moral judgment toward trees. While one study shows infants a scene where cues of animacy are removed (from both the smoocher and the thing being smooched), there is no control study where only one thing (the smoocher or the thing being smooched) is an object instead of an agent. If infants "punish" an object that smooches someone, it could reflect not punishment per se, but a desire to remove the negative thing. In other words, it is unclear whether we should think about infants' behavior as reflecting moral judgments or a desire to remove something that causes harm. Another tricky question is how these somatic responses eventually become integrated with infants' cognitive representations (refer to the embodiment section above).

While **Kanakogi** argues that somatic responses may underlie infants' moral judgment, **Yuen and colleagues** argue for a cognitive account of innate morality, suggesting that infants possess trait-based morality. They argue that when infants observe prosocial acts, such as helping or fair sharing, they infer a positive trait in the helper. When they observe antisocial acts, such as hindering or unfair distributions, they infer a negative trait. In the target article, I pointed out that infants do not seem to generalize traits from imitation (despite the evidence that infants reach for imitators). Yuen et al.'s response was that infants may interpret helping and imitation differently, leaving open the possibility that acts such as helping lead infants to make trait-based judgments. I shared this intuition. I predicted that infants would see imitation as more indicative of relationships, while helping would be more indicative of traits. However, recently completed studies do not support this proposal. A trait-based morality would predict that infants would use past helping or hindering behavior to update expectations of how the helper or hinderer would treat most or all people. In these new studies, 15-month-olds did not use helping or hindering toward someone to predict actions toward new characters (Pepe et al., 2025). In two studies, infants observed one character help a protagonist push a boulder uphill and another character hinder the protagonist. In new scenes, infants expected the helper (rather than the hinderer) to help the same protagonist, but this expectation did not generalize to a new character. Memory errors do not explain these results. The same infants successfully used transitive inference to make helping predictions: when shown A helping B over C and then C over D, they expected A to help B over D in a new scene. How do we reconcile these findings with the two other studies suggesting infants infer antisocial traits?

In one of these studies, infants observed a helper and a hinderer (in similar scenes to those above). Then, the helper and the hinderer were each shown in new scenes distributing resources. In response to

the scene where the helper distributed the resources, the infants behaved as they did in past experiments (where infants were given no information about the distributor) – looking longer at unequal distributions (Surian et al., 2018). However, when it was the hinderer in the scene, they no longer seemed to expect equal distributions, looking equally at both outcomes. These results were conceptually replicated in another study with infants aged 14 to 27 months. They went on to further demonstrate that infants looked longer when someone they had seen unequally distributed resources, then helped, as opposed to hindered, a new character (Gill & Sommerville, 2023). One explanation is that infants are more sensitive to negative events. However, this does not align with the data described above – infants were not surprised when a character who had previously hindered another character later helped a different character. Thus, the data are inconsistent with one another. To reconcile these differences, direct and conceptual replications of all three experiments, as well as testing narrower age ranges, may help shed light on whether and when, in development, infants make trait-based inferences or relationship-based inferences.

Suppose this further research supports the proposal that relational reasoning, rather than trait-based reasoning, is foundational to social cognition. In that case, it may represent a shift from the standard approach in social psychology. **Schubert and Seibt** stress this in their commentary. For example, the Stereotype Content Model proposes that we evaluate others based on whether they (or members of their group) tend to be warm or competent (though refer to S. T. Fiske, 2018 for a description whereby these traits are understood to be relational to the evaluator). Schubert and Seibt point out that an indiscriminately warm person may neither be seen as a good ally nor as very competent. They go on to argue that social evaluations should not be based on inferring traits from observing behaviors without considering the targets of those behaviors. Instead, people may primarily evaluate others based on whether they are acting in a manner that deviates from expectations about prescriptive norms within a relational model. This incorrect behavior, in practice, could arise from mismatches in the models people use in any given relationship. Their commentary also suggests a point that could help reconcile some of this dispute. They imply that people, including infants, may represent the value of relationships themselves rather than the value of individual people. Further studies could test predictions that follow this distinction.

As an interim summary, numerous empirical and theoretical puzzles remain to be solved. Currently, it is an intriguing possibility that infants possess prescriptive intuitions when observing social interactions. While most infant work has interpreted looking as a window into infants' predictions, it does not necessarily follow that this is the correct interpretation of the data in the proposal. One reason to suggest that infants make value judgments is the presence of evidence of evaluation in other domains. For example, infants crawl toward two crackers instead of one (even when they are not hungry!); Infants prefer looking at faces and listening to voices that seem to be engaging with them, compared to those that do not (refer to Spelke, 2022a for review). In fMRI studies, when infants (2 to 9 months) see faces, they show both face area responses, but also mPFC responses, an area that is associated with "value" in adults (Kosakowski et al., 2024). This leaves open the possibility that value plays a role in how infants interpret social interactions;

however, the current empirical support for early moral evaluations is unclear. Disambiguating predictions and evaluations will be tricky, but surely a worthy goal.

R1. Groups

In the target article, I argued that representations of groups and relationships are potentially distinct, and that many studies initially interpreted as evidence that infants represent ingroups and outgroups could also be interpreted as evidence that infants represent relationships. **Shutts and Kinzler** question whether it is reasonable to assume that a clear distinction exists between the two representations, ultimately arguing that social relationships are embedded within groups. First, I absolutely agree that relationships are embedded in groups. However, the key issue is not whether these representations may influence one another, but what the underlying representations are and when, in development, they become available to humans. In this paper, I argued that the distinction between a relationship representation and a group representation is the extent to which one generalizes to new people and how.

To expand on what I mean, consider an observer who represents two groups (the Montagues and the Capulets) and specific members of each group (Romeo, who is a Montague, and Juliet, who is a Capulet). If the observer learns the groups are at war, they may be surprised to learn Romeo and Juliet are in love. This is because, as **Shutts and Kinzler** point out, adults understand that relationships are embedded in groups. However, the observer can understand two things about this situation: first that the groups hate each other, and second, that individual members of each group love each other. In other words, to imagine the consequences of love across group lines, we need both group representation and dyadic representation. Adults can see how they are intertwined: after learning about Romeo and Juliet's affair, the audience could, in principle, update their expectations of whether other members of the warring groups will get along, but it need not be the case that they do. The tragedy of Romeo and Juliet, and the intuition that the audience can presumably easily grasp, is that Romeo and Juliet believe their families will forbid their love affair and may even deepen the conflict. Our ability to understand this play is only possible because we can hold these two representations separately. My argument is not that in the adult mind, these two kinds of representations operate with entirely different assumptions. For example, I think that relational models also apply to how we characterize groups. My argument is that the extent to which you generalize and individuate people implies *different* computations. There may also be special assumptions that go with one over the other. Ultimately, it is an empirical question whether infants have access to both representations. I stand by the claim that it is unclear from the current data whether infants represent "social group."

In asking this empirical question, we must also distinguish between ultimate and proximate causes of behavior. For example, let us assume that human infants are predisposed to notice when people speak the same or different languages, and this predisposition is the result of people being stratified by language groups in our ancestral past. It is still possible that the proximate representation that drives an infant's behavior in an experiment reflecting that predisposition is not a group representation. It may be limited to the two individuals in the interaction they see. Some evidence for this is the work showing

that 6-month-olds *only* predict friendliness after learning that two people prefer the same foods when those people appear together eating the same foods (Lieberman, Kinzler, & Woodward, 2021). In other words, the initial social interaction is important. **McCaffree** makes a stronger claim, arguing that the evidence for group categorization in infancy is "overwhelming." I obviously disagree⁶. While it is a worthy empirical question, it remains unsettled. For example, if infants have group representations, they should, at a minimum, make predictions about individuals they see coordinate beyond infants' attentional capacity to represent and track individual entities (i.e., four entities). To the best of my knowledge, this has not yet been tested, although my lab is currently investigating it.

R1. Social networks

In the target article, I focused on evidence that infants represent dyadic relationships; three of the commentaries (**Mascaro; Parkinson and Krems; Ozlu, Denison and Friedman**) show how this reasoning may also enable infants to represent social structures that extend beyond dyads. First, I agree with Mascaro when he argues that there is evidence that infants situate dyads into larger social structures. For example, he discusses his work wherein infants use transitive reasoning to infer dominance relationships (i.e., if A is higher ranked than B and B is higher ranked than C, the inference is that A is higher ranked than C) and wherein infants expect pyramidal hierarchies as opposed to a tree structure (i.e., it is easier for infants to recognize that A is higher ranked than B and C, but not that B and C are higher ranked than A; (Mascaro et al., 2023; Mascaro & Csibra, 2012). I agree with Mascaro that these abilities cast doubt on whether relationship representations can be reduced to core agency (c.f. **Powell and Tompkins'** commentary). While Mascaro does not argue for this himself, I believe that the evidence that infants represent larger structures also raises questions about whether infants reason about groups. If infants were primarily generalizing based on group markers, we might expect them to be less sensitive to structures inferred from individuals and the relationships between them. Of course, they may do both.

I also agree with **Parkinson and Krems** when they emphasize that more work needs to be done to understand the development of network representations. Recent work from my laboratory suggests that, by early childhood, children are accurately representing parts of their social networks. For example, 3- to 5-year-old children seem to know that their mother has a more intimate relationship with her own mother (the child's maternal grandmother) compared to the mother's mother-in-law (the child's paternal grandmother; (Steele & Thomas, 2025). Preschool-aged children, as young as three years old, are better at describing the social networks of their classroom than a randomly generated network, even when their own relationships are removed from the networks (Chuey, Gweon, Thomas, *in prep*). Finally, a recent small study conducted in my laboratory demonstrated that 24- to 36-month-old toddlers were nearly perfect at identifying the parents of their peers in a daycare classroom (Thomas, *in preparation*). While these studies are starting to shed light on what young children learn about their own social networks, very little is known about infants' representations of their social networks.

Moreover, as **Ozlu, Denison, and Friedman** highlight, another question is how young humans use general attributes of social networks to reason about relationships. In other words, do young humans use general properties of social networks to

understand relationships without directly observing them?⁷ For example, while children infer that people with more mutual connections are more likely to be friends, adults are also sensitive to other attributes of social networks. And, while there is some existing evidence that infants are sensitive to principles such as triadic closure, much more work is needed to investigate whether infants reason using assumptions based on the general attributes of social networks.

R2. How social relationship reasoning may affect other areas of cognition

In the target article, I focused on how infants come to understand their own social networks. **Woo and Laha** show how the ability to represent these relationships may have implications for other domains of cognition. They ask how other domains, such as mental state reasoning, may be affected by the relationship that the infant has with the person they are reasoning about. However, one question is by what mechanism? Woo and I collaborated on relevant work that provides evidence to support the claims made by Woo and Laha. Woo's studies built on the finding that infants aged 7–9 months fail to encode the target of a person's gaze. We replicated these past findings. However, in our studies, we find they do encode the targets of their caregivers' gaze – but only when the target is a social partner (a furry monster puppet who responds to their caregiver). If infants are inferring that their parents intended to look at one puppet rather than another, this supports Woo's proposal. However, the mechanism remains unclear: Are infants more motivated to process their caregiver's mental states? Is the caregiver's gaze easier to process? Or does longer looking when caregivers attend to new agents reflect infants' interest in potential additions to their social network? More work should be done to distinguish between different hypotheses that may drive these exciting results.

R1. Conclusions

Humans are not only dependent on others when we are born and throughout our long childhoods; we depend on others throughout our lives. Humans are also exceptionally good learners. In the target article, I proposed that one consequence of the centrality of social relationships to our survival and well-being is that we possess very early developing or innate knowledge that provides structure to the world of people into which infants are born. I proposed that this knowledge enables us to track and remember people and their relationships, as well as to understand the nature of these relationships. Implicit in this proposal was that understanding *enduring relationships* is essential. The wide array of commentaries expanded on and clarified this proposal. First, there are many open empirical questions about how we should think about the primitives of social relationship reasoning and what kinds of learning occur during the first few years of life. To what extent are infants' interactions with their primary caregivers central to learning about relationships? Regardless of the answer to these questions, there is much work to be done to understand how these primitives may help us learn the complex nature of institutional and cultural roles that govern much of adult human life. This is related to the question of how and when people come to understand large societal groups based on ethnic identities, race, or gender. Finally, there is much work to be done about where our moral reasoning comes from – do infants have a sense of “ought”? In summary, many open questions remain. I am optimistic that

this line of inquiry will inspire ongoing research and collaboration, deepening our understanding of how we think about one of the most important aspects of our lives: our social relationships.

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Notes

1 In 1982 Bowlby wrote the following, “The capacity to evoke and maintain enduring mental representations (images, or working models) of the mother-figure and of other familiar people is built up step by step and is dependent, among other things, on a level of intellectual development comparable with that which enables an infant to recall and search for a lost object.” It is not surprising that at the time of writing Bowlby would assume that infants are born without abstract representations! However, since the time of his writing, it has become clear that infants, as young as 3 months, do represent objects when they are out of sight as well as many other abstract representations (Spelke, 2022b).

2 Note, many of the other issues they raised in their commentary were either already addressed in the original proposal—e.g., why I favor relationship representations as opposed to group representations, or elsewhere in this response or both—e.g., evolutionary origins, moral intuitions versus relationship representations.

3 Infants are categorized as either securely attached, insecurely attached, or disorganized, by their behavior in the Strang Situation task which is structured observational assessment in which the infant experiences separations and reunions with their caregiver in an unfamiliar environment.

4 My graduate student, Christina Steele, has been working on this question for the past few years. Given the importance of caregiving to our survival, we hypothesized that infants may have a specific representation of caregiving. We began by asking whether infants expect larger, intimate social partners to respond to the distress of smaller individuals, as opposed to other smaller individuals. We failed to replicate our original positive result across several age ranges. We have even found evidence that children, up to age 6 years, do not, for example, expect mothers to respond to the distress of their children more often than those children's siblings.

5 In studies on how infants understand imitation, infants are shown imitation interactions that are repeated. For example, in one study, infants were shown an actor flanked by two puppets. Across conditions, we intended to manipulate whether the person in the middle was the target of imitation (one of the puppets vocally imitated her, and the other puppet made a different noise) or the actor in the imitation interaction (the person vocally imitated one of the puppets, but made a different noise than the other puppet). However, because we repeated the same scenes several times, with the same noises. This meant that the imitator was effectively also imitated. While there was a short pause between scenes, the infants may have interpreted the imitation to be symmetrical. We indeed found that infant's expectations about who would respond to distress were symmetrical – they expected both the targets of imitation and imitators to respond to their partner's distress.

6 McCaffree also argued that I should have framed this knowledge as “intuitive social psychology” instead of “intuitive sociology.” While I agree that sociology doesn't perfectly fit, I did not use “social psychology” because of social psychology's focuses on understanding individual minds in social contexts. As I noted, I don't think that humans need to represent the minds of others to represent relationships.

7 It is worth noting that the experiment they propose has effectively been done by Spokes and Spelke, where 17-month-olds expected that when two characters were comforted by the same character they would affiliate. It is possible that the 17-month-olds' inferences would have disappeared had the characters not had perceptual access to the interactions, but this seems unwell-being likely. This provides support for their hypothesis that infants may make inferences about network structure by observing individual relationships and inferring ones they have not observe dare born with knowledge that enables infants to structure.

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